



BUSS1020

Quantitative Business Analysis

*TURN BUSINESS DATA INTO DECISIONS – EVERY CONCEPT,
EVERY FORMULA, EVERY MARK.*

DESCRIPTIVE STATS

PROBABILITY

DISTRIBUTIONS

SAMPLING & CLT

CONFIDENCE INTERVALS

HYPOTHESIS TESTS

REGRESSION

The University of Sydney

Semester 1, 2026 · First-year undergraduate · Business School

Mapped to Berenson et al. (2023), Basic Business Statistics, 15e (Pearson) · closed-book · provided formula sheet · Excel as the everyday tool

Read it the way the exam asks you to think

A concept map, a worked method for every formula, and the traps that cost marks

IN ONE LINE

Every chapter follows the same rhythm: **one-line idea** → **definition** → **a diagram** → **the key formula** → **a worked example** → **the trap** → **the exam move**. BUSS1020 is a **method subject** — the formula sheet is handed to you, so the marks live in *choosing* and *applying* the right method, not in memorising algebra.

BUSS1020 takes you from raw business data to a defensible decision. The arc runs in one direction: **describe** the data you have (Ch 1–2), **model the uncertainty** in it (probability and distributions, Ch 3–5), understand **how samples behave** (Ch 6), then **infer** back to the population — confidence intervals, hypothesis tests and regression (Ch 7–11). Hold that map in your head and every formula has a home.

§ The repeating structure of each chapter

- **In one line** — the single idea, before any notation.
- **Define** — the terms and the conditions that make a method legal.
- **A diagram** — a decision tree or a curve, so you can *see* which tool fits.
- **Key formula** — in the dark box, exactly as it sits on the provided sheet.
- **Worked example** — a fresh business context, every step shown.
- **The trap** — the specific slip that loses the mark here.
- **Exam move** — how to attack this kind of question under time pressure.
- **Excel function** — the everyday tool BUSS1020 expects you to know (e.g. =NORM.DIST).

i Built to mirror the real unit

This bible is built to mirror BUSS1020 as it is actually taught and assessed at the University of Sydney: the week-by-week topic map, the Berenson 15e chapter coverage, the closed-book exam format and the provided formula sheet. Worked examples are **re-authored from scratch with new numbers and business contexts** — they teach the same skill the exam tests, never a real past-paper stem.

✓ Use it actively, not as a textbook

Cover the worked solution and try each example first. When you reach a decision tree (which distribution? which two-sample test?), **name the answer out loud before reading on** — naming the right method is exactly the judgement Part A multiple-choice and Part B written answers reward.

If you can classify the data, name the right model, and justify why the method is allowed — you have already earned most of the marks. The arithmetic is the easy part.

HOW TO READ EVERY BUSS1020 QUESTION



How BUSS1020 is assessed

Four components, two closed-book exams, one provided formula sheet

IN ONE LINE

Your mark is built from four pieces: weekly homework (15%), an in-semester test (20%), a group data-analysis assignment (25%) and a final exam (40%). Both the in-semester test and the final exam are closed-book, with a formula sheet (and an Excel-function page) supplied. You pass on a weighted average $\geq 50\%$ — the final exam is not a separate hurdle.

Component	Weight	Mode	Format & notes
Homework — MyLab, 11 weekly tasks	15%	Individual	Online in MyLab Statistics; 3 attempts per question, no penalty for retries; due most weeks. Task weights are unequal — four tasks are worth 2% each, the other seven 1% each.
In-semester test	20%	Individual	Closed-book, in person, 90 min. A mix of multiple-choice and short-answer; covers up to and including Week 6. Formula sheet + Excel-function page supplied; non-programmable calculator.
Group assignment	25%	Group (~5)	Analyse a real dataset with the course techniques. Three deliverables — team charter, a written report (20%) and a video presentation (5%). Peer assessment via FeedbackFruits; AI is allowed if acknowledged.
Final exam	40%	Individual	Closed-book, in person, 120 min + 10 min reading. Part A = 18 MCQ $\times$ 2 marks = 36; Part B = 5 written-answer = 64 marks (total 100). Part A spans Weeks 1–12; Part B covers Weeks 7–12. Formula sheet provided in the paper.

Book status & materials

- **Closed-book** for *both* the in-semester test and the final exam.
- A **formula sheet is provided** in each exam — including an Excel-function reference page.
- Allowed: an approved **non-programmable** (non-graphing) calculator, a physical bilingual dictionary, and blank paper supplied in the room.
- **No computer or Excel** in the exam itself — you read and interpret Excel output, you do not run it.

What the weighting tells you

- The **final exam alone is 40%** and the in-semester test 20% — **60% of your mark is closed-book exam performance**. Master the methods, not the software.
- Part B of the final (64 of 100 marks) is **written-answer** on Weeks 7–12 — inference and regression. That is where this bible spends most of its pages.
- The group assignment (25%) is the only place you *use* Excel and write for a business audience — the skill the unit ultimately certifies.

60%

CLOSED-BOOK EXAM WEIGHT

 $\geq 50\%$

WEIGHTED AVG TO PASS

Wk 1–12

EXAMINABLE SCOPE



The one-line hook

The first-year USyd unit where you learn to turn business data into decisions — **descriptive stats, probability, distributions, sampling, confidence intervals, hypothesis tests and regression** — all **closed-book on a provided formula sheet**, with **Excel as your everyday tool**.

Eleven chapters, one exam-ready map

From describing data to multiple regression – the whole subject in order

IN ONE LINE

The unit splits into three movements: **describe the data** (Ch 1–2), **model uncertainty** (Ch 3–6: probability, distributions, sampling), then **infer and predict** (Ch 7–11: intervals, tests, regression). The in-semester test covers Ch 1–6; the final exam covers all eleven.

#	Chapter	Week · text
01	Foundations: Data, Variables & Sampling Population vs sample, parameter vs statistic, variable types, levels of measurement, and how to collect a good sample.	Week 1 · Berenson Ch 1
02	Numerical Descriptive Measures Summarising centre, spread and shape, plus quartiles, boxplots and visualisation.	Week 2 · Berenson Ch 3.1–3.4, 3.6 & Ch 2
03	Basic Probability Probability rules, conditional probability, Bayes' theorem and counting.	Week 3 · Berenson Ch 4
04	Discrete Probability Distributions Random variables and the binomial, Poisson and hypergeometric models.	Week 4 · Berenson Ch 5
05	Continuous Probability Distributions The uniform, normal and exponential distributions and the standard normal.	Week 5 · Berenson Ch 6.1–6.5
06	Sampling Distributions & the CLT How sample means and proportions behave, and why the Central Limit Theorem matters.	Week 6 · Berenson Ch 7
07	Confidence Intervals Estimating population means and proportions, plus required sample sizes.	Week 7 · Berenson Ch 8.1–8.6
08	Hypothesis Testing: One-Sample Tests Testing a single mean or proportion, with the ethics of statistical reporting.	Week 8 · Berenson Ch 9
09	Hypothesis Testing: Two-Sample Tests Comparing two means (pooled, separate, paired) and two proportions.	Week 9 · Berenson Ch 10.1–10.3
10	Simple Linear Regression Covariance, correlation, the SLR model, fit, inference, and prediction vs confidence intervals.	Weeks 10–11 · Berenson Ch 13
11	Multiple Linear Regression Several predictors, model fit, the F-test, dummy variables and residual diagnostics.	Week 12 · Berenson Ch 14.1–14.4, 14.6

How the chapters map to the exams

Chapters 1–6 (data, descriptive measures, probability, discrete and continuous distributions, sampling distributions) are the **in-semester test** scope. **Chapters 7–11** (confidence intervals, one- and two-sample tests, simple and multiple regression) carry the **written-answer Part B** of the final — and are examined again in Part A across all twelve weeks.

A revision plan that follows the marks

The back half of the unit dominates the written-answer final

✓ Where the marks concentrate

Spend your revision time in proportion to the marks: the back half of the unit — **inference and regression (Ch 7–11)** — dominates the written-answer final. The front half is fast-moving foundations you must be fluent in, because everything later assumes them.

For the in-semester test (Ch 1–6)

- Be **fast and accurate** on descriptive measures and probability — these are MCQ-heavy and reward fluency, not depth.
- Drill the **distribution decision** (binomial vs Poisson vs hypergeometric vs normal) until naming the model is automatic.
- Know the **sampling-distribution intuition** ($SE = \sigma/\sqrt{n}$ shrinks with n) cold — it is the bridge into every later inference topic.

For the final exam (Ch 1–12)

- Part B (64 of 100 marks) lives in **Weeks 7–12** — confidence intervals, hypothesis tests and regression. This is where written method and interpretation earn marks.
- Practise **writing the conclusion in context**, not just computing a statistic — examiners reward the business interpretation.
- Rehearse **reading Excel/regression output**: identify coefficients, p-values, r^2 and the F-test from a printed table.

★ The proportionate-effort rule

If a topic carries **more marks on Part B**, it deserves more of your revision hours. A perfectly memorised Chapter 1 and a shaky Chapter 13 (regression) is exactly the wrong trade — the marks are not where the comfort is.

Foundations: Data, Variables & Sampling

Population vs sample, the parameter–statistic split, and the vocabulary the whole unit rests on

IN ONE LINE

We almost never measure a whole **population**; we take a **sample**, compute a **statistic** (a Latin letter — $\bar{X}$, S , p), and use it to estimate the unknown **parameter** (a Greek letter — μ , σ , π). Mnemonic: **Parameters describe a Population; Statistics come from a Sample.**

BUSS1020 is the first-year USyd unit where you turn business **data into decisions**. Everything that follows — descriptive measures, probability, distributions, sampling, confidence intervals, hypothesis tests and regression — sits on the Week 1 vocabulary. Getting these labels right is not pedantry: **the data type and the way you sampled decide which method is even legal later.**

1.1 Population vs sample · parameter vs statistic

The two collections

- A **population** is **every** item of interest — all N units (every customer, every transaction, every store). Usually too big, too costly, or impossible to measure in full.
- A **sample** is the **subset** you actually observe — n units, with $n \leq N$. You compute numbers from it and reason back to the population.

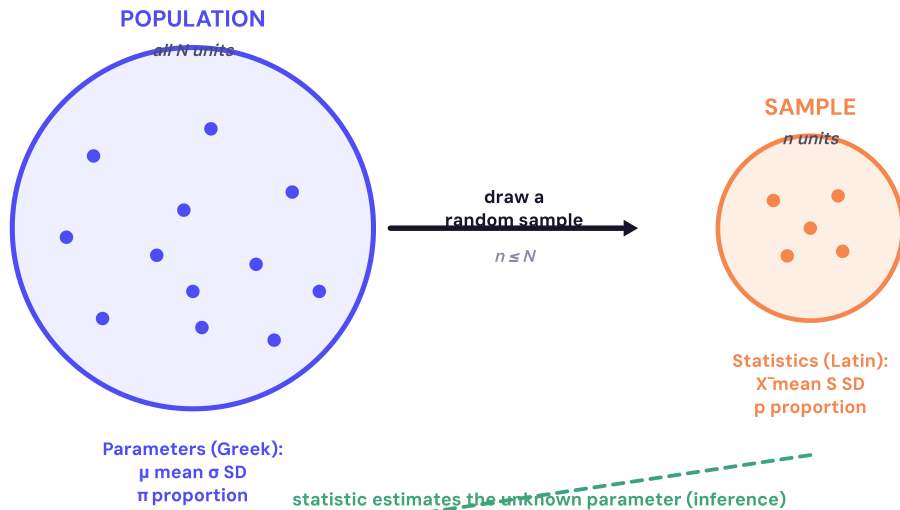
The two kinds of number

- A **parameter** is a fixed (usually unknown) number describing the **population** — population mean μ , SD σ , proportion π . Written with **Greek** letters.
- A **statistic** is computed from the **sample** — sample mean $\bar{X}$, SD S , proportion p . Written with **Latin** letters. It varies from sample to sample.

CHAPTER 1 (CONT.)

The inference picture

How a sample statistic stands in for the population parameter you cannot see



Population → sample → inference. We draw a random sample ($n \leq N$), compute a **statistic** (Latin: $\bar{X}$, S , p) and use it to estimate the unknown **parameter** (Greek: μ , σ , π). The dashed green arrow is the act of inference – the rest of the unit is machinery for doing it well.

i Greek for the population, Latin for the sample

The single most useful habit you can build in Week 1: as soon as you read a number, decide whether it describes the **whole population** (parameter, Greek – μ , σ , π) or a **sample you collected** (statistic, Latin – $\bar{X}$, S , p). Confusing the two is the root of most Week 6–9 errors (e.g. dividing by N vs $n-1$, or using σ vs S).

! “Sample” is a noun here, not the verb

In stats a **sample** is the subset of data; **sampling** is the process of choosing it. A large sample is still only a sample – it is **not** the population, and its mean $\bar{X}$ is still a statistic, never the parameter μ . Only a true **census** measures the whole population.

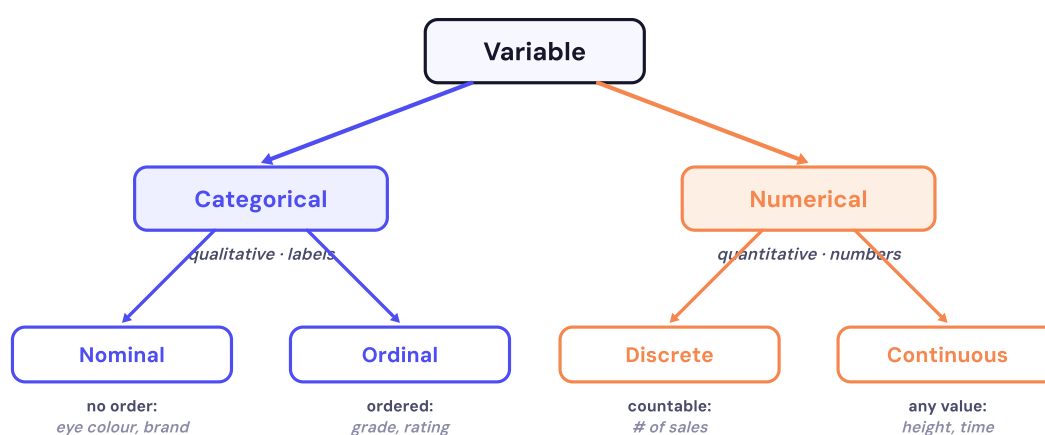
CHAPTER 1 (CONT.)

Variable types & levels of measurement

Categorical vs numerical, then the nominal < ordinal < interval < ratio ladder

At every step of an analysis you must know a variable's **data type** — it decides which chart, summary and test are valid. Two questions answer it: **(1)** is the variable **categorical** (labels) or **numerical** (counts/measurements)? and **(2)** what **level of measurement** does it carry?

1.2 Classifying a variable



Levels of measurement (information increases →)



First split: categorical (qualitative labels) vs numerical (quantitative numbers). **Categorical** is **nominal** (no order – brand, eye colour) or **ordinal** (ordered – rating, grade). **Numerical** is **discrete** (countable – number of sales) or **continuous** (any value in a range – height, time). The bottom strip ranks the four **levels of measurement** by how much information they carry.



CHAPTER 1 (CONT.)

Categorical vs numerical - the four levels

Why the data type decides every chart, summary and test you may use

Categorical vs numerical

- **Categorical (qualitative)**: values are **labels/categories** — gender, brand preference, payment method. Arithmetic is meaningless.
- **Numerical (quantitative)**: values are **numbers** you can do arithmetic on — counts, prices, times. Split into **discrete** (countable, e.g. number of employees) and **continuous** (any value in an interval, e.g. temperature, weight).

Why it matters downstream

Categorical data → counts, proportions, bar/pie charts, the χ^2 -style thinking. Numerical data → means, SDs, histograms, boxplots, correlation and regression.

Choosing a mean for a categorical variable, or a bar chart for a continuous one, is an instant error.

Level	Adds over the one below	Business example
Nominal	labels only — no order	brand chosen, store region
Ordinal	a meaningful order (gaps not equal)	satisfaction: low/med/high
Interval	equal gaps, but no true zero	temperature in °C, calendar year
Ratio	equal gaps and a true zero → ratios valid	revenue, weight, # of returns



Nominal < Ordinal < Interval < Ratio — information rises

Each level keeps everything the level below it has and adds one thing: order, then equal intervals, then a true zero. A **true zero** is what lets you say “twice as much”: \$40 revenue is twice \$20 (ratio), but 40 °C is **not** twice as hot as 20 °C (interval — its zero is arbitrary).



Numbers can still be categorical

A variable coded with digits is not automatically numerical. **Postcode 2006**, a **jersey number**, or “1 = card, 2 = cash, 3 = transfer” are **nominal labels** — averaging them is nonsense. Ask “*does arithmetic mean anything?*”, not “*are there digits?*”.

CHAPTER 1 (CONT.)

Where data comes from

Primary vs secondary sources, and structured vs unstructured form

1.3 Data sources & structure

Before you analyse anything you should know **who collected it** and **what shape it is in**. Two independent distinctions: the **source** (primary vs secondary) and the **structure** (structured vs unstructured).

Primary vs secondary source

- **Primary data** — you (the analyst) collect it for the question at hand: a survey you run, a designed experiment, an observational study, or transactional data captured inside the business (e.g. your own sales records).
- **Secondary data** — collected by **someone other than the analyst**, then reused: government statistical bureaus, internet-analytics providers, market-research firms, published syndicated datasets.

Primary data fits the question exactly but costs time and money; secondary data is fast and cheap but may not match your definitions, units or timeframe.

Structured vs unstructured

- **Structured data** — fits neatly into **rows and columns** (a database or Excel worksheet): **one column per variable, one row per item**. Easy to store, query and analyse.
- **Unstructured data** — irregular and not table-shaped: text messages, emails, social-media posts, images, audio, video. It needs interpretation (manual or automated) before analysis.

The “tidy” rectangle — one variable per column, one observation per row — is the target shape for everything you do in this unit.

Clean before you compute

Even correctly collected data may carry **invalid values** (text where a number belongs, out-of-range entries), **coding errors** (inconsistent category labels, stray characters) or **integration errors** (duplicate rows, redundant or mismatched columns, mixed units). **Data cleaning** — fixing, deleting the bad cell, deleting the whole row, or substituting a value — is mandatory: errors and extreme values make every later statistic unreliable.

“Garbage in, garbage out — the most elegant regression in the world cannot rescue a dirty, badly sampled dataset.”

BUSS1020 · THE WEEK 1 DISCIPLINE THAT PROTECTS EVERY LATER WEEK

Exam move — name the source AND the structure

A classification item often asks for **both** axes. Decide source by **who collected it** (you → primary; someone else → secondary) and structure by **does it fit rows × columns** (yes → structured; free text/media → unstructured). They are independent: a survey *you* run that records open-ended comments is **primary but unstructured**.

CHAPTER 1 (CONT.)

Sampling methods

Probability samples (random, systematic, stratified, cluster) vs non-probability

1.4 How to choose your sample

A **sampling frame** is the list of all N units you could draw from. How you draw from it splits sampling into two families. Only **probability samples** — where every unit has a known, non-zero chance of selection — support valid statistical inference; **non-probability samples** are quick but can be badly biased.

Method	How it works	Watch for
Probability samples — every unit has a known chance of selection		
Simple random (SRS)	every unit, and every sample of size n , is equally likely	needs a full, accurate frame
Systematic	pick $k = N/n$; choose one unit at random in the first group, then every k -th after	bias if the list has a hidden cycle of period k
Stratified	split the population into homogeneous strata (e.g. by age, region), sample randomly within each	need the strata variable for every unit
Cluster	split into clusters (e.g. suburbs, stores); randomly pick whole clusters and survey all units in them	cheap, but less precise per unit
Non-probability samples — selection chance unknown → inference not justified		
Convenience	take whoever is easiest to reach (people walking past, web volunteers)	strong selection bias
Judgement	an expert hand-picks the units they think are representative	depends entirely on the chooser

Stratified vs cluster — the classic mix-up

- **Stratified:** use **every** group, sample **some** units from each. Groups are meant to be **internally similar** and different from each other; boosts precision.
- **Cluster:** use **some** groups, then take **every** unit in the chosen ones. Clusters are meant to each look like a **mini-population**; chosen mainly to save cost.

A one-line test

Ask: “How many of the groups did I keep, and how much of each did I sample?” **All groups, part of each → stratified. Some groups, all of each → cluster.**
That single distinction is a perennial exam item.

✓ Systematic in one breath

For $N = 40$ and a target $n = 4$, the interval is $k = N/n = 10$: randomly pick a start among the first 10, then take every 10th unit. Fast and even-spread — but if the frame repeats every 10 (e.g. every 10th house is a corner block) the sample inherits that pattern.

CHAPTER 1 (CONT.)

Survey errors

Sampling vs non-sampling error, and the biases that no sample size can fix

1.5 What can go wrong

Two very different things can spoil a survey. **Sampling error** is the natural chance variation from observing a sample instead of the whole population — it **shrinks as n grows**. **Non-sampling errors** are flaws in the design or execution; a bigger sample only makes a biased result **more precisely wrong**.

Sampling error

Even a perfect random sample will not exactly match the population — different samples give slightly different $\bar{X}$ or p . This is **sampling error**: random, unavoidable, and **reducible by taking a larger sample**. It is the error the rest of the unit (standard errors, confidence intervals) actually quantifies.

Non-sampling error (bias)

Systematic problems that **do not** go away with more data — coverage error, selection bias, non-response, and measurement/recording error. These shift the result in one direction; only better **design** fixes them.

Error / bias	What it is	Example
Sampling error	random sample-to-sample variation (reducible by larger n)	$\bar{X}$ from one sample $\neq \mu$
Coverage error	the frame omits part of the target population	phone survey misses non-phone households
Selection bias	some units are more likely chosen than they should be	only loyal customers in the list
Non-response	those who don't answer differ from those who do	busy shoppers skip the survey
Measurement error	the recorded value is wrong (bad wording, mis-entry)	leading question, typo in entry

! A bigger sample cannot cure bias

The famous “**Dewey Defeats Truman**” poll had a **huge** sample but a biased frame — so it was confidently wrong. **Coverage and selection bias are non-sampling errors**: increasing n shrinks sampling error only. If the frame or selection is skewed, you just get a tighter estimate of the *wrong* number.

★ Exam move — diagnose the error in two steps

(1) Would the problem **shrink if you sampled more people**? Yes → **sampling error**. No → it's a **non-sampling** bias.
 (2) Then name the bias: who is **missing from the frame** (coverage), who is **over-/under-chosen** (selection), who **didn't reply** (non-response), or was the **value mis-recorded** (measurement).

CHAPTER 1 (CONT.)

Worked example: classify the design & the variables

Name the sampling method, then label four business variables by type and level

WE-1 A market-research plan, classified end to end

WE-1 · DESIGN + VARIABLE CLASSIFICATION

A market researcher studying small businesses in a city plans to **divide the city into its suburbs, randomly choose 3 suburbs, and then survey every business in each chosen suburb**. For each business they record four variables: **(a)** preferred bank brand, **(b)** number of employees, **(c)** the storefront temperature in °C on the survey day, and **(d)** the percentage of sales returned (returns ratio). Classify the **sampling method**, then give each variable its **type** and **level of measurement**.

Part 1 — name the sampling method

- 1 **Are the groups (suburbs) chosen, or sampled within?** The researcher keeps only **some** groups (3 suburbs) and then surveys **every** business inside each kept group.
- 2 **Some groups, all units within → cluster sampling.** (If they had kept *all* suburbs and sampled *part* of each, it would be **stratified**.)
- 3 It is a **probability sample** (suburbs picked at random), so statistical inference is justified — though cluster designs are usually less precise per unit than an SRS.

Part 2 — classify the four variables

Variable	Type	Level	Why
(a) preferred bank brand	Categorical	Nominal	unordered labels; arithmetic meaningless
(b) number of employees	Numerical · discrete	Ratio	a count with a true zero (0 employees) → ratios valid
(c) temperature (°C)	Numerical · continuous	Interval	equal gaps but no true zero (0 °C is arbitrary)
(d) returns ratio (%)	Numerical · continuous	Ratio	true zero (0% returns); 6% is twice 3%

Cluster

SAMPLING METHOD

4 / 4

VARIABLES CLASSIFIED

°C = Interval

THE EASY-TO-MISS ONE

! Temperature in °C is interval, not ratio

It is tempting to call every measured number “ratio”. But **°C has no true zero** — 0 °C does not mean “no temperature”, so 40 °C is **not** twice 20 °C. That makes it **interval**. A **count** (employees) and a **percentage of a total** (returns) both have a genuine zero, so they are **ratio**.

★ Exam move — the Week 1 classification drill

(1) **Sampling**: ask “how many groups kept, how much of each sampled?” — all groups/part each → **stratified**; some groups/all each → **cluster**; one even-spaced run → **systematic**; whoever’s easy → **convenience** (non-probability).
 (2) **Variable**: categorical vs numerical, then nominal < ordinal < interval < ratio — and remember **°C = interval**, **counts & percentages = ratio**.

Numerical Descriptive Measures

Summarising a column of numbers by its centre, its spread and its shape

IN ONE LINE

Describe any numerical variable with three questions: where is the **centre** (mean, median, mode), how much do values **spread** (range, S, CV), and what is the **shape** (skew, kurtosis). The mean = $\sum X_i/n$ is the workhorse, but it is pulled toward outliers — the median is not.

Week 2 turns a raw column of data into a handful of summary numbers a manager can act on. Every measure here is either a **centre** (a typical value), a **spread** (how consistent the data are) or a **shape** (how lopsided). Remember the Week 1 mnemonic: a **parameter** describes a **population** (Greek: μ, σ); a **statistic** describes a **sample** (Latin: $\bar{X}, S$). Almost everything below is a sample statistic.

2.1 Measures of central tendency

Mean — the balance point

The **arithmetic mean** $\bar{X}$ is the sum of the values divided by how many there are. It uses every observation, which is its strength and its weakness: one extreme value drags it noticeably.

Median & mode

The **median** is the middle of the *ranked* data (average the two middle values if n is even) — it ignores how far away the extremes are, so it **resists outliers**. The **mode** is the most frequent value; data can have no mode or several.

CENTRAL TENDENCY

$\bar{X} = \frac{\sum X_i}{n}$ median = middle of ranked data mode = most frequent value

i Which centre to report

Use the **mean** for roughly symmetric data with no wild outliers. Switch to the **median** when the data are **skewed** or contain outliers (incomes, house prices, delivery times). The **mode** is the only centre that works for *categorical* data (the most common category).

CHAPTER 2 (CONT.)

The geometric mean

The honest average when quantities multiply across periods

2.2 Geometric mean — for multiplicative data

When values **multiply** across periods — investment returns, growth rates — the ordinary mean overstates performance. Use the **geometric mean** of the n values, and for a series of returns the **geometric rate of return** built from the growth factors $(1 + R_i)$.

GEOMETRIC MEAN & RATE OF RETURN

$$\bar{X}_G = (X_1 \cdot X_2 \cdot \dots \cdot X_n)^{1/n}$$

$$\bar{R}_G = [(1+R_1)(1+R_2)\dots(1+R_n)]^{1/n} - 1 \quad \text{Excel: } =\text{GEOMEAN}(\text{range})$$

WE-2B

Geometric rate of return on a managed fund

WE-2b · WHY GEOMETRIC, NOT ARITHMETIC

A fund returns **+30%** in year 1, **-10%** in year 2 and **+5%** in year 3. What single annual rate, compounded, would give the same end wealth?

- 1 Turn each return into a growth factor $(1 + R)$: 1.30, 0.90, 1.05.
- 2 Multiply the factors: $1.30 \times 0.90 \times 1.05 = 1.22850$.
- 3 Take the n -th root and subtract 1: $\bar{R}_G = (1.22850)^{1/3} - 1 = 1.0710 - 1 \approx 7.10\% \text{ p.a.}$

! Never arithmetic-average percentage returns

The arithmetic mean of +30%, -10%, +5% is +8.33% — but that **overstates** the true compounded growth of 7.10%. Whenever quantities **multiply** across periods (returns, growth rates, index changes) the **geometric** mean is the honest one; arithmetic only fits quantities that **add**.

CHAPTER 2 (CONT.)

Measures of variation

Range, variance, standard deviation, the coefficient of variation and the Z-score

IN ONE LINE

Spread says how consistent the data are. The **standard deviation S** is the typical distance from the mean; the **coefficient of variation CV** = $(S/\bar{X}) \times 100\%$ makes spreads comparable across different units; the **Z-score Z** = $(X - \bar{X})/S$ says how many SDs a single value sits from the mean.

Range — quick but fragile

The **range** = $X_{\text{largest}} - X_{\text{smallest}}$. It is instant but uses only the two extreme values, so a single outlier blows it up. Prefer the variance / SD for a measure that uses *every* observation.

Variance & standard deviation

The **variance** averages the squared deviations from the mean; the **standard deviation** is its square root, back in the original units. Squaring stops positive and negative deviations from cancelling and penalises large gaps more.

SAMPLE VS POPULATION VARIANCE

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} \quad s = \sqrt{s^2} \quad \text{population: } \sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

! Sample divides by n-1, population by N

A **sample** variance divides by **n - 1** (the degrees of freedom — it corrects for estimating $\bar{X}$ from the same data, giving an unbiased S^2). A **population** variance divides by **N**. Excel: **=STDEV.S** uses n-1; **=STDEV.P** uses N. In this course you almost always have a *sample*, so reach for STDEV.S and n-1.

CHAPTER 2 (CONT.)

Relative spread & the Z-score

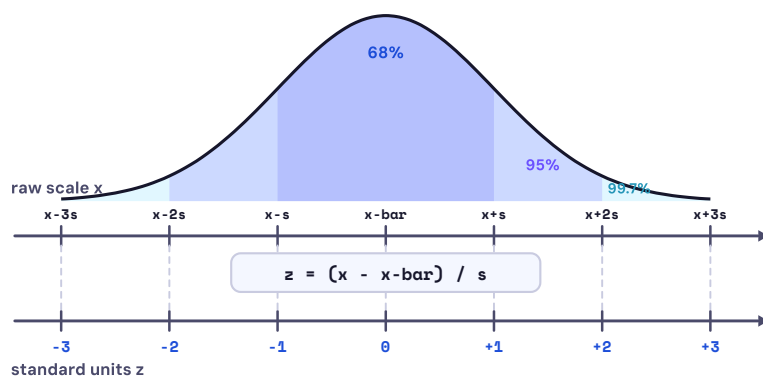
Comparing spreads across units, and how far a single value sits from the mean

Coefficient of variation — spread, scale-free

The **CV** expresses S as a percentage of the mean, so you can compare the relative variability of variables measured in different units or of very different sizes (e.g. a \$-priced stock vs a cents-priced one). The bigger the CV, the more variable the data *relative* to its own average.

COEFFICIENT OF VARIATION & Z-SCORE

$$CV = \frac{S}{\bar{x}} \times 100\% \quad Z = \frac{x - \bar{x}}{s} \quad CV = \text{relative spread}; Z = \text{standard units}$$



Standard units (the Z-score). Each value is re-expressed by how many SDs it sits from the mean: $Z = (x - \bar{x})/s$. The raw scale $(\bar{x} \pm k \cdot S)$ maps tick-for-tick onto the z axis ($-3 \dots +3$). A positive Z is above the mean, a negative Z below; $|Z| \geq 2$ is unusual.

✓ Read a Z-score in one sentence

$Z = +1.5$ means “this value is **1.5 standard deviations above the mean**”; $Z = -0.4$ is “a bit below average”. Because Z strips out the units, you can compare a mark on one test with a mark on another even when their means and SDs differ.

CHAPTER 2 (CONT.)

Worked: centre, spread & a Z-score

One small dataset, all five measures – the bread-and-butter Week-2 question

WE-2A Daily café orders over seven days

WE-2a · MEAN / MEDIAN / S / CV / Z

A small Sydney café records online orders on seven days: **12, 18, 15, 9, 21, 14, 17**. Find the mean, median and mode, the standard deviation, the coefficient of variation, and the Z-score of the busiest day.

- 1 Mean.** $\bar{X} = (12+18+15+9+21+14+17) / 7 = 106 / 7 = \mathbf{15.14}$ orders.
- 2 Median & mode.** Rank the data: 9, 12, 14, **15**, 17, 18, 21. The 4th value is the middle, so median = **15**. No value repeats, so there is **no mode**.
- 3 Standard deviation.** Sum the squared deviations $\sum (X_i - 15.14)^2 \approx 94.86$, divide by $n - 1 = 6$ to get $S^2 \approx 15.81$, then $S = \sqrt{15.81} \approx \mathbf{3.98}$ orders.
- 4 Coefficient of variation.** $CV = (3.98 / 15.14) \times 100\% \approx \mathbf{26.3\%}$ — moderate relative spread.
- 5 Z-score of the 21-order day.** $Z = (21 - 15.14) / 3.98 \approx \mathbf{+1.47}$ — about 1.5 SD above the mean, a busy but not extraordinary day.

15.14

MEAN ORDERS/DAY

3.98

STD DEVIATION S

26.3%

COEFF. OF VARIATION

+1.49

Z OF BUSIEST DAY

! n-1 and squared deviations are where marks are lost

Two classic slips: (1) dividing the squared-deviation sum by **n** instead of **n - 1** — for a sample always use **n - 1**; (2) forgetting to **square-root** the variance, so you report S^2 (15.48) where **S** (3.93) was asked. Keep at least one extra decimal in $\bar{X}$ before computing deviations — rounding $\bar{X}$ early distorts **S**.

★ Exam move — the “describe this data” checklist

Asked to **summarise a dataset**? Quote a **centre** (mean, plus median if it might be skewed), a **spread** (**S**, and **CV** if comparing to another variable), and a **shape** word (symmetric / right-skewed / left-skewed). For a single value, give its **Z-score** to say how unusual it is. Excel one-liners: **=AVERAGE**, **=MEDIAN**, **=STDEV.S**.

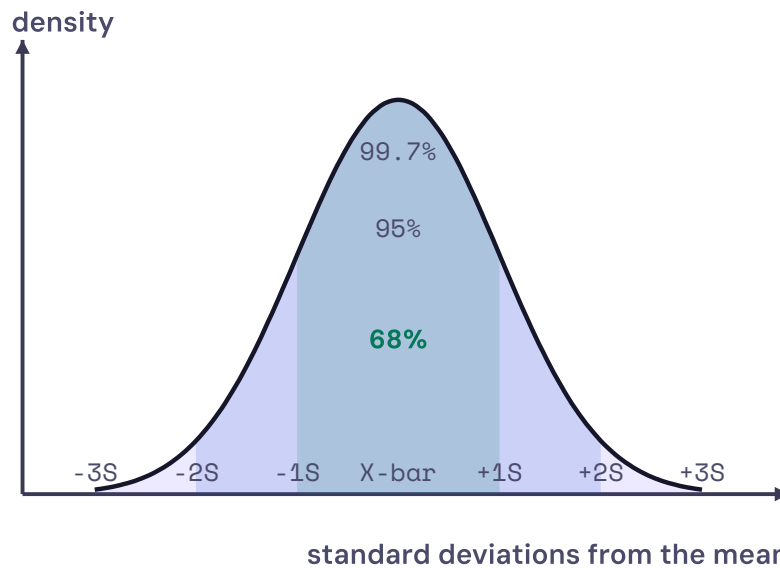
CHAPTER 2 (CONT.)

Bands, skewness & kurtosis

How much sits within k SDs, and what the shape of the data tells you

IN ONE LINE

For bell-shaped data the Empirical Rule gives $\approx 68\%$ / 95% / 99.7% within ± 1 / ± 2 / ± 3 SD. For any shape, Chebyshev's Rule guarantees a weaker floor: at least $(1 - 1/k^2)$ of values within k SD. Skew is read off the mean-vs-median gap.



Empirical Rule (bell-shaped data): about **68%** / **95%** / **99.7%** of values lie within ± 1 / ± 2 / ± 3 standard deviations of the mean. **Chebyshev** is the weaker but universal floor for *any* shape: at least 0% within $\pm 1S$, $\geq 75\%$ within $\pm 2S$, $\geq 88.9\%$ within $\pm 3S$.

Empirical Rule (bell-shaped only)

If the data are roughly **normal**: $\approx 68\%$ lie within ± 1 SD, $\approx 95\%$ within ± 2 SD, $\approx 99.7\%$ within ± 3 SD. This is the “almost everything inside 3 SD” intuition behind Z-scores and (later) the normal distribution.

Chebyshev (any shape)

For *any* distribution, at least $(1 - 1/k^2) \times 100\%$ of values fall within k SD ($k > 1$): so $\geq 75\%$ within 2 SD, $\geq 88.9\%$ within 3 SD. It is a guaranteed floor, weaker than the Empirical Rule but valid without assuming bell shape.

EMPIRICAL RULE VS CHEBYSHEV

Empirical (bell): **68%** / **95%** / **99.7%** within ± 1 / ± 2 / ± 3 S

Chebyshev (any): at least $(1 - 1/k^2)$ within $k \cdot S$ ($k > 1$)

! Empirical Rule needs a bell; Chebyshev does not

Only quote **68/95/99.7** when the data are roughly **symmetric and bell-shaped**. If the distribution is skewed or unknown, fall back to **Chebyshev** — its bounds ($\geq 75\%$, $\geq 88.9\%$) hold for every shape but are deliberately loose. Note Chebyshev says *nothing* for $k = 1$ ($1 - 1/1^2 = 0\%$).

CHAPTER 2 (CONT.)

Shape: skewness & kurtosis

Reading lopsidedness off the mean-vs-median gap

2.3 Shape: skewness & kurtosis

Skewness measures lopsidedness; **kurtosis** measures how heavy the tails are (how prone the data are to extreme values) relative to a normal curve. The fastest read on skew is the **gap between the mean and the median**.

Shape	Tail	Mean vs median	Excel
Symmetric	balanced	mean $\approx$ median	=SKEW $\approx$ 0
Right-skewed	long right tail	mean $>$ median	=SKEW $>$ 0
Left-skewed	long left tail	mean $<$ median	=SKEW $<$ 0

i The mean chases the longer tail

A few large values (a long **right** tail) drag the mean **above** the median $\rightarrow$ **right-skew**; a long **left** tail pulls the mean **below** the median $\rightarrow$ left-skew. The median, immune to extremes, stays put — which is why the **mean-median gap** is your instant skew detector. Excel: **=SKEW**, **=KURT**.

Kurtosis — how heavy the tails are

Kurtosis describes the **tails**, not the peak: how prone a distribution is to **extreme** values relative to a normal curve. **Leptokurtic** (positive excess kurtosis) data have **heavier tails** — more outliers than normal; **platykurtic** (negative) data have **lighter tails**. A normal curve is the **mesokurtic** benchmark (excess kurtosis $\approx$ 0). In business data, heavy tails (financial returns are a classic case) mean extreme events are more common than a normal model would suggest.

✓ Skew and kurtosis answer different questions

Skew asks “*is it lopsided?*” (which side is the long tail). **Kurtosis** asks “*how often do extremes happen?*” (how heavy the tails). A distribution can be perfectly **symmetric yet heavy-tailed** — so always report shape on **both** axes, not just skew.

CHAPTER 2 (CONT.)

Quartiles, the boxplot & good charts

Position rules, the five-number summary, reading skew off a box, and an IQR normality check

IN ONE LINE

Quartiles cut ranked data into four. With position rules $Q_1 = (n+1)/4$, $Q_2 = (n+1)/2$, $Q_3 = 3(n+1)/4$, the $IQR = Q_3 - Q_1$ is an outlier-resistant spread, and the five-number summary draws the boxplot.

Quartiles & IQR

Rank the data, then locate each quartile by its **position**. If a position is fractional, interpolate between neighbours. The **IQR** is the spread of the middle 50% — unlike the range it ignores the extremes, so it is **resistant to outliers**.

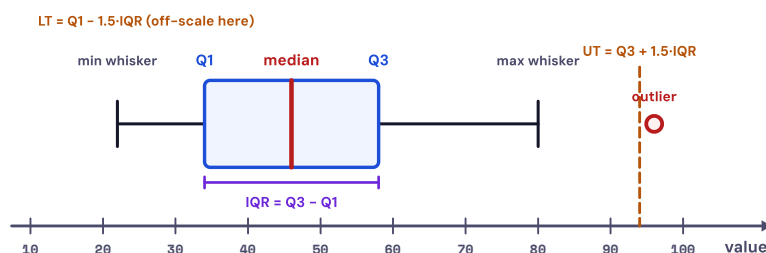
Five-number summary

The five numbers $X_{\text{smallest}} \cdot Q_1 \cdot \text{median} \cdot Q_3 \cdot X_{\text{largest}}$ are exactly what a boxplot draws: a box from Q_1 to Q_3 with the median inside, and whiskers to the extremes.

QUARTILES, IQR & FIVE-NUMBER SUMMARY

$$Q_1 = (n+1)/4 \quad Q_2 = (n+1)/2 \quad Q_3 = 3(n+1)/4 \quad IQR = Q_3 - Q_1$$

5-number summary: $X_{\min} \cdot Q_1 \cdot \text{median} \cdot Q_3 \cdot X_{\max}$



The boxplot from a five-number summary. The **box** spans Q_1 – Q_3 (the IQR) with the **median** inside; **whiskers** reach the extremes inside the $1.5 \cdot IQR$ fences, and points beyond a fence are **outliers**. A longer upper whisker / median nearer Q_1 signals a **right skew**.

! Position rules give a POSITION, not the value

$Q_1 = (n+1)/4$ tells you **where** in the ranked list the quartile sits, not its size. For $n = 9$, Q_1 position = 2.5 means “halfway between the 2nd and 3rd values” — interpolate. Also rank the data **first**; quartiles of an unsorted list are meaningless. (Excel =QUARTILE.EXC / =PERCENTILE.EXC use this $(n+1)$ convention.)

CHAPTER 2 (CONT.)

Worked: a five-number summary & visualising well

Quartiles, IQR and boxplot skew, the IQR normality check, and the rules of a good chart

WE-2C Nine weekly delivery times

WE-2C · QUARTILES / IQR / 5-NUMBER SUMMARY

A courier records nine weekly delivery times (minutes): **22, 25, 28, 30, 31, 33, 35, 40, 52** (already ranked). Build the quartiles, IQR and five-number summary, then read the skew.

- 1 **Q_1 position** = $(9+1)/4 = 2.5$ → halfway between the 2nd (25) and 3rd (28) values → $Q_1 = (25+28)/2 = 26.5$.
- 2 **Median (Q_2) position** = $(9+1)/2 = 5$ → the 5th value → median = **31**.
- 3 **Q_3 position** = $3(9+1)/4 = 7.5$ → halfway between the 7th (35) and 8th (40) values → $Q_3 = (35+40)/2 = 37.5$.
- 4 **IQR** = $Q_3 - Q_1 = 37.5 - 26.5 = 11$ min.
- 5 **Five-number summary**: 22 · 26.5 · 31 · 37.5 · 52. The long upper reach to 52 (40, 52 stretch the top whisker) ⇒ **right-skewed**.

26.5

 Q_1 (MIN)

31

MEDIAN (MIN)

37.5

 Q_3 (MIN)

11

IQR (MIN)

Assessing normality with the IQR

A quick rule of thumb: for a **normal** distribution the IQR is about **1.33 standard deviations**. Compare a dataset's IQR against $1.33 \cdot S$ — a large mismatch (or a long whisker) warns that the data are **not** normal, which matters before you lean on the Empirical Rule or normal-based methods later in the course.

ASSESSING NORMALITY

Normal rule of thumb: $IQR \approx 1.33 \times S$ big mismatch → not normal → prefer median & IQR

Organising & visualising data — principles of a good chart

- **Pick the chart for the data type**: bar/Pareto/pie for *categorical*; histogram/boxplot/scatter for *numerical*.
- **Start bar-chart axes at zero** and keep scales honest — a truncated axis exaggerates differences.
- **Maximise data-ink**: drop 3-D effects, heavy gridlines and chart-junk that hide the message.
- **Label clearly**: title, axis labels with units, and a source — the reader should not have to guess.

★ Exam move — the quartile & boxplot checklist

- (1) **Rank first**, then apply the position rules $Q_1 = (n+1)/4$, $Q_2 = (n+1)/2$, $Q_3 = 3(n+1)/4$; **interpolate** fractional positions.
- (2) **IQR = $Q_3 - Q_1$** ; quote the **five-number summary** to draw the box. (3) Read **skew** from whisker length / median position. (4) For shape vs normal, compare **IQR with $1.33 \cdot S$** . Excel: **=QUARTILE.EXC**, **=PERCENTILE.EXC**.



Basic Probability

Week 3 · Berenson Ch 4 – the language of uncertainty, before any distribution

IN ONE LINE

Probability is a number between 0 and 1 that measures how likely an event is. You can get it three ways — **a priori** (count equally-likely outcomes), **empirical** (relative frequency from data), or **subjective** (informed judgement). Everything else this week — addition, conditioning, Bayes, counting — is just bookkeeping on those numbers over a well-defined sample space.

This chapter is the hinge of BUSS1020. Weeks 1–2 *described* data you already had; from here on you reason about data you *have not yet seen*. Every distribution in Weeks 4–6 and every test in Weeks 7–12 is built on the four rules below, so a shaky Week 3 quietly costs marks all semester.

3.1 Three approaches to assigning a probability

Approach	How P(event) is set	Worked feel
A priori (classical)	Outcomes are equally likely; $P = (\text{favourable outcomes}) \div (\text{total outcomes})$. Known <i>before</i> any data.	A fair die shows a 4: $P = 1/6$.
Empirical (relative frequency)	$P = (\text{times the event occurred}) \div (\text{total observations})$. Comes <i>from</i> data.	247 of 1,000 visitors bought: $P(\text{buy}) \approx 0.247$.
Subjective	An informed personal degree of belief, used when outcomes are neither equally likely nor repeatable.	An analyst: $P(\text{rate cut next meeting}) = 0.30$.



All three obey the same axioms

However you assign it, a probability must satisfy: $0 \leq P(A) \leq 1$, the probability of the whole sample space is **1**, and the probabilities of all **mutually exclusive, collectively exhaustive** outcomes sum to 1. The approach only changes *how you get the number*, never the rules you then apply to it.

Sample space, events & the complement

The vocabulary of an experiment, and the rule that turns 'at least one' inside out

3.2 Sample space, simple and joint events

- The **sample space** (S) is the complete list of every possible outcome of an experiment.
- An **event** is any subset of S; a **simple event** is one characteristic (e.g. "Approved"), a **joint event** combines two or more (e.g. "Approved *and* used a broker").
- The **complement** A' is "A does not happen" — everything in S outside A.

KEY EQUATION

Complement rule $P(A') = 1 - P(A)$
use it whenever "at least one" is easier as "1 minus none"

EX 3.1 A priori vs empirical vs complement

Wk3 · approaches

A subscription app finds that, of **1,000** trial users last month, **360** converted to paid. A new trial user starts today.

- 1 Empirical estimate.** $P(\text{convert}) = 360 \div 1,000 = \mathbf{0.36}$ — a relative frequency drawn from data, the right approach here (outcomes are not equally likely).
- 2 Complement.** $P(\text{does NOT convert}) = 1 - 0.36 = \mathbf{0.64}$. Reading it straight off the data ($640 \div 1,000$) gives the same 0.64 — a quick consistency check.
- 3 Why not a priori?** There is no symmetry making 'convert' and 'not convert' equally likely, so the classical $1/2$ would be wrong. **Match the approach to the situation.**

Naming the approach is itself worth a mark in Part B — examiners want "empirical / relative frequency", not just the number.

! 0.36 is not a 'one-in-three' a-priori chance

A common slip is to treat an empirical probability as if outcomes were equally likely, or to forget that probabilities must stay in **[0, 1]**. If your working ever produces a probability above 1 or below 0, you have added overlapping events without subtracting the overlap — stop and recheck.

★ Exam move — classify, then compute

Before any arithmetic, label the question: **which approach** (a priori / empirical / subjective) and **which event** (simple, joint, or complement)? In Part A multiple-choice the distractors are built from the wrong approach; naming it correctly eliminates two options before you touch the calculator.

Tables, trees and Venn diagrams

One joint distribution, three ways to see it – and the general addition rule

IN ONE LINE

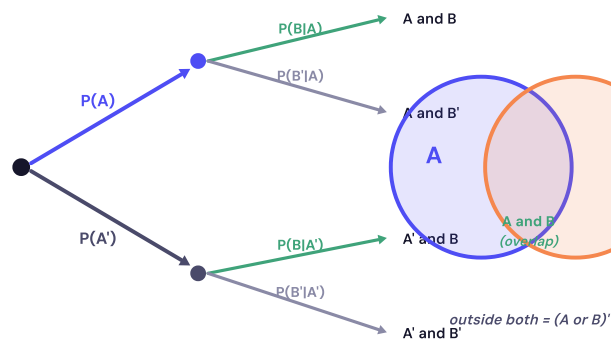
A two-way classification can be drawn three ways: a **contingency table** (joint counts in the cells, marginal totals on the edges), a **probability tree** (multiply a marginal by a conditional along each branch), and a **Venn diagram** (circles whose overlap is the joint event). Whichever you draw, the **general addition rule** stops you double-counting the overlap.

Contingency table

	B	B'	Tot
A	n11	n12	row A
A'	n21	n22	row A'
Tot	col B	col B'	N

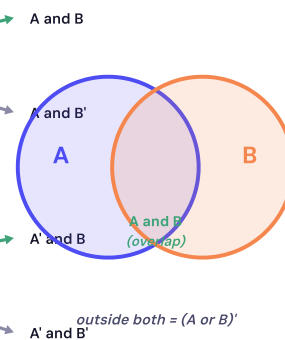
cells = joint · margins = marginal

Probability tree



branch product = joint P

Venn diagram



Three pictures of the SAME two-way classification of events A and B. The **contingency table** stores joint counts (cells) and marginal totals (margins); the **probability tree** multiplies a marginal by a conditional along each branch to get a joint probability; the **Venn diagram** shows the overlap A and B that the general addition rule must subtract. Pick whichever view the question hands you.

3.3 General addition rule — "A or B"

"A or B" means A happens, B happens, or both. If you simply add $P(A) + P(B)$ you count the overlap A and B **twice**, so you subtract it back out once:

KEY EQUATION

General addition rule $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
mutually exclusive events cannot both occur, so $P(A \text{ and } B) = 0 \Rightarrow P(A \text{ or } B) = P(A) + P(B)$

Marginal vs joint vs conditional

- **Marginal** $P(A)$: one event, ignore everything else — read it off the *margin* total of the table.
- **Joint** $P(A \text{ and } B)$: both at once — read it off a single *cell*.
- **Conditional** $P(A|B)$: A given that B already happened — a cell divided by its *row or column* total.

Mutually exclusive vs independent

- **Mutually exclusive** = cannot co-occur ($P(A \text{ and } B) = 0$). A statement about *overlap*.
- **Independent** = one tells you nothing about the other ($P(A|B) = P(A)$). A statement about *information*.
- These are **different ideas** — two events can be one without being the other (see the worked example).

The addition rule in action

One card draw shows the subtraction term — and why exclusive $\neq$ independent

EX 3.2 Addition rule on a card draw

Wk3 · addition rule

Draw one card from a standard 52-card deck. Let A = "a King" and B = "a heart".

- 1 $P(A) = 4/52$, $P(B) = 13/52$. The overlap A and B = the King of hearts, so $P(A \text{ and } B) = 1/52$.
- 2 $P(\text{King or heart}) = 4/52 + 13/52 - 1/52 = \mathbf{16/52} \approx \mathbf{0.308}$. Dropping the $-1/52$ would wrongly give $17/52$ — the double-count.
- 3 Now let C = 'a King' and D = 'a Queen'. These are **mutually exclusive** (no card is both), so $P(\text{King or Queen}) = 4/52 + 4/52 = 8/52$, with nothing to subtract.

The single subtraction term is the whole point of the rule — it is exactly what separates "A or B" from a naive sum.

! Mutually exclusive is NOT the same as independent

If A and B are mutually exclusive and both have positive probability, then knowing B happened tells you A *cannot* have happened — so they are **highly dependent**, not independent. Mutually exclusive events with $P > 0$ are never independent. Examiners love this distractor: read whether the question is about *overlap* (exclusive) or about *information* (independent).

★ Exam move — build the table first

When a worded problem gives you totals and a joint count, **draw the 2×2 contingency table and fill the margins immediately**. Almost every marginal, joint, conditional, addition-rule and Bayes answer is then a direct read or a one-line ratio off that grid — far faster and safer than juggling formulas in your head.



$P(A | B)$ — and when B tells you nothing

The single most examined idea in Ch 4: divide a cell by its conditioning total

IN ONE LINE

A conditional probability $P(A|B)$ shrinks the world to the cases where B is true, then asks how often A also holds: $P(A|B) = P(A \text{ and } B) \div P(B)$. If conditioning on B changes nothing — $P(A|B) = P(A)$ — then A and B are independent, equivalently $P(A \text{ and } B) = P(A)P(B)$.

KEY EQUATION

Conditional $P(A | B) = P(A \text{ and } B) / P(B)$ **Multiplication** $P(A \text{ and } B) = P(A | B) P(B)$

Independence $P(A | B) = P(A) \Leftrightarrow P(A \text{ and } B) = P(A) P(B)$

EX 3.3 Loan approvals — marginal, joint, conditional, independence

Wk3 · the core skill

A lender classifies **600** loan applications by outcome (Approved / Declined) and by whether the applicant used a mortgage broker. The counts:

Loan outcome	Broker	No broker	Total
Approved	216	144	360
Declined	84	156	240
Total	300	300	600

(a) Marginal. $P(\text{Approved}) = 360 \div 600 = \mathbf{0.60}$; $P(\text{Broker}) = 300 \div 600 = 0.50$. (Read straight off the margins.)

(b) Joint. $P(\text{Approved and Broker}) = 216 \div 600 = \mathbf{0.36}$ — one cell over the grand total.

(c) Conditional. $P(\text{Approved} | \text{Broker}) = 216 \div 300 = \mathbf{0.72}$ — the broker *column* total is the new denominator. Likewise $P(\text{Approved} | \text{No broker}) = 144 \div 300 = \mathbf{0.48}$.

(d) Independence test. $P(\text{Approved} | \text{Broker}) = 0.72 \neq P(\text{Approved}) = 0.60$, so broker use and approval are **NOT independent** — going through a broker is associated with a higher approval rate. Cross-check with the product rule: $P(A)P(B) = 0.60 \times 0.50 = 0.30$, but the joint is $0.36 \neq 0.30$ — same verdict, dependence.

! Divide by the RIGHT total

$P(\text{Approved} | \text{Broker})$ divides by the **broker total (300)**, giving 0.72. The reverse, $P(\text{Broker} | \text{Approved})$, divides by the **approved total (360)**, giving $216 \div 360 = 0.60$ — a *different* number. $P(A|B)$ and $P(B|A)$ are not interchangeable; mixing them is the classic conditional-probability error (and the seed of every Bayes question).

Declaring independence cleanly

Two equivalent tests, and letting the word 'given' set your denominator

The dependence we just found

In the loan table, brokered applications approved at **0.72** while the overall approval rate was only **0.60**. Because $P(\text{Approved} | \text{Broker}) \neq P(\text{Approved})$, the two variables are **dependent** — knowing a broker was used *changes* your belief about approval. That is exactly the kind of association regression (Ch 10–11) will later quantify.

Why the two tests must agree

Start from the multiplication rule $P(A \text{ and } B) = P(A|B) \cdot P(B)$. If $P(A|B) = P(A)$, substitute to get $P(A \text{ and } B) = P(A) \cdot P(B)$ — and the reverse holds too. So the conditional test and the product test are **algebraically the same statement**; use whichever the data hand you.

Two ways to declare independence — pick the easy one

You can test either $P(A|B) = P(A)$ or $P(A \text{ and } B) = P(A)P(B)$; they always agree. If the table already gives you a clean conditional, compare it to the marginal. If it gives you the joint, compare it to the product of marginals. Either route, ONE mismatch anywhere in the table is enough to declare the variables dependent.

Exam move — "given" sets your denominator

Underline the word after "given" (or after the conditioning bar). That event is your **denominator**; the "and" event is your **numerator**. Get those two right and the conditional is a single division — no formula juggling, no reversed ratio.

Conditioning shrinks the world

The intuition behind every conditional: $P(A|B)$ **throws away** the cases where B did not happen and re-computes A's share inside what remains. That is why the conditioning event becomes the new denominator — you are taking a fraction *of B's world*, not of the whole sample space. Hold that picture and you will never invert the ratio.

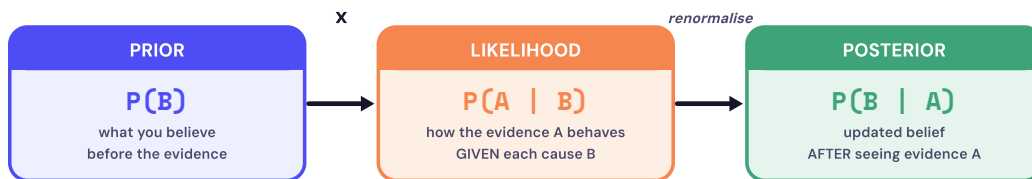
Reversing the conditioning

Marginal (total) probability, then flipping $P(A|B)$ into the $P(B|A)$ you want

IN ONE LINE

When you know how evidence behaves under each cause — $P(A|B)$ and $P(A|B')$ — but want the cause given the evidence, Bayes' theorem reverses it. The denominator is the marginal (total) probability of the evidence, summed over every cause: $P(A) = P(A|B)P(B) + P(A|B')P(B')$.

Bayes' theorem: reverse the conditioning



$$P(B | A) = P(A | B) P(B) / P(A)$$

denominator (law of total probability): $P(A) = P(A|B) P(B) + P(A|B') P(B')$

The numerator is ONE joint path; the denominator adds up ALL paths that reach the evidence A.

Bayes flips a known conditional $P(A|B)$ into the one you want, $P(B|A)$. **Prior** belief about the cause B, times the **likelihood** of the evidence under that cause, divided by the **total** probability of the evidence across every cause, gives the **posterior**. On a contingency table this is just a row/column ratio — Bayes is the engine when no table is given.

KEY EQUATION

Law of total probability $P(A) = \sum_i P(A | B_i) P(B_i)$

Bayes $P(B_j | A) = P(A | B_j) P(B_j) / \sum_i P(A | B_i) P(B_i)$

numerator = one path · denominator = all paths to the evidence

Working a reversal end to end

Map the numbers onto prior, likelihood and posterior — the arithmetic is one division

EX 3.4 Bayes on the loan table — $P(\text{Broker} \mid \text{Approved})$

Wk3 · the reversal

From the same 600-application table, suppose you are told only the conditional rates and the broker split, with no table in front of you — exactly the Part B setup. $P(\text{Broker}) = 0.50$, $P(\text{Approved} \mid \text{Broker}) = 0.72$, $P(\text{Approved} \mid \text{No broker}) = 0.48$. Find $P(\text{Broker} \mid \text{Approved})$.

- 1 **Prior** for the cause: $P(\text{Broker}) = 0.50$, so $P(\text{No broker}) = 0.50$.
- 2 **Likelihoods** of the evidence 'Approved' under each cause: 0.72 (broker) and 0.48 (no broker).
- 3 **Total probability** of the evidence: $P(\text{Approved}) = 0.72(0.50) + 0.48(0.50) = 0.36 + 0.24 = \mathbf{0.60}$.
- 4 **Bayes:** $P(\text{Broker} \mid \text{Approved}) = [0.72 \times 0.50] \div 0.60 = 0.36 \div 0.60 = \mathbf{0.60}$.

Sanity check against the table: $216 \text{ approved-with-broker} \div 360 \text{ approved} = 0.60$. ✓ Bayes is just the row/column ratio when the table is hidden.

Read the update: the prior $P(\text{Broker})$ was 0.50; after learning the loan was approved, the posterior $P(\text{Broker} \mid \text{Approved})$ rose to 0.60 — approval is *evidence* that a broker was probably used, because brokers approve at a higher rate.

! The denominator is NOT $P(B)$ — it is total $P(\text{evidence})$

Students plug the prior into the denominator by reflex. Bayes divides by **$P(A)$, the total probability of the EVIDENCE** across all causes, not by $P(B)$. Build the denominator with the law of total probability first (every branch that lands on A), then divide. A denominator below the numerator is an instant red flag.

✓ Tree first, formula second

Bayes problems are far safer drawn as a two-stage **probability tree**: split on the cause (broker / no broker), then on the evidence (approved / declined). Multiply along each branch to get joint probabilities; the posterior is the target joint divided by the sum of the joints that reach the evidence. Same numbers, almost no chance of inverting the ratio.

★ Exam move — label prior, likelihood, posterior

Write the three words on your page before computing: **prior** $P(\text{cause})$, **likelihood** $P(\text{evidence} \mid \text{cause})$, **posterior** $P(\text{cause} \mid \text{evidence})$. Mapping the question's numbers onto those three slots is most of the marks; the arithmetic is one division.

How many ways can it happen?

Multiplication, factorial, permutations and combinations – the size of the sample space

IN ONE LINE

A priori probability needs the size of the sample space, and listing is hopeless once choices stack up. Four rules count for you: the multiplication rule for independent stages, factorial $n!$ for arranging all n , permutations when order matters, and combinations when it does not.

Rule	Formula	Use it when...
Multiplication	$k_1 \cdot k_2 \cdot \dots \cdot k_n$	n independent stages, each with its own number of choices.
Same k repeated	k^n	n stages that each have the SAME k options (e.g. n digits, 10 each).
Factorial	$n! = n(n-1)\dots(2)(1)$	arranging all n distinct items in order; note $0! = 1$.
Permutations	${}_nP_k = n! / (n-k)!$	choose k of n AND the ORDER matters (rankings, schedules).
Combinations	${}_nC_k = n! / [k!(n-k)!]$	choose k of n and order does NOT matter (committees, samples).

KEY EQUATION

Permutations ${}_nP_k = n! / (n - k)!$ **Combinations** ${}_nC_k = n! / [k!(n - k)!]$
 order matters $\rightarrow P$ · order ignored $\rightarrow C$ $\Rightarrow {}_nP_k = k! \cdot {}_nC_k$ (Excel: =COMBIN(n, k))

EX 3.5 A cafe build, a tasting flight, and a sampler box

Wk3 · counting

(a) Multiplication rule. A barista builds one drink by choosing exactly one of each: **5** cup sizes, **4** milks, and **6** syrups. Number of distinct drinks = $5 \times 4 \times 6 = 120$.

(b) Permutations — order matters. A tasting flight pours **4** of the cafe's **8** single-origin coffees *in a fixed tasting order*. Arrangements = ${}_8P_4 = 8! \div 4! = 8 \times 7 \times 6 \times 5 = 1,680$.

(c) Combinations — order ignored. Instead, a take-home sampler box just needs **4** of those **8** coffees, in any order. Selections = ${}_8C_4 = 8! \div [4! 4!] = 70$.

Sanity link: ${}_8P_4 = 4! \times {}_8C_4 = 24 \times 70 = 1,680$. ✓ Each unordered selection of 4 can be arranged in $4! = 24$ orders — exactly why permutations exceed combinations by a factor of $k!$.

Picking permutations vs combinations

The single question that fixes the rule – and the Chapter 3 takeaway

! Order matters? — that one question picks P vs C

The single decision that fixes everything is **does order matter?** Rankings, schedules, podium finishes and passwords are **permutations**; committees, samples, hands of cards and sampler boxes are **combinations**. Choosing P when the answer wants C overcounts by exactly $k!$, and is the most common counting mistake in Part A.

★ Exam move — stage it, then label it

Break the problem into stages. If each stage is an independent choice → **multiply**. If you are selecting a subset from one pool → ask **order?** — yes is ${}_nP_k$, no is ${}_nC_k$. Counting answers usually feed straight into an a-priori probability (favourable ÷ total), so a wrong count quietly wrecks the probability that follows.

A two-question decision

- **Several independent stages?** Multiply the per-stage counts — same k each stage gives k^n .
- **One pool, choose a subset?** Ask whether **order matters**: yes → ${}_nP_k$, no → ${}_nC_k$.
- Always sanity-check with ${}_nP_k = k! \cdot {}_nC_k$ — the two must reconcile.

Where counting feeds back in

Counts are rarely the final answer. Most often they become the **numerator and denominator** of an a-priori probability — favourable arrangements ÷ total arrangements. They also set the sample-space size behind the binomial and hypergeometric models you meet in Chapter 5, so a clean count here pays off all the way into Week 4.

Classify the data, name the right rule, divide by the right total. In Ch 4 the marks live in choosing the picture — table, tree, Venn, or a counting rule — not in the arithmetic that follows.

CHAPTER 03 — THE ONE-LINE TAKEAWAY

Discrete Probability Distributions

Random variables, their expected value and variance, and the three named models

IN ONE LINE

A **discrete random variable** X lists mutually exclusive outcomes with probabilities that sum to 1. Its centre is the **expected value** $\mu = E(X) = \sum x_i P(X=x_i)$ and its spread is the **variance** $\sigma^2 = E(X^2) - [E(X)]^2$. Three named models — **binomial, Poisson, hypergeometric** — cover most exam questions.

A **random variable** assigns a number to each outcome of a random experiment. It is **discrete** when it can only take a countable list of values (0, 1, 2, ... tickets, defects, calls) — as opposed to *continuous* (Week 5: any value in a range, like time or weight). A **probability distribution** pairs each possible value x_i with its probability $P(X=x_i)$. Two rules always hold: every probability is between 0 and 1, and they **sum to exactly 1**.

4.1 Expected value and variance of X

Expected value — the long-run mean

The **expected value** $\mu = E(X)$ is a probability-weighted average of the outcomes: multiply each value by its probability and add. It is the average you would see if the experiment ran many times — and need **not** be a value X can actually take ($E(X) = 2.05$ tickets).

Variance & standard deviation

The **variance** σ^2 is the probability-weighted average squared distance from μ . The exam shortcut is $\sigma^2 = E(X^2) - [E(X)]^2$: take E of X -squared, subtract the square of $E(X)$. The **standard deviation** $\sigma = \sqrt{\sigma^2}$ brings it back to the original units.

MEAN & VARIANCE OF A DISCRETE RV

$$\mu = E(X) = \sum x_i P(X=x_i)$$

$$\sigma^2 = E(X^2) - [E(X)]^2 = \sum [x_i - E(X)]^2 P(X=x_i) \quad \sigma = \sqrt{\sigma^2}$$

i $E(X^2)$ is not $[E(X)]^2$

To get $E(X^2)$ you **square each outcome first, then take the weighted average** — you do not square $E(X)$. The gap between these two, $E(X^2) - [E(X)]^2$, is precisely the variance. Mixing them up (using $[E(X)]^2$ in place of $E(X^2)$) is the single most common discrete-distribution slip.

✓ Build a working table

For any custom distribution, set out columns x , $P(x)$, $x \cdot P(x)$ and $x^2 \cdot P(x)$. The third column sums to $E(X)$; the fourth sums to $E(X^2)$. Then $\sigma^2 = (\text{sum of column 4}) - (\text{sum of column 3})^2$. One tidy table earns every mark on the question.

CHAPTER 4 (CONT.)

Worked: mean & SD; and sums of two variables

The table method on a real distribution, then combining X and Y

WE-4A Help-desk tickets closed per hour

WE-4a · $E(X)$ AND σ VIA $E(X^2) - [E(X)]^2$

A help desk closes X support tickets per hour with this distribution. Find $E(X)$, the variance and the standard deviation.

x (tickets)	0	1	2	3	4
$P(X=x)$	0.05	0.25	0.40	0.20	0.10

- Check it is a distribution:** $0.05 + 0.25 + 0.40 + 0.20 + 0.10 = 1.00$. ✓
- Expected value** $E(X) = \sum x \cdot P(x) = 0(0.05) + 1(0.25) + 2(0.40) + 3(0.20) + 4(0.10) = 0 + 0.25 + 0.80 + 0.60 + 0.40 = \mathbf{2.05}$ tickets.
- $E(X^2) = \sum x^2 \cdot P(x) = 0 + 1(0.25) + 4(0.40) + 9(0.20) + 16(0.10) = 0.25 + 1.60 + 1.80 + 1.60 = \mathbf{5.25}$.**
- Variance** $\sigma^2 = E(X^2) - [E(X)]^2 = 5.25 - (2.05)^2 = 5.25 - 4.2025 = \mathbf{1.0475}$.
- Standard deviation** $\sigma = \sqrt{1.0475} \approx \mathbf{1.02}$ tickets.

2.05
 $E(X)$ TICKETS/HR

1.0475

VARIANCE Σ^2
1.02

STD DEVIATION Σ

CHAPTER 4 (CONT.)

Sums of two random variables

Expectation always adds; variance adds only after the covariance term

4.2 Sums of two random variables

Often the quantity of interest is a **combination** of two random variables — total sales = store A + store B, or a weighted portfolio $aX + bY$. Expectation is always additive; variance is additive **only after** accounting for how the two variables move together, via the **covariance** σ_{XY} .

COMBINING TWO RANDOM VARIABLES

$$E(X+Y) = E(X) + E(Y) \quad E(aX+bY) = aE(X) + bE(Y)$$

$$\sigma_{XY} = E(XY) - E(X)E(Y) \quad \text{covariance}$$

$$\text{Var}(X+Y) = \sigma_X^2 + \sigma_Y^2 + 2\sigma_{XY}$$

$$\text{Var}(aX+bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY}$$

! Variances do not simply add — covariance is the catch

It is tempting to write $\text{Var}(X+Y) = \sigma_X^2 + \sigma_Y^2$, but that is only true when X and Y are **independent** ($\sigma_{XY} = 0$). In general you must add $+2\sigma_{XY}$: positively correlated variables make the sum *more* variable; negatively correlated ones (diversification) make it *less* variable. And mind the **2ab** on the cross term for weighted combinations $aX + bY$.

i Covariance sign tells the story

A **positive** σ_{XY} means X and Y tend to rise and fall together; a **negative** σ_{XY} means one rises as the other falls; **zero** means no linear co-movement (which independence guarantees, though zero covariance alone does not prove independence).

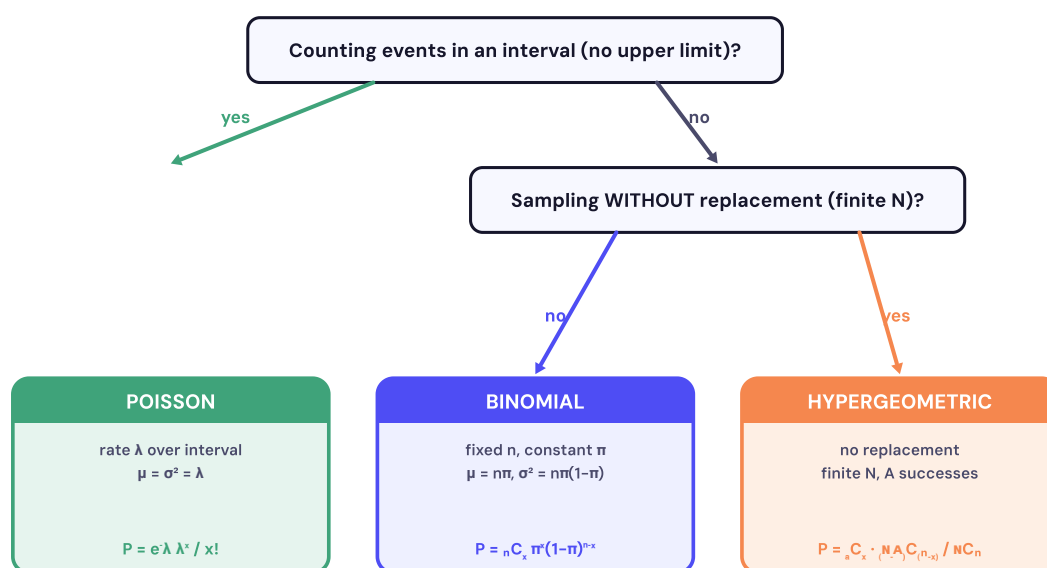
CHAPTER 4 (CONT.)

Which model? & the binomial

A decision card for binomial / Poisson / hypergeometric, then the binomial in full

IN ONE LINE

Read the question structure, not the words. **Binomial**: a fixed number n of independent success/failure trials at constant π . **Poisson**: counts of events in a continuous interval with no upper limit, rate λ . **Hypergeometric**: sampling without replacement from a finite population.



Binomial sampling WITH replacement $\rightarrow \pi$ constant; without $\rightarrow$ hypergeometric.

Choosing a discrete model. **BINOMIAL** – a fixed number n of independent trials, each a success/failure with constant π ; count the successes. **POISSON** – count events in a continuous interval (time / area / volume) with no fixed upper limit, mean rate λ . **HYPERGEOMETRIC** – sample without replacement from a finite population of size N with A successes, so π changes from draw to draw.

4.3 The binomial distribution

The **binomial** counts the number of **successes** X in a fixed number n of independent trials, where each trial is success/failure with the *same* probability π . Its four conditions are easy to check: **fixed** n , exactly **two outcomes** per trial, **constant** π , and **independent** trials.

BINOMIAL PMF, MEAN & VARIANCE

$$P(X=x) = {}_n C_x \pi^x (1-\pi)^{n-x} \quad {}_n C_x = n! / [x!(n-x)!]$$

$$\mu = n\pi \quad \sigma^2 = n\pi(1-\pi) \quad \sigma = \sqrt{n\pi(1-\pi)}$$

Excel: =BINOM.DIST(x, n, π , 0) exact; (... ,1) cumulative; =COMBIN(n,x)

CHAPTER 4 (CONT.)

Inside the binomial PMF

What each factor counts, and getting the mean and SD without the full PMF

The PMF term by term

The combination ${}_nC_x$ counts the *ways* to place x successes among n trials; π^x is the probability those x trials all succeed; $(1-\pi)^{n-x}$ is the probability the other $n-x$ all fail. Multiply the three.

Mean and SD straight away

You rarely need the full PMF for the centre: the mean is just $\mu = n\pi$ and the variance $\sigma^2 = n\pi(1-\pi)$. The variance is largest at $\pi = 0.5$ (most uncertainty) and shrinks toward 0 as π approaches 0 or 1.

“Success” is just a label

A binomial “success” is simply the outcome you are counting — it can be a sale, a defect, a click or a no-show. Define it once, fix π as *its* probability, and stay consistent: if you switch to counting failures part-way, swap π for $1-\pi$ everywhere.

The four conditions, checked in one breath

- **Fixed n** — you know in advance how many trials there are (12 visits, 20 items).
- **Two outcomes** — each trial is a clean success/failure (sale / no sale).
- **Constant π** — the success probability is the *same* on every trial; this is what fails under sampling without replacement.
- **Independent trials** — one trial’s result does not change another’s odds.

Variance peaks at $\pi = 0.5$

Because $\sigma^2 = n\pi(1-\pi)$, a binomial is **most variable when $\pi = 0.5$** (maximum uncertainty) and tightens toward 0 as π nears 0 or 1. That is a quick sanity check: a near-certain process ($\pi = 0.97$) should return a small SD.

CHAPTER 4 (CONT.)

Worked binomial; and the Poisson model

Checkout successes over 12 visits, then the 'window of opportunity' count

★ Exam move — “at least / at most” via the complement

Binomial “at least k ” questions are answered with the **cumulative** function and the complement: $P(X \geq k) = 1 - P(X \leq k-1)$. In Excel that is `=1 - BINOM.DIST(k-1, n, π, 1)`. “At most k ” is just `BINOM.DIST(k, n, π, 1)`. Adding many individual terms by hand is the slow, error-prone route — reach for 1 – cumulative.

WE-4B Checkout success on 12 independent visits

WE-4b · BINOMIAL MEAN / SD / AT-LEAST

A website’s checkout succeeds on **85%** of attempts. Over **$n = 12$** independent visits ($\pi = 0.85$), find the mean and SD of the number of successes, and the chance of **at least 10** successes.

- 1 Identify the model.** Fixed $n = 12$, two outcomes (success / fail), constant $\pi = 0.85$, independent visits $\rightarrow$ **binomial**.
- 2 Mean.** $\mu = n\pi = 12 \times 0.85 = \mathbf{10.2 \text{ successes}}$.
- 3 Standard deviation.** $\sigma^2 = n\pi(1-\pi) = 12 \times 0.85 \times 0.15 = 1.53$, so $\sigma = \sqrt{1.53} \approx \mathbf{1.24}$.
- 4 Exactly 10:** $P(X=10) = {}_{12}C_{10}(0.85)^{10}(0.15)^2$ — Excel `=BINOM.DIST(10, 12, 0.85, 0)`.
- 5 At least 10** via the complement: $P(X \geq 10) = 1 - P(X \leq 9) = 1 - \text{BINOM.DIST}(9, 12, 0.85, 1)$.

10.2

MEAN $\mu = n\pi$

1.24

STD DEVIATION σ

1 – F(9)

 $P(X \geq 10)$

4.4 The Poisson distribution

The **Poisson** model counts events in a **continuous “window of opportunity”** — a stretch of time, an area, a volume, a distance — where there is **no natural upper limit** on the count and events occur at a constant average **rate λ** independently. Calls per 5 minutes, flaws per metre of cable, accidents per month: all Poisson.

POISSON PMF — MEAN = VARIANCE = λ

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots \text{ (no upper limit)}$$

$$\mu = \lambda \quad \sigma^2 = \lambda \quad \sigma = \sqrt{\lambda} \quad \text{Excel: } =\text{POISSON.DIST}(x, \lambda, 0 \text{ or } 1)$$

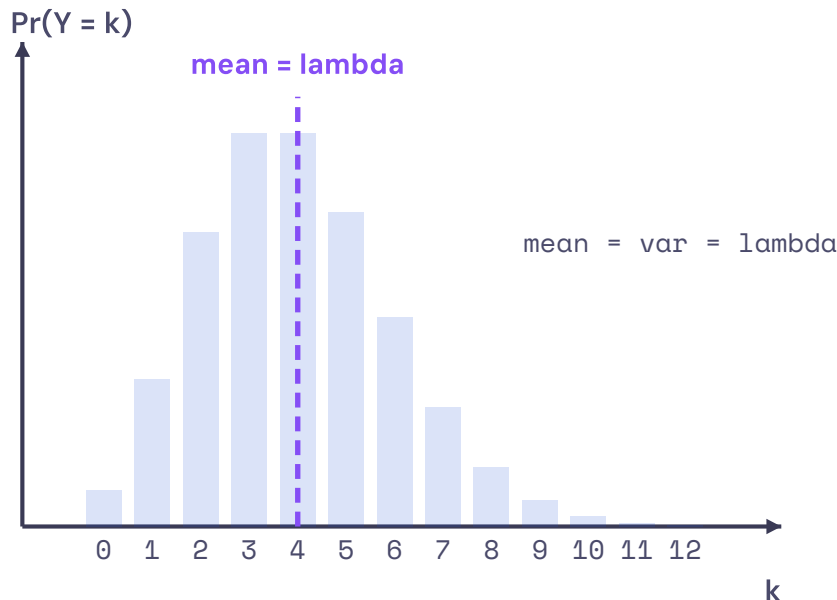
i The Poisson signature: mean = variance

What makes Poisson distinctive is that the **mean equals the variance**, both $= \lambda$. If a count variable’s sample mean and variance are roughly equal, Poisson is a good model; if the variance is much larger (“overdispersion”), it is not. Always make λ match the **interval in the question** — if the rate is 4 per 5 min and you are asked about 10 min, rescale to $\lambda = 8$.

CHAPTER 4 (CONT.)

Worked Poisson; and the hypergeometric

The PMF shape, calls per window, then sampling without replacement



Poisson PMF for $\lambda = 4$ (the call-centre rate below). The bars sit on the integer support $x = 0, 1, 2, \dots$; the distribution is **right-skewed** for small λ , with the mode near $\lfloor \lambda \rfloor = 4$. Its signature is **mean = variance = λ** , so a Poisson count has $\sigma = \sqrt{\lambda}$.

WE-4C

Calls arriving at a call centre

WE-4c · POISSON 'WINDOW OF OPPORTUNITY'

A call centre receives on average **$\lambda = 4$ calls per 5-minute window**. Calls are independent counts with no upper limit, so $X \sim \text{Poisson}(4)$. Find $P(\text{exactly 2 calls})$, $P(\text{at least one call})$, and σ .

- Exactly 2 calls:** $P(X=2) = e^{-4} \cdot 4^2 / 2! = (0.01832 \times 16) / 2 \approx \mathbf{0.1465}$. Excel **=POISSON.DIST(2, 4, 0)**.
- At least one** via the complement: $P(X \geq 1) = 1 - P(X=0) = 1 - e^{-4} = 1 - 0.0183 \approx \mathbf{0.9817}$.
- Standard deviation:** $\sigma = \sqrt{\lambda} = \sqrt{4} = \mathbf{2 \text{ calls}}$ (since mean = variance = λ).

0.1465

P(EXACTLY 2)

0.9817

P(AT LEAST 1)

2 $\Sigma = \sqrt{\lambda}$

4.5 The hypergeometric distribution

When you sample **without replacement** from a **finite** population, the probability of a success *changes* with every draw — so the binomial (which needs constant π) no longer applies. The **hypergeometric** handles exactly this: a population of size **N** containing **A** successes, from which you draw **n** items and count the **x** successes caught.

HYPERGEOMETRIC PMF (WITHOUT REPLACEMENT)

$$P(X=x) = \frac{A^C_x \cdot N-A^C_{n-x}}{N^C_n} \quad \text{draw } n \text{ from } N, A \text{ successes in the population}$$

$$\mu = \frac{nA}{N} \quad (\text{finite-population factor appears in } \sigma^2)$$

CHAPTER 4 (CONT.)

Worked hypergeometric & the model recap

Sampling defectives without replacement, and the decision checklist

WE-4D Defective USB drives sampled without replacement

WE-4d · HYPERGEOMETRIC (FINITE POPULATION)

A box holds **N = 20** USB drives, of which **A = 6** are defective. An auditor draws **n = 5** drives **without replacement**. What is $P(\text{exactly 2 defective})$?

- 1 Why hypergeometric.** Drawing without replacement from a finite box means π changes each draw — not binomial.
- 2 Ways to get 2 of the 6 defectives:** ${}_6C_2 = 15$. **Ways to get the other 3 from the 14 good ones:** ${}_{14}C_3 = 364$.
- 3 Total ways to draw 5 from 20:** ${}_{20}C_5 = 15,504$. Excel: **=COMBIN(20,5)**.
- 4 Combine:** $P(X=2) = (15 \times 364) / 15,504 = 5,460 / 15,504 \approx 0.352$.

15
 ${}_6C_2$ DEFECTIVES

364
 ${}_{14}C_3$ GOOD ONES

0.352
 $P(\text{EXACTLY 2 DEFECTIVE})$

! Without replacement → hypergeometric, not binomial

The tell is the phrase **“without replacement”** (or a small, finite population where draws clearly affect later odds). The binomial assumes **constant π** , which holds only *with* replacement or for an effectively infinite population. As a rule of thumb, if the sample is under ~5% of the population the binomial is a fine approximation; otherwise use the hypergeometric.

i Counting rules carry over from Week 3

Every PMF here is built from **combinations ${}_nC_x$** — the same counting machinery as Week 3. The binomial uses one combination; the hypergeometric multiplies two (successes chosen $\times$ non-successes chosen) over the total ways to draw. Excel **=COMBIN(n, x)** evaluates each one if your calculator chokes on large factorials.

★ Exam move — name the model first, then compute

Before any arithmetic, classify: **fixed n trials, constant π** → binomial ($\mu = n\pi$); **count in an interval, no upper limit** → Poisson ($\mu = \sigma^2 = \lambda$); **without replacement from finite N** → hypergeometric. For “at least one” reach for the **complement**. The formula sheet gives the PMFs — the marks are for picking the **right model** and reading n, π, λ, N, A off the stem correctly.

Continuous probability — density and the uniform

Probability is now an AREA, not a sum — and the simplest continuous model

IN ONE LINE

For a **continuous random variable**, probability is the **area under the density curve** over an interval — never the height, and never a single point: $P(X = \text{exact value}) = 0$. The **uniform** on $[a, b]$ is the flat starter: mean $= (a + b)/2$, variance $= (b - a)^2/12$.

A **continuous random variable** can take *any* value in an interval — a waiting time, a weight, a dollar amount measured finely. Because there are infinitely many possible values, we no longer attach a probability to each one. Instead a **probability density function $f(x)$** describes the distribution, and probability is the **area under $f(x)$** between two values.

5.1 Density, area, and why $P(\text{exact}) = 0$

Area = probability

The total area under any density curve is **1**. The chance the variable lands between a and b is the area of the slice from a to b : **$P(a < X < b)$** . There is no gap between “ $<$ ” and “ $\leq$ ” here — for a continuous variable they give the *same* answer.

A single point has zero area

The area over a point with *no width* is zero, so **$P(X = c) = 0$** for any exact c . That is why $P(X \leq c) = P(X < c)$: including or excluding the endpoint changes nothing. Contrast this with **discrete** variables (Week 4), where each outcome carries a real, positive probability.

CONTINUOUS PROBABILITY = AREA

Total area under $f(x) = 1$ $P(a < X < b) = \text{area of } f(x) \text{ from } a \text{ to } b$
 $P(X = c) = 0$ so $P(X \leq c) = P(X < c)$

! “Exactly” on a continuous scale means zero

If a question asks for the probability a waiting time is **exactly** 5.00 minutes, the answer is **0** — a single point has no area. Real questions always ask for a **range** (more than, less than, between). When you do see “at least” or “at most”, the $\leq/<$ distinction is irrelevant for continuous models, so do not waste time worrying about endpoints.

5.2 The uniform distribution on $[a, b]$

The **uniform** distribution spreads probability *evenly* over an interval $[a, b]$ — every sub-interval of equal width is equally likely. Its density is the flat constant **$1/(b - a)$** , so probabilities are just **length ratios**: the chance of landing in a window is that window’s width divided by the total width.

UNIFORM ON $[A, B]$

$f(x) = \frac{1}{(b - a)}$ for $a \leq x \leq b$ $\mu = \frac{(a + b)}{2}$ $\sigma^2 = \frac{(b - a)^2}{12}$ $\sigma = (b - a)/\sqrt{12}$
 $P(X < x) = \frac{(x - a)}{(b - a)}$ (linear — a straight-line CDF)

CHAPTER 5 (CONT.)

Worked: the uniform train arrival

Mean, SD and a probability from a flat distribution on $[0, 12]$ minutes

i The mean sits dead centre; the SD comes from a fixed recipe

Because the rectangle is symmetric, the **mean is the midpoint** $(a + b)/2$. The spread does not depend on where the interval sits, only on its **width**: variance is always $(b - a)^2/12$, so $SD = (b - a)/\sqrt{12} \approx 0.289(b - a)$. Memorise the “over 12” — the formula sheet gives it, but knowing it speeds the arithmetic.

WE-5A Train arrival ~ Uniform $[0, 12]$ minutes

WE-5a · UNIFORM $[0, 12]$

A commuter train arrives **uniformly** between **0 and 12 minutes** after the scheduled time, so $a = 0$, $b = 12$. Find the mean delay, the standard deviation, and the chance the train arrives **within the first 4 minutes**.

- 1 Mean (midpoint):** $\mu = (a + b)/2 = (0 + 12)/2 = \mathbf{6 \text{ minutes}}$ — the centre of the interval.
- 2 Standard deviation:** $\sigma = (b - a)/\sqrt{12} = 12/\sqrt{12} = 12/3.464 = \mathbf{3.46 \text{ minutes}}$. (Equivalently $\sigma^2 = 12^2/12 = 12$, so $\sigma = \sqrt{12}$.)
- 3 Probability within the first 4 minutes:** use the length ratio $P(X < 4) = (4 - 0)/(12 - 0) = 4/12 = \mathbf{0.333}$.
- 4 Read it back:** there is about a **33%** chance the train is no more than 4 minutes late, since 4 minutes is one-third of the 12-minute window.

6 min

MEAN DELAY μ

3.46 min

 $SD \ \Sigma = (b - a) / \sqrt{12}$

0.333

 $P(\text{ARRIVES IN FIRST 4 MIN})$

✓ Uniform probabilities are just ratios of lengths

For any window $[c, d]$ inside $[a, b]$, $P(c < X < d) = (d - c)/(b - a)$. No standardising, no table — just **window width over total width**. The same ratio gives the linear CDF $P(X < x) = (x - a)/(b - a)$, which rises in a straight line from 0 at a to 1 at b .

! Do not put the whole interval into “over 12” twice

A frequent slip is computing σ as $(b - a)^2/12$ (that is the **variance**) and forgetting the square root, or using $\sqrt{(b - a)}$ instead of $(b - a)/\sqrt{12}$. Keep them straight: **variance** = $(b - a)^2/12$, then **SD** = $\sqrt{\text{variance}} = (b - a)/\sqrt{12}$.

★ Exam move — spotting a uniform

Trigger words: “equally likely anywhere between”, “arrives uniformly”, “chosen at random from the interval”. Once you see a **flat** spread over $[a, b]$: (1) mean = midpoint, (2) SD = $(b - a)/\sqrt{12}$, (3) any probability = **length ratio**. Three one-line answers — no calculator-heavy work.

CHAPTER 5 (CONT.)

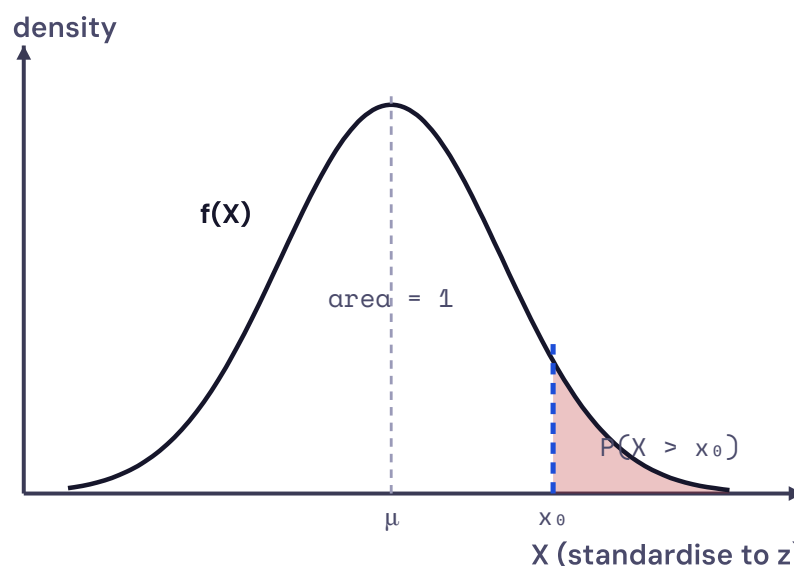
The normal distribution & the standard normal Z

The bell, its symmetry, and the four-step machine that prices any normal area

IN ONE LINE

The normal $N(\mu, \sigma^2)$ is the symmetric bell where **mean = median = mode**. To find a probability, **standardise** with $Z = (X - \mu)/\sigma$ — which has mean 0 and SD 1 — then read the area from Table E.2.

The **normal distribution** is the workhorse continuous model: bell-shaped, symmetric about its mean μ , with spread set by σ . It is fully described by just two parameters, the **mean μ** and the **standard deviation σ** (variance σ^2), written $X \sim N(\mu, \sigma^2)$. Because every normal curve has the same shape, we convert any of them to one common ruler.



A Normal probability is an area. To find $P(X > x_0)$, standardise x_0 to $z = (x_0 - \mu)/\sigma$, then read the tail area from Table E.2. The whole curve encloses area 1; it is **symmetric** about μ , where the **mean = median = mode** all coincide.

Properties of the bell

- **Symmetric** about μ — the left and right halves mirror, so **mean = median = mode** all sit at the centre.
- **Total area = 1**; the curve never touches the axis (infinite tails).
- Spread is set by σ : a larger σ gives a flatter, wider bell; the mean μ just slides the whole curve left or right.
- The **Empirical Rule** (Week 2) is the normal in action: $\approx 68\%$ within $\pm 1\sigma$, $\approx 95\%$ within $\pm 2\sigma$, $\approx 99.7\%$ within $\pm 3\sigma$.

The standard normal Z

Standardising re-expresses X as **how many SDs it sits from the mean**. The result, Z , follows the **standard normal $N(0, 1)$** . Table E.2 is the **cumulative** standard normal: it returns the area to the *left* of a z value, $P(Z < z)$.

CHAPTER 5 (CONT.)

Standardising & the four-step procedure

$Z = (X - \mu)/\sigma$, Table E.2, and the routine that prices any normal area

STANDARD NORMAL

$$Z = \frac{(X - \mu)}{\sigma} \quad Z \sim N(0, 1) \quad \text{mean } 0, \text{ SD } 1$$

Table E.2 gives $P(Z < z)$; by symmetry $P(Z > z) = 1 - P(Z < z)$ and $P(Z < -z) = P(Z > z)$

i The four-step procedure for any normal probability

(1) Draw the bell, mark μ and shade the region you want. **(2) Standardise** each X to $z = (X - \mu)/\sigma$. **(3) Look up** the cumulative area $P(Z < z)$ in Table E.2. **(4) Combine** by symmetry/complement to match your shaded picture (subtract for “greater than”; subtract two left-areas for “between”). The picture is your check that you combined the areas correctly.

s Excel does steps 2–4 in one call

=NORM.DIST(x, μ , σ , 1) returns $P(X \leq x)$ directly (the final 1 = cumulative). **=NORM.S.DIST(z, 1)** does the same for a z value on the standard scale. In the exam there is no Excel — you standardise by hand and read Table E.2 — but knowing the function names is examinable.

Three area shapes you will combine

- **Less than** $P(X < x)$: read $P(Z < z)$ straight off the table.
- **Greater than** $P(X > x)$: complement, $1 - P(Z < z)$.
- **Between** $P(a < X < b)$: two left-areas subtracted, $P(Z < z_b) - P(Z < z_a)$.

Symmetry shortcuts

Table E.2 only needs the cumulative area, because the bell is symmetric: **$P(Z > z) = P(Z < -z)$** and **$P(Z < -z) = 1 - P(Z < z)$** . So a negative- z lookup or a right tail is always one step from a single table read — never look for a separate “negative” table.

CHAPTER 5 (CONT.)

Worked: normal probabilities & the inverse normal

From X to an area (WE-5b), and from an area back to X (WE-5c)

WE-5B Electricity bills ~ $N(\$180, \$35^2)$ — $P(\text{bill} > \$230)$

WE-5b · NORMAL TAIL AREA

Monthly household electricity bills are **normal with $\mu = \$180$ and $\sigma = \$35$** . What fraction of households pay **more than \$230**? Follow the four-step procedure.

- 1 Draw & shade:** a bell centred at \$180; shade the right tail beyond \$230.
- 2 Standardise:** $z = (230 - 180)/35 = 50/35 = 1.43$ — the \$230 bill sits about 1.43 SD above the mean.
- 3 Look up:** Table E.2 gives $P(Z < 1.43) = 0.9236$ (area to the left).
- 4 Combine (complement):** $P(Z > 1.43) = 1 - 0.9236 = 0.0764$. So about **7.6%** of households pay more than \$230. Excel: **=1 - NORM.DIST(230, 180, 35, 1)**.

WE-5C The 90th-percentile bill (inverse normal)

WE-5c · INVERSE NORMAL

Now run it backwards. The bill that is **exceeded by only the top 10%** of households is the **90th percentile** — the value with 90% of the area to its left. Go from the *area* to the *X*.

- 1 Find the z first:** we need z with $P(Z < z) = 0.90$. Reading Table E.2 *inwards*, the closest area is 0.8997, giving **$z = 1.28$** .
- 2 Un-standardise:** rearrange $Z = (X - \mu)/\sigma$ to **$X = \mu + Z \cdot \sigma$** .
- 3 Substitute:** $X = 180 + 1.28 \times 35 = 180 + 44.8 =$ **\$224.80**.
- 4 Interpret:** only 10% of households pay more than about **\$224.80** a month. Excel: **=NORM.INV(0.90, 180, 35)**.

 $z = 1.43$

STANDARDISED \$230 BILL

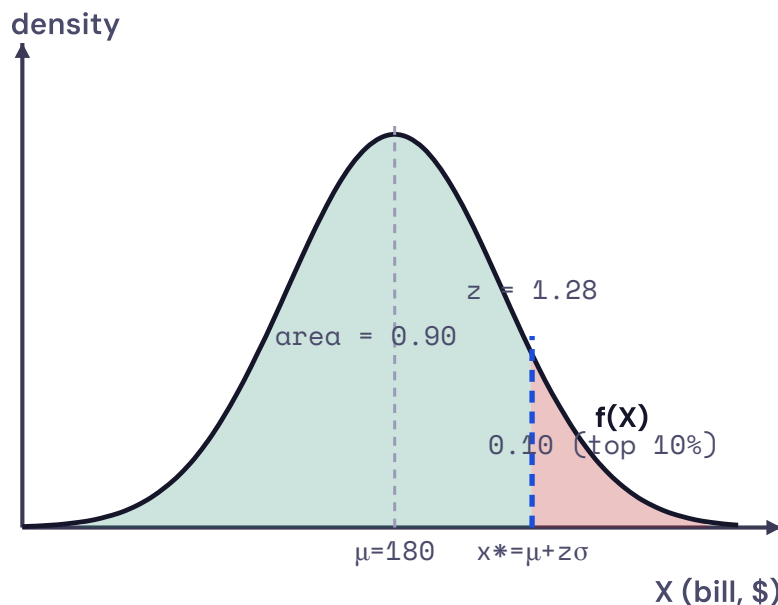
0.0764 $P(\text{BILL} > \$230)$ **\$224.80**

90TH-PERCENTILE BILL

CHAPTER 5 (CONT.)

Forward vs inverse — one machine, two directions

Fix the cut and read the area, or fix the area and read back the value



Inverse-normal logic in one picture. A forward question fixes the cut and reads the shaded area; an **inverse** question fixes the area and reads back the value. Here is WE-5c: the **90th-percentile bill** on $X \sim N(\mu = \$180, \sigma = \$35)$. The single cut leaves **0.90** of the area to its left and the **top 10%** in the right tail; the matching $z = 1.28$ un-standardises to $X = \mu + z \cdot \sigma = \224.80 . Same machine, run in reverse.

! Forward vs inverse — which way is the table read?

A **forward** problem gives you X (or z) and asks for an **area**: standardise, then look the z up in Table E.2. An **inverse** problem gives you the **area** and asks for X : find the z whose cumulative area matches, then $X = \mu + z \cdot \sigma$. Mixing up the direction — or forgetting that a “top 10%” value needs the **0.90** left-area, not 0.10 — is the classic percentile error.

Forward: value → area

- **Given** a value X (or z).
- **Standardise** $z = (X - \mu) / \sigma$.
- **Read** $P(Z < z)$ from Table E.2.
- **Combine** by complement / symmetry to match the shaded region.

Inverse: area → value

- **Given** an area (a percentile, or a tail / two-tailed α).
- **Find** the z whose cumulative area matches — read Table E.2 *inwards*.
- **Un-standardise** $X = \mu + z \cdot \sigma$.
- **Interpret** the value in the question's units.

CHAPTER 5 (CONT.)

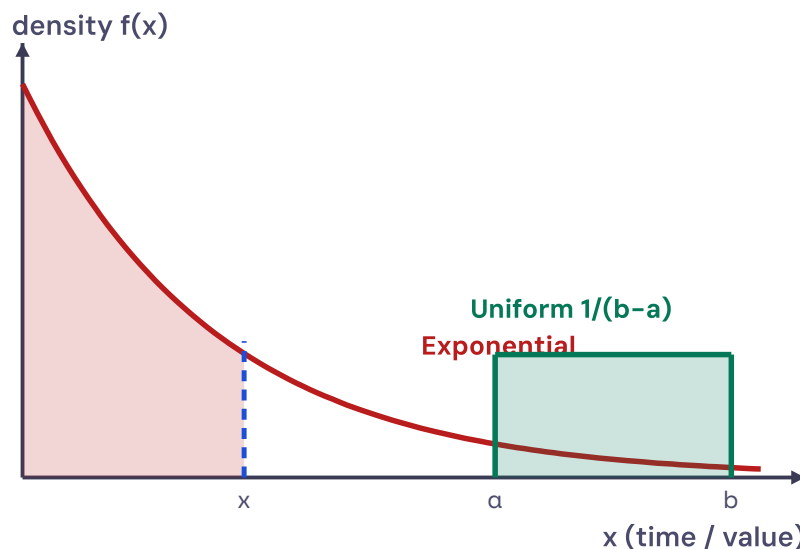
The exponential distribution & the Poisson link

Time *BETWEEN* rare events – one parameter λ , mean $1/\lambda$, a clean CDF

IN ONE LINE

The exponential models the time between events that arrive at a constant rate λ . It has one parameter: mean $= 1/\lambda$, and a tidy CDF $P(X \leq x) = 1 - e^{(-\lambda x)}$. Link: if counts are $\text{Poisson}(\lambda)$, the gaps between events are $\text{Exponential}(\lambda)$.

Where the Poisson (Week 4) counts *how many* events occur in an interval, the **exponential** measures *how long you wait* until the next one. It is a continuous, right-skewed curve that decays from a peak at $x = 0$: short waits are common, long waits are rare but possible. A single rate parameter λ (events per unit time) controls everything.



Two shapes you must recognise. The **exponential** (red, decaying) models the time between rare events – most gaps are short, a few are long. The **uniform** (green rectangle) spreads probability evenly across $[a, b]$, so its density is the constant $1/(b - a)$. Both enclose a total area of 1. **Link:** if events arrive as $\text{Poisson}(\lambda)$, the gaps between them are $\text{Exponential}(\lambda)$.

EXPONENTIAL (RATE λ)

$P(X \leq x) = 1 - e^{(-\lambda x)}$ $P(X > x) = e^{(-\lambda x)}$
mean $= 1/\lambda$ SD $= 1/\lambda$ $\text{Poisson}(\lambda)$ counts $\leftrightarrow$ $\text{Exponential}(\lambda)$ gaps

§ Match the units of λ and x

λ is a **rate per unit** (per week, per hour); x must be measured in the **same unit**. If $\lambda = 0.5$ per week, then x is in weeks. To convert, multiply: a rate of 0.5/week is also 2/month, so re-scale λ and x consistently. Excel: **=EXPON.DIST(x, λ , 1)** returns the cumulative $P(X \leq x)$.

i Why the exponential and Poisson are two views of one process

Imagine events landing randomly at average rate λ . **Count them** in a fixed window $\rightarrow$ that count is **$\text{Poisson}(\lambda)$** . **Time the gaps** between consecutive events $\rightarrow$ each gap is **$\text{Exponential}(\lambda)$** . So “4 calls per hour on average” (Poisson) and “mean 15 minutes between calls” (Exponential, $1/\lambda$) describe the **same** stream of events.

CHAPTER 5 (CONT.)

Worked exponential, assessing normality & recap

Breakdown times (WE-5d), the IQR $\approx 1.33S$ check, and the exam moves

WE-5D Breakdowns ~ Exponential($\lambda = 0.5$ per week)

WE-5d · EXPONENTIAL CDF

Equipment breakdowns occur at an average rate of $\lambda = 0.5$ per week, so the time to the next breakdown is **Exponential(0.5)**. Find the mean time between breakdowns and the chance the next one happens **within one week**.

- 1 **Mean time:** mean = $1/\lambda = 1/0.5 = 2$ weeks between breakdowns — consistent with a rate of half a breakdown per week.
- 2 **Probability within one week ($x = 1$):** use the CDF $P(X \leq 1) = 1 - e^{(-\lambda x)} = 1 - e^{(-0.5 \times 1)}$.
- 3 **Evaluate:** $e^{(-0.5)} = 0.6065$, so $P(X \leq 1) = 1 - 0.6065 = 0.393$ — about a **39%** chance of a breakdown within the week. Excel: **=EXPON.DIST(1, 0.5, 1)**.
- 4 **Poisson cross-check:** $P(\text{no breakdown in 1 week}) = e^{(-0.5)} = 0.6065$ matches the Poisson $P(X = 0)$ with mean 0.5 — the two views agree.

2 weeks

MEAN TIME $1/\lambda$

0.393

$P(\text{WITHIN ONE WEEK})$

0.6065

$P(\text{NONE}) = e^{(-0.5)}$

CHAPTER 5 (CONT.)

Assessing normality & the chapter recap

Two quick checks for a bell, and the four continuous models on one card

5.5 Assessing normality

Many tools ahead (Weeks 6–12) assume the data are roughly normal, so you must be able to **check** it. Two quick diagnostics:

Rule-of-thumb on the IQR

For a true normal distribution the interquartile range is about **1.33 standard deviations**. Compare the data's IQR with 1.33S: if **IQR $\approx$ 1.33S**, normality is plausible; a big mismatch is a warning sign.

QUICK NORMALITY CHECK

Normal rule of thumb: **IQR $\approx$ 1.33 $\times$ S**

Normal probability plot

Plot the ordered data against the values you would expect from a normal. If the points fall on a **roughly straight line**, normality holds; a curved or S-shaped pattern flags **skew** or heavy tails. Combine it with the mean-vs-median gap and a boxplot for a fuller picture.

Continuous model	Mean	Spread / key formula	Excel
Uniform [a, b]	$(a + b)/2$	$\sigma^2 = (b - a)^2/12$	—
Normal $N(\mu, \sigma^2)$	μ	$Z = (X - \mu)/\sigma$; SD σ	NORM.DIST / NORM.INV
Standard normal	0	SD 1; Table E.2 = $P(Z < z)$	NORM.S.DIST / NORM.S.INV
Exponential(λ)	$1/\lambda$	$P(X \leq x) = 1 - e^{-(\lambda x)}$	EXPON.DIST

★ Exam move — the continuous-distribution checklist

(1) Name the model: flat over an interval $\rightarrow$ **uniform**; symmetric bell $\rightarrow$ **normal**; waiting time / gap between events $\rightarrow$ **exponential**. **(2) Uniform** $\rightarrow$ mean = midpoint, SD = $(b-a)/\sqrt{12}$, probability = length ratio. **(3) Normal** $\rightarrow$ four steps: draw, standardise $z = (X-\mu)/\sigma$, read Table E.2, combine; inverse $\rightarrow X = \mu + z \cdot \sigma$. **(4) Exponential** $\rightarrow$ mean $1/\lambda$, $P(X \leq x) = 1 - e^{-(\lambda x)}$, match units of λ and x . **(5) Remember P(exact) = 0** for every continuous model.



Sampling Distributions & the CLT

Why one sample mean is enough to reason about a whole population

IN ONE LINE

A statistic ($\bar{X}$, p) changes from sample to sample. Its **sampling distribution** is how it behaves over all possible samples of size n . Two facts run the rest of the course: the sample mean is **unbiased** — $E(\bar{X}) = \mu$ — and its spread, the **standard error** $\sigma/\sqrt{n}$, shrinks as n grows.

Weeks 1–5 described data and single random variables. This week is the **hinge**: it explains why a number from *one* sample — your $\bar{X}$ — can be trusted to say something about an unknown population μ you will never fully observe. Every **confidence interval** (Week 7) and **hypothesis test** (Weeks 8–9) is built directly on the sampling distribution and the Central Limit Theorem you meet here.

6.1 What is a sampling distribution?

A statistic is itself random

Draw a random sample and you get one $\bar{X}$. Draw a *different* sample and you get a slightly different $\bar{X}$. Because the value depends on which units happened to be chosen, a **statistic is a random variable** — it has its own distribution.

The distribution over all samples

The **sampling distribution** of a statistic is the probability distribution of *all* the values it could take across every possible sample of size n from the population. It tells us how far a typical $\bar{X}$ lands from the truth — the basis of all later inference.

THE BIG IDEA

A **statistic** ($\bar{X}$, p , S) estimates a **parameter** (μ , π , σ)
sampling distribution = the spread of that statistic over **all possible samples of size n**

i Parameter (fixed, unknown) vs statistic (random, observed)

Recall the Week 1 mnemonic: a **P**arameter describes a **P**opulation (Greek: μ , σ , π) and is a *fixed* number we usually never see; a **S**tatistic comes from a **S**ample (Latin: $\bar{X}$, S , p) and *varies* from sample to sample. Inference uses the observed statistic, plus its sampling distribution, to make a calibrated statement about the fixed parameter.

CHAPTER 6 (CONT.)

Sampling distribution of the sample mean $\bar{X}$

Unbiased centre $E(\bar{X}) = \mu$, and the standard error $\sigma/\sqrt{n}$ that shrinks with n

6.2 Sampling distribution of the sample mean $\bar{X}$

Suppose every possible sample of size n is taken and the mean $\bar{X}$ computed for each. Two properties of that collection of means matter, and they hold for **any** population shape:

- **Centre — $\bar{X}$ is unbiased:** the mean of all the sample means equals the population mean, $E(\bar{X}) = \mu$. On average the sample mean hits the target.
- **Spread — the standard error:** the sample means vary less than the raw data, by a factor of $\sqrt{n}$: $\sigma_{\bar{X}} = \sigma/\sqrt{n}$. Averaging cancels out high and low values.

MEAN & STANDARD ERROR OF $\bar{X}$

$$E(\bar{X}) = \mu \quad \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \quad \text{"standard error of the mean" — SE for short}$$

✓ Standard error $\neq$ standard deviation

The population **SD** σ measures spread of the *raw values*. The **standard error** $\sigma/\sqrt{n}$ measures spread of the *sample mean* — it is always smaller, and it keeps shrinking as n rises. Quadruple the sample size and you halve the standard error ($\sqrt{4} = 2$). This diminishing return is why huge samples buy ever-smaller gains in precision.

The $\sqrt{n}$ law of diminishing returns

Because $SE = \sigma/\sqrt{n}$, precision improves with the **square root** of effort, not in step with it. Going from $n = 100$ to $n = 400$ (four times the data, four times the cost) only halves the standard error. That is why beyond a point a bigger survey is rarely worth its price — the next increment of certainty gets expensive.

Two properties, any population

Note what we have *not* assumed yet: $E(\bar{X}) = \mu$ and $SE = \sigma/\sqrt{n}$ hold for *any* population shape — symmetric, skewed, bimodal, anything. They fix the centre and the spread of the sampling distribution. The one thing still missing is its **shape** — and that is exactly what the Central Limit Theorem supplies next.

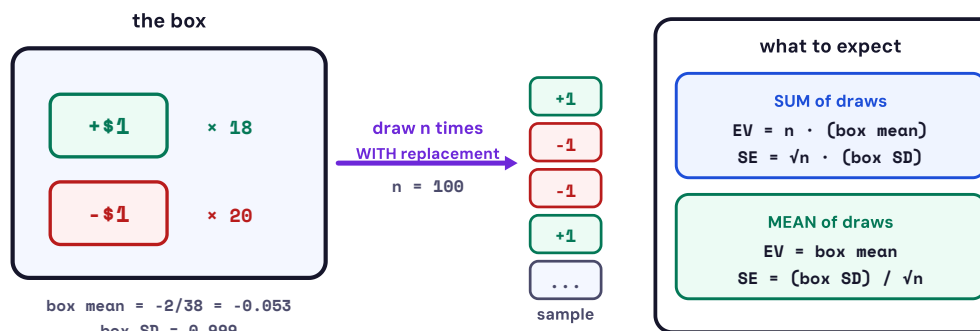
CHAPTER 6 (CONT.)

The box model & the Central Limit Theorem

Where $\sigma/\sqrt{n}$ comes from, and why $\bar{X}$ goes Normal whatever the population looks like

IN ONE LINE

Picture drawing tickets from a box. For the mean of n draws, the centre stays at the box mean and the spread is $\sigma/\sqrt{n}$. The Central Limit Theorem then adds the magic: for large enough n the sampling distribution of $\bar{X}$ is approximately Normal — *no matter* the population shape.



the Normal approximation then turns $EV \pm SE$ into chances

The box model (standard-error intuition). Tickets in a box have a known mean and SD. Draw n at random with replacement: for the **sum**, $EV = n \cdot (\text{box mean})$ and $SE = \sqrt{n} \cdot (\text{box SD})$; for the **mean**, $EV = \text{box mean}$ and $SE = (\text{box SD})/\sqrt{n}$. Chance variability grows like $\sqrt{n}$ for the sum but shrinks like $1/\sqrt{n}$ for the mean — which is exactly why a larger sample gives a more reliable $\bar{X}$.

6.3 The Central Limit Theorem (CLT)

Unbiasedness and the standard error fix the *centre* and *spread* of the sampling distribution, but not its *shape*. The Central Limit Theorem supplies the shape, and it is the single most important result in the unit:

"As the sample size n gets large, the sampling distribution of the sample mean $\bar{X}$ becomes approximately Normal — regardless of the shape of the population from which the sample was drawn."

THE CENTRAL LIMIT THEOREM

CLT — THE SAMPLING DISTRIBUTION OF $\bar{X}$

For large n : $\bar{X} \approx \text{Normal}(\mu, \sigma^2/n)$ so $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx \text{standard normal}$

How large is "large enough"? — the rules of thumb

Population shape	Sample size needed for $\bar{X} \approx \text{Normal}$
Already Normal	any n — $\bar{X}$ is exactly Normal
Fairly symmetric (mild skew, no heavy tails)	$n \geq 15$ is usually enough
General / unknown / skewed	$n \geq 30$ is the safe working rule

CHAPTER 6 (CONT.)

Reading the CLT correctly

What it claims about $\bar{X}$ – and the rule-of-thumb caveats markers reward

i What the CLT does and does not say

The CLT is about the **sampling distribution of $\bar{X}$** , not the raw data. It does **not** claim the population becomes Normal, nor that one big sample of values looks Normal. It says the *average* of a large sample behaves Normally. For a population that is already Normal, $\bar{X}$ is Normal for *any* n ; the $n \geq 30$ / $n \geq 15$ thresholds are only needed when the population is non-Normal.

! $n \geq 30$ is a rule of thumb, not a law

Markers reward the right justification, not a memorised number. The more skewed or heavy-tailed the population, the larger n must be before the bell is a good approximation. State the population shape (or “shape unknown”), then quote the matching threshold — that is the sentence that earns the mark.

Why averaging tames any shape

Each sample mean blends n independent draws, so the highs and lows partly cancel. As n grows the rough edges of a skewed population are averaged away and the distribution of $\bar{X}$ smooths toward the bell — the centre stays at μ while the spread tightens to $\sigma/\sqrt{n}$. The population itself does not change; only the behaviour of its *averages* does.

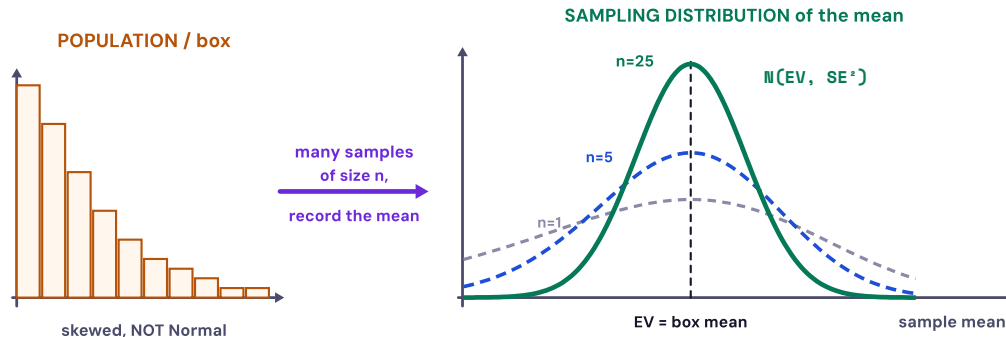
The sentence that earns the mark

Whenever you invoke the CLT, write the justification explicitly: “the population is *[shape]*, and with $n =$ *[value]* $\geq$ *[threshold]* the sampling distribution of $\bar{X}$ is approximately Normal.” Examiners are checking that you matched the **threshold to the shape**, not that you recalled “30”.

CHAPTER 6 (CONT.)

Seeing the CLT, then using it

The odd-population → tightening-bell picture, then a probability about a sample mean



CLT: the sampling distribution → Normal even if the box isn't; $SE = (\text{box SD})/\sqrt{n}$ shrinks as n grows

The CLT in one picture. Start from a decidedly *non-Normal* population (here a skewed shape).

Take many samples and record each sample mean. The histogram of those means is **Normal**, centred at μ with $SE = \sigma/\sqrt{n}$ – and it gets **narrower and more bell-shaped as n grows** ($n = 1, 5, 25$ shown). Bigger samples → smaller standard error → a tighter, more reliable estimate of μ .

The four-step recipe for a probability about $\bar{X}$

“What is the chance the *sample mean* exceeds (or falls below) some value?” is the classic Week 6 question. It is just the Week 5 normal-probability procedure, run on $\bar{X}$ instead of a single value, with $\sigma/\sqrt{n}$ in place of σ :

- 1 **Check the CLT applies** — population Normal, or $n \geq 30$ ($n \geq 15$ if fairly symmetric).
- 2 **Compute the standard error** $SE = \sigma/\sqrt{n}$.
- 3 **Standardise the sample mean:** $Z = (\bar{X} - \mu) / SE$.
- 4 **Read the area** from Table E.2 (the cumulative standard normal). Excel out of the exam: **=1-NORM.DIST($\bar{x}$, μ , $\sigma/\sqrt{n}$, 1)** for an upper tail.

CHAPTER 6 (CONT.)

Worked: a probability about a sample mean

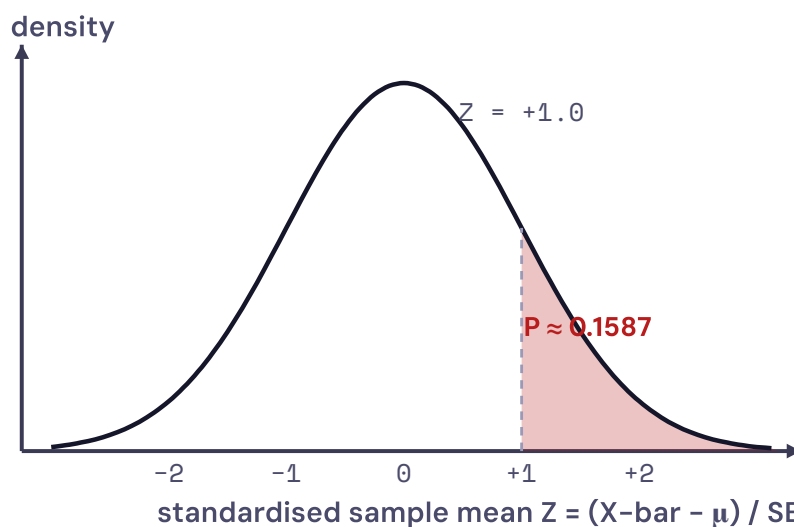
Standard error, CLT check, standardise with $\sigma/\sqrt{n}$ – then read the tail

WE-6A Probability about a supermarket's average basket value

WE-6a · STANDARD ERROR + CLT PROBABILITY

A supermarket's **daily basket value** has population mean $\mu = \$48$ and SD $\sigma = \$14$; the population shape is **unknown**. A sample of $n = 49$ days is taken. (a) Find $E(\bar{X})$ and the standard error. (b) Find the probability the sample mean basket value **exceeds \$50**.

- 1 **Centre:** $E(\bar{X}) = \mu = \$48$ — the sample mean is unbiased.
- 2 **Standard error:** $\sigma_{\bar{X}} = \sigma/\sqrt{n} = 14/\sqrt{49} = 14/7 = \2 .
- 3 **CLT check:** shape is unknown, but $n = 49 \geq 30$, so $\bar{X} \approx \text{Normal}(\$48, \$2^2)$.
- 4 **Standardise the cut-off \$50:** $Z = (50 - 48)/2 = 1.0$.
- 5 **Upper-tail area:** $P(\bar{X} > \$50) = P(Z > 1.0) = 1 - 0.8413 = 0.1587$ (about a **16%** chance). Excel: `=1-NORM.DIST(50, 48, 2, 1)`.



The standard normal curve with the **right tail beyond $Z = +1.0$** shaded. Standardise the sample mean with $Z = (\bar{X} - \mu)/SE$, then read the tail area from Table E.2: $P(\bar{X} > \$50) = P(Z > 1.0) \approx 0.1587$.

\$48

$E(\bar{X}) = \mu$

\$2

$SE = \sigma/\sqrt{n}$

1.0

$Z = (50-48)/2$

0.1587

$P(\bar{X} > \$50)$

! Divide by SE, never by σ

The single most common Week 6 error is standardising a **sample mean** with the raw SD: $Z = (50 - 48)/14 = 0.14$ (wrong). A sample mean is far less variable than one day, so you must use the **standard error $\sigma/\sqrt{n} = 2$** , giving $Z = 1.0$. Ask yourself: "is this value a single observation or an average?" — averages get the $\sqrt{n}$.

CHAPTER 6 (CONT.)

Finite population correction

When the sample is a sizeable slice of a small population, the SE shrinks

IN ONE LINE

$\sigma/\sqrt{n}$ assumes an effectively infinite population (or sampling with replacement). When you sample a large fraction of a finite population, multiply the SE by the finite population correction fpc = $\sqrt{[(N-n)/(N-1)]}$, which is < 1 and so makes the estimate more precise.

If you sampled every unit ($n = N$) there would be no sampling error at all. The plain $\sigma/\sqrt{n}$ never reaches zero; the **finite population correction (fpc)** patches the standard error so it collapses to 0 as n approaches N .

STANDARD ERROR WITH FINITE POPULATION CORRECTION

$$\sigma_{\bar{x}} = \sigma_{\sqrt{n}} \cdot \sqrt{(N-n)/(N-1)} \quad \text{fpc} = \sqrt{[(N-n)/(N-1)]}$$

When the fpc matters

The fpc only bites when the **sampling fraction n/N is large**. The usual working rule: apply it when $n/N > 0.05$ (you have drawn more than 5% of a finite population). Below that, $\text{fpc} \approx 1$ and the correction is negligible — which is why most textbook problems quietly ignore it.

- **$n = N$:** numerator $(N - n) = 0 \rightarrow \text{fpc} = 0 \rightarrow \text{SE} = 0$ (a census has no sampling error).
- **n tiny vs N :** $(N - n)/(N - 1) \approx 1 \rightarrow \text{fpc} \approx 1 \rightarrow \text{SE} \approx \sigma/\sqrt{n}$.
- The fpc is always ≤ 1 , so it only ever *reduces* the standard error.

Quick numerical feel

n/N	$\text{fpc} = \sqrt{[(N-n)/(N-1)]}$	Effect on SE
1% (e.g. 50 of 5000)	≈ 0.995	ignore it
5% (e.g. 50 of 1000)	≈ 0.975	borderline
20% (e.g. 40 of 200)	≈ 0.896	noticeable
50% (e.g. 100 of 200)	≈ 0.708	large

WE-6B Correcting the SE for a small audited population

WE-6b · APPLYING THE fpc

An auditor samples $n = 40$ invoices from a finite batch of $N = 200$, where invoice amounts have $\sigma = \$90$. Find the corrected standard error of the mean.

- Uncorrected SE:** $\sigma/\sqrt{n} = 90/\sqrt{40} = 90/6.325 = \14.23 .
- Sampling fraction:** $n/N = 40/200 = 0.20 > 0.05 \rightarrow$ the fpc is worth applying.
- fpc:** $\sqrt{[(200 - 40)/(200 - 1)]} = \sqrt{(160/199)} = \sqrt{0.804} = 0.897$.
- Corrected SE:** $14.23 \times 0.897 = \$12.76$ — about a 10% tighter standard error, because 20% of the population has already been measured.

★ Exam move — decide on the fpc before you compute

Always check n/N first. If the question gives a **finite N** and $n/N > 0.05$ (or says “without replacement” from a small population), **multiply $\sigma/\sqrt{n}$ by the fpc**; otherwise use the plain $\sigma/\sqrt{n}$. Most Week 6 items are the plain case — but a stated finite N is the examiner signalling the fpc.

CHAPTER 6 (CONT.)

Sampling distribution of the proportion p

The same machinery for a percentage, and the $n\pi$ conditions that license the bell

IN ONE LINE

For a **category** (recommend / not, defective / not), the statistic is the sample proportion $p = X/n$. It is unbiased — $E(p) = \pi$ — with standard error $\sqrt{[\pi(1-\pi)/n]}$, and it is approximately Normal when $n\pi \geq 5$ and $n(1-\pi) \geq 5$.

Means are not the only statistic that has a sampling distribution. When the variable is **categorical** — does a customer recommend the brand, is an item defective — we summarise it with a **proportion**. Because a proportion is just the mean of 0/1 outcomes, it inherits the same structure: unbiased centre, a $\sqrt{n}$ -shrinking standard error, and a CLT-driven Normal approximation.

SAMPLING DISTRIBUTION OF P

$$p = \frac{X}{n} \quad E(p) = \pi \quad \sigma_p = \sqrt{\frac{\pi(1-\pi)}{n}}$$

$$Z = \frac{p - \pi}{\sqrt{[\pi(1-\pi)/n]}} \approx \text{standard normal when the conditions below hold}$$

The two CLT conditions for a proportion

The Normal approximation is only safe when the expected count of *each* outcome is at least 5. Both must hold:

CONDITIONS FOR $P \approx \text{NORMAL}$

$$n\pi \geq 5 \quad \text{AND} \quad n(1-\pi) \geq 5$$

If π is near 0 or 1 you need a **larger n** before the bell is a good fit — a rare event needs many trials before its count is reliably ≥ 5 .

π vs p — keep them straight

- π = the *population* proportion (a fixed parameter, often unknown).
- p = the *sample* proportion X/n (a random statistic you observe).
- The SE uses π (sampling-distribution / hypothesis-test setting); the Week 7 confidence interval will instead use the observed p because π is unknown.

CHAPTER 6 (CONT.)

Worked: a probability about a proportion

Check both conditions, find $SE = \sqrt{[\pi(1-\pi)/n]}$, then standardise and read the tail

WE-6C Probability about a sample recommendation rate

WE-6C · PROPORTION CONDITIONS + PROBABILITY

Across all customers, the true proportion who would recommend a brand is $\pi = 0.40$. A random sample of $n = 200$ is surveyed. (a) Check the conditions and find the standard error of p . (b) Find the probability the sample proportion is **above 0.45**.

- 1 **Conditions:** $n\pi = 200(0.40) = 80 \geq 5$ and $n(1-\pi) = 200(0.60) = 120 \geq 5 \rightarrow p \approx \text{Normal}$.
- 2 **Centre:** $E(p) = \pi = 0.40$.
- 3 **Standard error:** $\sqrt{[\pi(1-\pi)/n]} = \sqrt{[0.40 \cdot 0.60/200]} = \sqrt{0.0012} = 0.0346$.
- 4 **Standardise 0.45:** $Z = (0.45 - 0.40)/0.0346 = 1.44$.
- 5 **Upper-tail area:** $P(p > 0.45) = P(Z > 1.44) = 1 - 0.9251 = 0.0749$ (about a **7.5%** chance).

0.40
 $E(P) = \pi$
0.0346
 $SE = \sqrt{[\pi(1-\pi)/N]}$
1.44
 $Z = (0.45 - 0.40)/SE$
0.0749
 $P(P > 0.45)$

! Check BOTH conditions, and don't confuse SE with σ of the count

A frequent slip is checking only $n\pi \geq 5$ and forgetting $n(1-\pi) \geq 5$ — the binding one when π is close to 1. A second slip: the SD of the **count** X is $\sqrt{[n\pi(1-\pi)]}$ (Week 4 binomial), but the SD of the **proportion** p is $\sqrt{[\pi(1-\pi)/n]}$. Divide the count variance by n^2 to get the proportion variance.

★ Exam move — the Week 6 decision in one breath

Identify the statistic: **mean** $\rightarrow SE = \sigma/\sqrt{n}$, CLT needs $n \geq 30$ (15 if symmetric); **proportion** $\rightarrow SE = \sqrt{[\pi(1-\pi)/n]}$, CLT needs $n\pi \geq 5$ and $n(1-\pi) \geq 5$. Then it is always the same three moves: **standardise** $\rightarrow Z \rightarrow \text{Table E.2 tail area}$. State the condition you checked — that justification is examined, not assumed.

Confidence Intervals

Turning a single point estimate into an honest range — and saying what 'confidence' actually means

IN ONE LINE

Every confidence interval has the same shape: **point estimate $\pm$ (critical value) $\times$ (standard error)**. The point estimate ($\bar{X}$ or p) is your best single guess; the margin of error widens it into a range. '95% confident' is a statement about the long-run reliability of the method, not a probability that this one interval contains μ .

A sample gives one number — a sample mean $\bar{X}$ or a sample proportion p — but that number will land in a slightly different place every time you re-sample. Reporting it alone hides that uncertainty. A **confidence interval (CI)** reports a **range** built around the estimate, wide enough that the procedure captures the true population parameter a stated percentage of the time. Week 6 gave us the standard error (how much $\bar{X}$ jumps around); Week 7 spends it to build the interval.

7.1 The one form behind every interval

Every interval in this chapter — for a mean or a proportion, with Z or with t — is the same three ingredients. Learn the skeleton once and you only swap the parts.

GENERAL FORM OF A CONFIDENCE INTERVAL

point estimate $\pm$ (critical value) $\times$ (standard error)
 $\text{estimate} = \bar{X} \text{ or } p \quad \cdot \quad \text{critical} = Z \text{ or } t \quad \cdot \quad \text{SE} = \sigma/\sqrt{n}, S/\sqrt{n} \text{ or } \sqrt{[p(1-p)/n]}$

The three parts

- **Point estimate** — the centre, your single best guess ($\bar{X}$ for μ , p for π).
- **Critical value** — how many standard errors wide, set by the **confidence level** ($Z_{\alpha/2}$ or $t_{\alpha/2}$).
- **Standard error** — the spread of the estimate over repeated samples.

The **margin of error** $e = (\text{critical value}) \times (\text{standard error})$ is the half-width. The interval is estimate $\pm e$.

What moves the width

- **Higher confidence** (95% $\rightarrow$ 99%) $\Rightarrow$ larger critical value $\Rightarrow$ **wider**.
- **Larger n** $\Rightarrow$ smaller SE ($\propto 1/\sqrt{n}$) $\Rightarrow$ **narrower** (the only free lunch).
- **More variable data** (bigger σ or S) $\Rightarrow$ **wider**.

There is always a **trade-off**: you buy more confidence with a wider, less useful interval — or buy precision by collecting more data.

i Same skeleton, three swappable parts

Do not memorise three separate formulas. Memorise **one**: estimate $\pm$ critical $\times$ SE. Then ask only two questions — **mean or proportion?** and (for a mean) **is σ known?** — to fill in which point estimate, which critical value (Z or t) and which standard-error formula to use. The rest of the chapter just walks the branches of that single template.

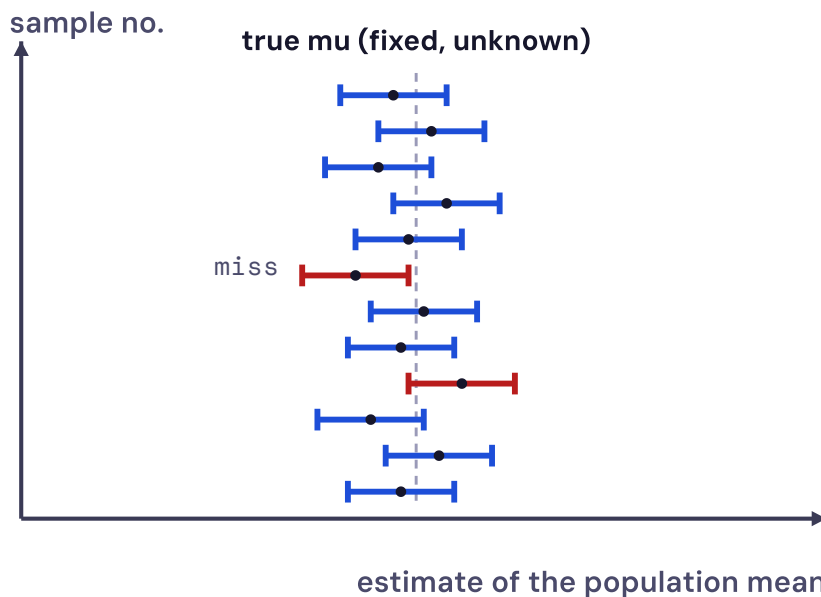
CHAPTER 7 (CONT.)

What "95% confident" really means

The single most-tested idea in the chapter – and the wording that loses marks

Before any arithmetic, get the **interpretation** right, because Part B of the exam awards marks for the sentence as much as the numbers. The confidence level is a property of the **procedure that generates intervals**, not of any one interval you have already calculated.

7.2 The capture picture



Each blue bracket is one sample's interval $\bar{X} \pm (\text{critical} \times \text{SE})$; the dot is its point estimate $\bar{X}$. Samples vary, so the brackets slide left and right, but **about 95% of them straddle the fixed true μ** (the vertical line). A **red** bracket is one of the ~5% that miss. You never know which kind you drew – that is exactly what '95% confident' means.

The right sentence

"We are 95% confident that the interval (L, U) contains the true mean μ " – meaning **the method that produced (L, U) captures μ in 95% of all possible samples**. The μ is fixed; the interval is the random thing that varies from sample to sample.

The wrong sentence

"There is a 95% probability that μ lies between L and U." Once L and U are numbers, μ is either in or out – **no probability remains to attach to a fixed parameter**. This single rewording is the most common Part-B deduction.

! Confidence is about the method, not this one interval

Once you compute, say, (56.3, 67.8), the true μ is a **fixed number** that is either in that interval or not – there is no '95% probability' left to assign to it. The 95% describes the **procedure: if you repeated the whole sampling-and-interval process many times, about 95% of the intervals would capture μ** . Saying 'there is a 95% chance μ is between 56.3 and 67.8' is the classic exam error.

✓ A higher confidence level buys a wider net

Want to be 99% sure the net catches μ ? Use $Z = 2.58$ instead of 1.96 – the brackets in the figure grow longer, so **fewer of them miss**, but each is **less precise**. 100% confidence would need an infinitely wide (useless) interval. The art is choosing a level (usually 95%) that is reliable enough yet still narrow enough to be useful.

CHAPTER 7 (CONT.)

CI for a mean μ when σ is known

The Z-interval, the common confidence levels, and where each critical value comes from

Start with the textbook (if unrealistic) case where the **population standard deviation σ is known**. Then the sampling distribution of $\bar{X}$ is normal with standard error $\sigma/\sqrt{n}$, so the critical value is read straight off the **standard normal Z** table. This is the cleanest place to meet the machinery before t complicates it.

CONFIDENCE INTERVAL FOR μ - σ KNOWN

$$\bar{X} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \quad \sigma \text{ known} \cdot \bar{X} \text{ normal by CLT } (n \geq 30) \text{ or normal population}$$

Reading the critical value

The confidence level is $1 - \alpha$. The interval is two-sided, so the leftover α is split into **two tails of $\alpha/2$ each**. The critical value **$Z_{\alpha/2}$** is the Z that leaves $\alpha/2$ in the upper tail. For 95% confidence, $\alpha = 0.05$, $\alpha/2 = 0.025$, and $Z_{0.025} = 1.96$.

Common critical values

Confidence	α	$\alpha/2$	$Z_{\alpha/2}$
90%	0.10	0.05	1.645
95%	0.05	0.025	1.96
98%	0.02	0.01	2.33
99%	0.01	0.005	2.58

WE-7A0

A quick σ -known interval

WE-7a0 · σ KNOWN → USE Z

A logistics firm knows from years of data that parcel weights have **$\sigma = 1.2$ kg**. A random sample of **$n = 64$** parcels has mean $\bar{X} = 4.5$ kg. Build a 95% CI for the mean parcel weight μ .

- Standard error:** $\sigma/\sqrt{n} = 1.2/\sqrt{64} = 1.2/8 = 0.15$ kg.
- Critical value:** 95% $\Rightarrow Z_{0.025} = 1.96$.
- Margin of error:** $e = 1.96 \times 0.15 = 0.294$ kg.
- Interval:** $4.5 \pm 0.294 \rightarrow (4.21 \text{ kg}, 4.79 \text{ kg})$. We are 95% confident the mean parcel weight lies in this range.

i Why σ -known is mostly a teaching device

If you genuinely knew σ you would usually know μ too, so the σ -known interval is rare in practice. It matters because it isolates one idea — **read the critical value off the normal table** — before the t-distribution arrives to handle the realistic case on the next page. The **structure is identical**; only the critical value changes.

✓ Width is symmetric and predictable

Because the interval is $\bar{X} \pm e$, the point estimate always sits **exactly in the middle**. To halve the margin of error you must **quadruple n** (the $\sqrt{n}$ in the denominator) — precision gets expensive fast, which is the whole motivation for the sample-size formulas a couple of pages on.

CHAPTER 7 (CONT.)

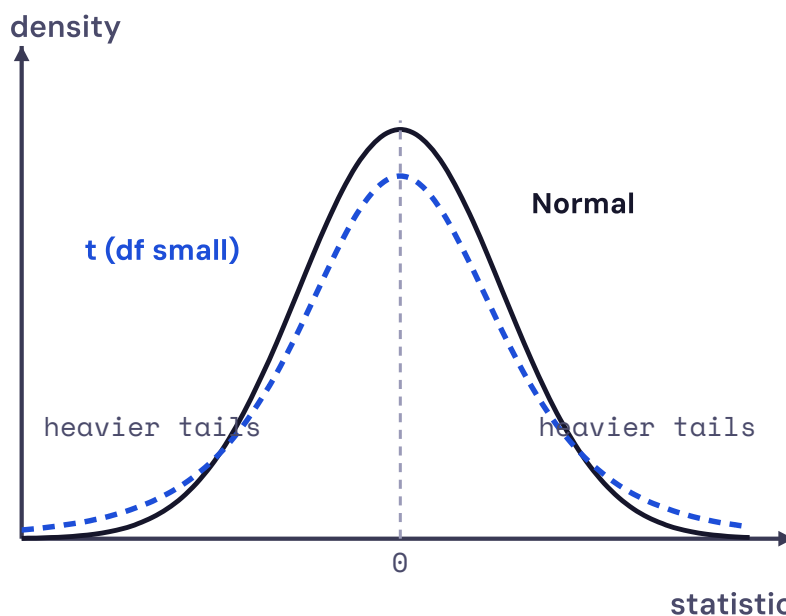
CI for a mean μ when σ is unknown

The realistic case: estimate σ with S , switch from Z to the t -distribution

In real problems σ is **unknown**, so we estimate it with the sample standard deviation S . Replacing a fixed σ with a noisy S adds extra uncertainty, so the normal Z is no longer correct — we use the **Student t -distribution with $df = n - 1$ degrees of freedom**. Its fatter tails give a slightly larger critical value, making the interval a little wider to pay for not knowing σ .

CONFIDENCE INTERVAL FOR μ — σ UNKNOWN

$$\bar{X} \pm t_{\alpha/2, n-1} \cdot \frac{S}{\sqrt{n}} \quad \sigma \text{ unknown} \rightarrow \text{use } S \cdot df = n - 1 \cdot \text{Excel: } =T.INV.2T(\alpha, n-1)$$



Standard normal Z (solid) vs Student t (dashed). The t has a **lower peak and heavier tails**, so its critical value is **larger** than Z for the same confidence — that is the price of estimating σ with S . As df grows (bigger n) the t curve tightens toward the normal: by $df \approx 100$ they are almost identical.

t critical values shrink toward Z

$df = n-1$	$t_{0.025}$ (95%)
10	2.228
20	2.086
30	2.042
39	≈ 2.02
$\infty (= Z)$	1.96

When is t valid?

The t -interval assumes the **population is roughly normal**, OR the sample is large enough ($n \geq 30$) for the **CLT** to make $\bar{X}$ approximately normal. For small, badly skewed samples, neither holds and the interval is unreliable. Always default to t when σ is unknown — even for large n , since $t \rightarrow Z$ anyway.

i Degrees of freedom = $n - 1$, not n

The t critical value depends on the sample size only through **$df = n - 1$** . We lose one degree of freedom because S itself is computed from the same n observations (the deviations must sum to zero). Read the t -table row labelled **$n - 1$** , not n — an off-by-one here picks the wrong critical value.

CHAPTER 7 (CONT.)

Worked: a t-interval for mean spend

The realistic σ -unknown calculation, start to finish (WE-7a)

This is the chapter's flagship mean calculation: σ is unknown, so it runs on the t-distribution with $df = n - 1$. Follow the four moves — standard error, critical value, margin of error, interval — then read the answer back in plain business language, which is where Part B awards its marks.

WE-7A 95% interval for mean spend at a bookshop

WE-7a · σ UNKNOWN $\rightarrow$ t, $df = 39$

A random sample of $n = 40$ receipts at a campus bookshop has mean $\bar{X} = \$62$ and sample standard deviation $S = \$18$. The population σ is unknown, so use t with $df = 40 - 1 = 39$. Build a 95% confidence interval for the mean spend per receipt μ .

- 1 **Standard error:** $S/\sqrt{n} = 18/\sqrt{40} = 18/6.3246 = \2.846 .
- 2 **Critical value:** $df = 39, 95\% \Rightarrow t_{0.025, 39} \approx 2.02$ (from the table; close to 1.96 because n is fairly large).
- 3 **Margin of error:** $e = 2.02 \times 2.846 \approx \5.75 .
- 4 **Interval:** $62 \pm 5.75 \rightarrow (\$56.25, \$67.75)$.
- 5 **Read it back:** we are 95% confident the mean spend per receipt is between \$56.25 and \$67.75 — i.e. the method captures μ in 95% of such samples.

\$2.846

STANDARD ERROR $S/\sqrt{n}$

±\$5.75

MARGIN OF ERROR

\$56.25–67.75

95% INTERVAL FOR μ

! Use t whenever you only have S — even when n is large

Students reach for $Z = 1.96$ the moment $n \geq 30$. With σ **unknown** the correct critical value is $t_{\alpha/2, n-1}$; it is only *approximately* 1.96 for large n . On a closed-book exam, decide first: σ **given** $\rightarrow$ Z; **only S given** $\rightarrow$ t. Then count $df = n - 1$ and look up the t-row.

✓ Sanity-check the width against the data scale

A $\pm \$5.75$ margin on a \$62 mean is about 9% — sensible for $n = 40$ fairly variable receipts. If your interval comes out absurdly wide (e.g. $\pm \$60$) or impossibly tight, suspect a slip: forgot the $\sqrt{n}$, used S instead of $S/\sqrt{n}$, or grabbed the wrong t-row. Always pause to ask whether the interval is a plausible size for the numbers.

CHAPTER 7 (CONT.)

How big a sample do I need?

Invert the margin-of-error formula to find n before you collect any data

Run the logic backwards: instead of taking n and getting a margin of error e , **fix the e you can tolerate and solve for the n that delivers it**. This is a planning calculation — done *before* sampling — so we use Z (the level you want) and a best guess of the variability.

Sample size for a mean

FOR A MEAN

$$n = \frac{Z^2 \sigma^2}{e^2}$$

Needs an estimate of σ (a pilot study, a prior sample, or the range/4 rule). e is the desired half-width in the **same units** as the data.

Sample size for a proportion

FOR A PROPORTION

$$n = \frac{Z^2 \pi(1-\pi)}{e^2}$$

If π is unknown, use the **worst case $\pi = 0.5$** — it maximises $\pi(1-\pi) = 0.25$ and so gives the largest, safest n .

✓ Always round a sample size UP

n must be a whole number, and rounding *down* would miss the target margin of error. So **always round n up** to the next integer ($78.0001 \rightarrow 79$; $77.8 \rightarrow 78$). Rounding to nearest or down is the standard mistake.

WE-7B Sample size for a \$4 margin of error on mean spend

WE-7b · PLAN n BEFORE SAMPLING

Continuing the bookshop study (WE-7a): the manager now wants a 95% interval for mean spend with a margin of error of just **$e = \$4$** . Using the pilot estimate of variability $\sigma \approx S = \$18$ and $Z = 1.96$, how many receipts must be sampled?

- Write the formula:** $n = Z^2 \sigma^2 / e^2 = (1.96^2 \times 18^2) / 4^2$.
- Numerator:** $1.96^2 = 3.8416$; $18^2 = 324$; $3.8416 \times 324 = 1244.7$.
- Divide by e^2 :** $4^2 = 16$; $1244.7 / 16 = 77.8$.
- Round UP:** $n = 78$ receipts. (The original 40 was not enough to pin the mean down to $\pm \$4$ at 95% confidence.)

\$4

TARGET MARGIN OF ERROR e
\$18
 Σ ESTIMATE (= S)

78

RECEIPTS REQUIRED

! Use Z (not t) and σ^2 (not σ) in the sample-size formula

Two slips cost marks: (1) the planning formula uses **Z** , because you do not yet have a sample to count df from; (2) the numerator has the **variance σ^2** , not σ — forgetting to square σ (and to square Z and e) is the most common arithmetic error.

CHAPTER 7 (CONT.)

CI for a proportion π , and auditing

Estimating a yes/no rate – and the classic business use: auditing

When the variable is **categorical** (recommend / would not, defective / fine, error / correct) we estimate the **population proportion π** with the sample proportion $p = X/n$. By the CLT the sampling distribution of p is approximately normal, so the interval again uses Z – with the standard error $\sqrt{[p(1-p)/n]}$.

CONFIDENCE INTERVAL FOR A PROPORTION π

$$p \pm Z_{\alpha/2} \cdot \sqrt{p(1-p)/n} \quad p = X/n \quad \text{valid when } np \geq 5 \text{ and } n(1-p) \geq 5$$

WE-7C

95% interval for the proportion who would recommend

WE-7C · PROPORTION → Z, SE = $\sqrt{[p(1-p)/n]}$

In a survey of **$n = 250$** customers, **90** said they would recommend the brand. Build a 95% confidence interval for the true proportion π of all customers who would recommend.

- 1 **Point estimate:** $p = X/n = 90/250 = 0.36$.
- 2 **Check the CLT conditions:** $np = 90 \geq 5$ and $n(1-p) = 160 \geq 5$ – normal approximation OK.
- 3 **Standard error:** $\sqrt{[p(1-p)/n]} = \sqrt{[0.36 \times 0.64 / 250]} = \sqrt{0.0009216} = 0.03036$.
- 4 **Margin of error:** $e = 1.96 \times 0.03036 = 0.0595$.
- 5 **Interval:** $0.36 \pm 0.0595 \rightarrow (0.300, 0.420)$, i.e. about **30% to 42%** of customers would recommend the brand.

! Check $np \geq 5$ and $n(1-p) \geq 5$ before trusting the interval

The normal approximation underpinning the proportion interval needs **at least about 5 expected successes and 5 expected failures**. With a rare event (tiny p) and a small n it can fail, and the symmetric $\pm$ interval may even stray below 0 or above 1. Always state the check – it is an easy mark and flags when the method is invalid.

i A proportion is just a mean of 0s and 1s

Code each yes as 1 and each no as 0; the sample proportion p is literally the mean of that 0/1 column. That is why the proportion interval has the same estimate $\pm$ critical $\times$ SE skeleton as the mean – only the standard-error formula, $\sqrt{[p(1-p)/n]}$, is special to the binary case.

CHAPTER 7 (CONT.)

Application: auditing, and the exam recap

Sampling invoices for an error rate — and the decision tree for any CI question

7.6 Application: auditing

Auditors cannot recheck every invoice, so they **sample** and report a confidence interval for a population total or, more often, an **error rate** — the proportion of records containing a mistake. The proportion interval is exactly the tool: sample n records, count the errors, and report $p \pm Z_{\alpha/2} \sqrt{[p(1-p)/n]}$ as the plausible range for the population error rate. Planning the sample size in advance uses the proportion sample-size formula with the conservative $\pi = 0.5$.

WE-7D Auditing: sample size for an error-rate estimate

WE-7d · AUDIT PLANNING WITH $\pi = 0.5$

An auditor wants to estimate the proportion of invoices with a posting error to within **e = ±3%** (0.03) at 95% confidence. With **no prior estimate of the error rate**, how many invoices should be sampled?

- 1 **No prior $\pi \rightarrow$ use the worst case $\pi = 0.5$** (maximises $\pi(1-\pi) = 0.25$).
- 2 **Formula:** $n = Z^2 \pi(1-\pi) / e^2 = (1.96^2 \times 0.25) / 0.03^2$.
- 3 **Compute:** $3.8416 \times 0.25 = 0.9604$; $0.03^2 = 0.0009$; $0.9604 / 0.0009 = 1067.1$.
- 4 **Round UP:** $n = 1068$ invoices to guarantee the $\pm 3\%$ margin at 95% confidence.

$\pi = 0.5$

WORST-CASE VARIABILITY

±3%

TARGET MARGIN OF ERROR

1068

INVOICES TO SAMPLE

! Proportion CI vs proportion sample-size use DIFFERENT variability

When you already have data, the interval uses the **observed p** in the standard error $\sqrt{[p(1-p)/n]}$. When you are **planning n in advance** with no data yet, use the conservative **$\pi = 0.5$** . Mixing them — planning with a hopeful low π , or building the interval with 0.5 — is a common slip.

★ Exam move — the confidence-interval decision tree

(1) **Mean or proportion?** Numerical data $\rightarrow$ mean; yes/no data $\rightarrow$ proportion. (2) **For a mean, is σ known?** Given $\sigma \rightarrow Z$, $SE = \sigma/\sqrt{n}$; only $S \rightarrow t$ with $df = n-1$, $SE = S/\sqrt{n}$. (3) **For a proportion** $\rightarrow Z$, $SE = \sqrt{[p(1-p)/n]}$, check $np \geq 5$ and $n(1-p) \geq 5$. (4) Interval = estimate $\pm$ (critical)(SE); state it in context. (5) **Sample size?** Invert with Z , use σ^2 (mean) or $\pi = 0.5$ (proportion), and **round up**. (6) Never say ' $P(\mu \text{ in interval}) = 0.95$ ' — confidence is a property of the **method**.

Hypothesis Testing: One-Sample Tests

The courtroom logic of H_0 vs H_1 , the significance level, and the two ways to be wrong

IN ONE LINE

A hypothesis test asks whether sample evidence is strong enough to reject a claim about a population parameter. Set up H_0 (status quo) and H_1 (what you must prove), fix a significance level α , compute a test statistic, then reject H_0 if it falls in the rejection region (or if $p < \alpha$). Rejecting a true H_0 is a Type I error (prob α); missing a real effect is a Type II error (prob β).

Hypothesis testing lets data **overturn a default belief**. You never *prove* H_0 true; you only ask whether the sample gives enough evidence to **reject** it. Think of a courtroom: H_0 is "innocent", kept unless the evidence convicts beyond your chosen standard of doubt α .

8.1 Setting up H_0 and H_1

The null H_0 — the status quo

The **null hypothesis H_0** is the claim of **no change / no effect**, always stated with an **equals sign** on the parameter: $H_0: \mu = \mu_0$ (or $\pi = \pi_0$). It is the benchmark we assume true *while computing the test statistic*.

The alternative H_1 — what you must show

The **alternative H_1** carries the research claim and never an equals sign: it uses $\neq$ (two-tailed), or a direction $< / >$ (one-tailed). Whatever you are trying to demonstrate goes in H_1 ; the burden of proof rests on it.

THE HYPOTHESIS PAIR

$H_0: \mu = \mu_0$ (always =, the status quo)

$H_1: \mu \neq \mu_0$ two-tailed or $H_1: \mu < \mu_0 / \mu > \mu_0$ one-tailed

i The claim you want to prove goes in H_1

If QC *suspects under-filling*, the thing to demonstrate is $\mu < 250$, so that becomes H_1 and $H_0: \mu = 250$. The test can only ever **reject H_0** ('significant evidence') or **fail to reject H_0** ('not enough evidence') — it never accepts or proves H_0 . 'Fail to reject' is a verdict of *not guilty*, not of *innocent*.

8.2 Significance level α and the two errors

The **significance level α** is the risk you are willing to take of **rejecting a true H_0** — conventionally 0.05, sometimes 0.01 or 0.10. It is chosen **before** seeing the data. There are exactly two ways a test can go wrong, and they trade off against each other.

Error	What happens	Probability
Type I	Reject H_0 when it is actually TRUE (a false alarm)	α
Type II	Fail to reject H_0 when it is actually FALSE (a miss)	β
Power	Correctly reject a false H_0 (detect a real effect)	$1 - \beta$

! α and β pull in opposite directions

For a fixed sample size n , making the test **stricter** (smaller α) *raises* β — you reject less often, so you also miss more real effects. You cannot drive both errors to zero at once. The only clean way to lower α and β together is to **increase n** , which raises power.

CHAPTER 8 (CONT.)

Errors, power, and choosing the tails

The 2x2 truth table, what raises power, and one- vs two-tailed tests

		TRUTH about H_0	
		H_0 is TRUE	H_0 is FALSE
Your DECISION	Do NOT reject H_0	Correct prob $1 - \alpha$	Type II error prob β
	Reject H_0	Type I error prob α	Correct – POWER prob $1 - \beta$

The two ways to be wrong. A **Type I error** (probability α) is convicting an innocent H_0 – rejecting a true null. A **Type II error** (probability β) is acquitting a guilty H_0 – failing to reject a false null. **Power** = $1 - \beta$ is the chance of correctly detecting a real effect. Shrinking α (a stricter test) raises β for a fixed n ; the clean way to cut both is a **larger sample**.

8.3 Power — the chance of catching a real effect

Power = $1 - \beta$ is the probability the test **correctly rejects a false H_0** . A high-powered test reliably detects effects that are really there. Four levers raise power:

- **Bigger sample n** — shrinks the standard error, so the sampling distributions of H_0 and the truth overlap less. This is the lever you control.
- **Larger true effect** — the further the real parameter is from μ_0 , the easier it is to detect (not under your control).
- **Larger α** — a looser test rejects more readily, raising power but also Type I risk.
- **Less variability σ** — a cleaner measurement leaves less noise to hide the signal.

i Power is why sample size matters

Underpowered studies (small n) routinely **fail to reject H_0** even when an effect is real — a Type II miss, not evidence of 'no effect'. 'Not statistically significant' with a tiny sample usually means *we could not see it*, not *it is not there*.

8.4 One-tailed vs two-tailed tests

Two-tailed ($H_1: \neq$)

Use when you care about a difference in **either direction**. The rejection region splits into **two tails**, each holding $\alpha/2$. At $\alpha = 0.05$ the critical Z values are **± 1.96** .

One-tailed ($H_1: < \text{or } >$)

Use when only **one direction** matters (under-filling, an *increase* in conversion). The whole α sits in a **single tail**, so the critical value is closer to zero ($Z = \pm 1.645$ at $\alpha = 0.05$) — a one-tailed test is **more powerful** in that direction.

! Decide the tail from H_1 BEFORE seeing the data

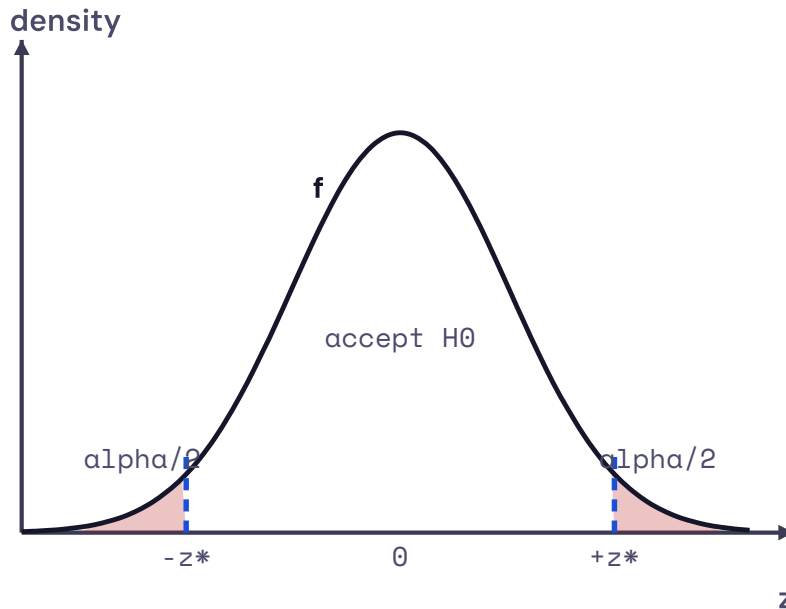
Choosing a one-tailed test *after* peeking at which way the sample fell is a form of **p-hacking** — it doubles your real Type I rate. The direction of H_1 must come from the research question, set in advance. If the question is genuinely 'is it different?', use two tails.

CHAPTER 8 (CONT.)

Two roads to the same decision

The critical-value approach and the p-value approach – and the rejection region

Once H_0 , H_1 and α are fixed and the test statistic is computed, there are **two equivalent ways** to reach a verdict. They always agree; pick whichever the question asks for (Excel naturally gives you the p-value).



Rejection region (two-tailed). Under H_0 the test statistic is approximately standard normal. The two shaded tails beyond the critical values $\pm z^*$ each hold $\alpha/2$; together they are the **rejection region**. A statistic landing in the unshaded centre (“accept H_0 ”) gives **fail to reject**; one in a shaded tail gives **reject H_0** . For a one-tailed test the whole α goes into a single tail.

Critical-value approach

- Find the **critical value** (z^* or t^*) that cuts off α (one-tailed) or $\alpha/2$ (two-tailed) in the tail(s).
- Compute the **test statistic** from the sample.
- **Reject H_0** if the statistic is more extreme than the critical value (lands in the rejection region).

p-value approach

- The **p-value** is the probability, *assuming H_0 is true*, of a test statistic at least as extreme as the one observed.
- **Reject H_0 if $p < \alpha$** ; otherwise fail to reject.
- Smaller p = stronger evidence against H_0 . The p-value is the exact tail area; the critical value is just the α cut-off.

THE DECISION RULE, TWO WAYS

Critical-value: reject if $|stat| > \text{critical value}$

p-value: reject if $p < \alpha$ (the two rules always agree)

! The p-value is NOT the probability that H_0 is true

A p-value of 0.03 does **not** mean ‘there is a 3% chance H_0 is true’. It means: *if H_0 were true*, a result this extreme would occur 3% of the time. H_0 is either true or false — it has no probability. Confusing these two is the most-penalised written-answer slip.

CHAPTER 8 (CONT.)

Testing a single mean

Z when σ is known, t with $df = n - 1$ when σ is unknown

IN ONE LINE

Both statistics measure how many standard errors the sample mean $\bar{X}$ sits from the hypothesised μ_0 . Use Z only when the population σ is known (rare); when you must estimate σ with the sample S — almost always — use t with $n - 1$ degrees of freedom.

σ known $\rightarrow$ Z-test

If the population standard deviation σ is genuinely known, the standardised mean is exactly **standard normal** under H_0 . The standard error is $\sigma/\sqrt{n}$.

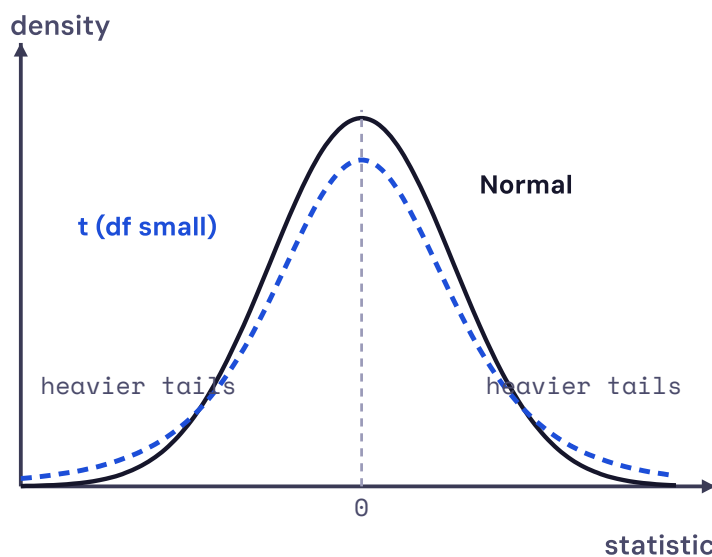
σ unknown $\rightarrow$ t-test

In practice σ is unknown, so we plug in the sample SD S. The extra uncertainty in estimating σ makes the statistic follow the **Student t-distribution with $df = n - 1$** , not the normal.

TEST STATISTIC FOR A MEAN

σ known: $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$

σ unknown: $t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$ $df = n - 1$



Why t, not Z, when σ is unknown. The Student t (dashed) has a lower peak and **heavier tails** than the standard normal (solid), reflecting the extra uncertainty from estimating σ with S. So t critical values are **larger** (the test is more conservative) for small n, and $t \rightarrow$ normal as $df \rightarrow \infty$ (by about $n = 30$ they nearly coincide).

i Assumptions of the one-mean test

The mean test assumes the sample is **random** and the data are **numerical**. The sampling distribution of $\bar{X}$ must be approximately **normal**: this holds if the **population is normal**, or by the **CLT** if n is large enough ($n \geq 30$ generally; $n \geq 15$ suffices for a fairly symmetric population). For small n from a clearly skewed population the t-test is unreliable.

CHAPTER 8 (CONT.)

Worked: one-tailed t-test for a mean

A vending machine that should pour 250 ml – is it under-filling?

! Use t unless σ is truly given

Examiners hand you a **sample S** and expect a **t-test**. Reaching for Z just because n is large is wrong in principle — if σ is estimated, the distribution is t. (They will be numerically close for large n, but the **named procedure** earns the mark.)

WE-15 Vending machine — testing for under-filling

WE-15 · ONE-TAILED t-TEST FOR μ

A vending machine is set to pour **250 ml** per cup. Quality control suspects it is **under-filling**. A random sample of **n = 36** cups gives **$\bar{X} = 246$ ml** with **S = 9 ml**. Test at **$\alpha = 0.05$** . (σ is unknown $\rightarrow$ t-test, $df = n - 1 = 35$.)

- Hypotheses.** The claim to prove is under-filling, so it goes in H_1 : $H_0: \mu = 250$ vs $H_1: \mu < 250$ — a **one-tailed (left) test**.
- Standard error.** $SE = S/\sqrt{n} = 9/\sqrt{36} = 9/6 = 1.5$ ml.
- Test statistic.** $t = (\bar{X} - \mu_0)/SE = (246 - 250)/1.5 = -4/1.5 = -2.67$, with $df = 35$.
- Critical value.** For a left-tailed test at $\alpha = 0.05$ with $df = 35$, $t^* \approx -1.69$. The rejection region is $t < -1.69$.
- Decision.** $-2.67 < -1.69$, so the statistic is in the rejection region $\rightarrow$ **reject H_0** .
- Conclusion.** At the 5% level there is **significant evidence the machine under-fills** (true mean below 250 ml).

-2.67

TEST STATISTIC T

35

DEGREES OF FREEDOM

Reject

 H_0 AT $\alpha = 0.05$

The Excel p-value

Excel gives the **left-tail** probability directly. T.DIST with the cumulative flag returns $P(T \leq t)$:

EXCEL P-VALUE

p-value (left tail) = **`=T.DIST(-2.67, 35, 1)`** ≈ 0.0058

syntax: **`=T.DIST(t, df, cumulative=1)`** $\rightarrow$ area to the LEFT of t

i p-value confirms the same verdict

$p \approx 0.0058 < \alpha = 0.05$, so we **reject H_0** — identical to the critical-value decision. For a *right*-tailed test you would use **`=T.DIST.RT(t, df)`**; for a *two*-tailed test, **`=T.DIST.2T(|t|, df)`** (which needs a positive t).

! Match the Excel function to the tail

T.DIST is the **left** tail, T.DIST.RT the **right**, T.DIST.2T **both** (and it rejects a negative input). Picking the wrong one is a classic exam slip — the calculator gives the wrong tail area but a plausible-looking number.

CHAPTER 8 (CONT.)

Testing a single proportion, & the ethics of reporting

A Z-test for π , its CLT conditions, and the rules of honest statistical practice

For a categorical outcome (convert / not, defective / not) the parameter is a **proportion π** . By the CLT the sample proportion p is approximately normal, so the test statistic is a **Z** — built on the **hypothesised π_0** , not the sample p , inside the standard error.

TEST STATISTIC FOR A PROPORTION

$$Z = \frac{p - \pi_0}{\sqrt{(\pi_0(1-\pi_0)/n)}} \quad p = X/n \text{ (sample proportion)}$$

CLT conditions: $n\pi_0 \geq 5$ and $n(1 - \pi_0) \geq 5$

! Use π_0 (not p) in the standard error

The proportion test uses the **null value π_0** inside the standard error — $\sqrt{(\pi_0(1-\pi_0)/n)}$ — because the whole calculation is done *assuming H_0 is true*. (A confidence interval, by contrast, uses the sample p , since there is no null to assume. Mixing these two standard errors is a common error.)

WE-16 Did a redesign lift trial-to-paid conversion?

WE-16 · ONE-SAMPLE Z-TEST FOR π

Historically **20%** of trial users convert to paid. After a product redesign, **64 of 280** trial users convert ($p = 64/280 = 0.2286$). Test at $\alpha = 0.05$ whether conversion has **improved**.

- Hypotheses.** The claim is an *increase*, so $H_0: \pi = 0.20$ vs $H_1: \pi > 0.20$ — a **one-tailed (right) test**.
- CLT check.** $n\pi_0 = 280(0.20) = 56 \geq 5$ and $n(1-\pi_0) = 280(0.80) = 224 \geq 5$ — the normal approximation is valid.
- Standard error (under H_0).** $\sqrt{(0.20 \times 0.80 / 280)} = \sqrt{(0.16 / 280)} = \sqrt{0.0005714} = 0.0239$.
- Test statistic.** $Z = (0.2286 - 0.20) / 0.0239 = 0.0286 / 0.0239 = 1.20$.
- Critical value.** Right-tailed at $\alpha = 0.05 \rightarrow z^* = 1.645$.
- Decision.** $1.20 < 1.645$, so the statistic is **not** in the rejection region $\rightarrow$ **fail to reject H_0** .
- Conclusion.** There is **not** enough evidence at the 5% level that the redesign raised conversion — the lift to 22.86% could plausibly be sampling noise.

1.20

TEST STATISTIC Z

1.645

CRITICAL Z^* (RIGHT)

Fail to reject

 H_0 AT $\alpha = 0.05$

i “Fail to reject” is not “no effect”

A non-significant result does not prove the redesign did nothing. With $n = 280$ the test simply lacked the **power** to confirm so small a lift. A larger trial might well reach significance — absence of evidence is not evidence of absence.

CHAPTER 8 (CONT.)

Ethics & the exam playbook

Honest reporting, statistical vs practical significance, and a clean answer template

8.5 The ethics of statistical reporting

Because a hypothesis test can be nudged into “significance” by selective choices, BUSS1020 treats **honest reporting as part of the method**. Three duties recur in written answers:

Duty	What it means	The abuse it prevents
Report every test	Disclose all tests run, not just the ones that came out significant	cherry-picking results
No p-hacking	Fix α and the tail in advance; do not tweak them to cross 0.05	manufactured significance
Stat $\neq$ practical	Report effect size and cost, not just “ $p < 0.05$ ”	trivial effects oversold

“A result can be statistically significant and practically worthless — with a huge sample, even a 0.1% lift clears $p < 0.05$. Always ask: is the effect *big enough to act on*?”

BUSS1020 · ETHICS OF HYPOTHESIS TESTING

Statistical vs practical significance

Statistical significance ($p < \alpha$) says an effect is **unlikely to be pure chance**. **Practical significance** asks whether the effect is **large enough to matter** for the business decision. Large n makes tiny, meaningless effects significant; small n can leave large, important effects non-significant. A complete answer reports **both** the test result and the effect size.

! p-hacking — the cardinal sin

Running many tests and reporting only the “winners”, switching to a one-tailed test after seeing the data, or stopping data collection the moment p dips below 0.05 all **inflate the true Type I rate** far above the stated α . The fix: **pre-specify** H_1 , α and the tail, and **report all tests run**.

8.6 The exam playbook

- 1 State H_0 and H_1** in symbols, putting the claim to prove in H_1 ; this fixes the **tail(s)**.
- 2 Pick the procedure:** mean with σ known $\rightarrow$ **Z**; mean with σ unknown $\rightarrow$ **t** (**df = n - 1**); proportion $\rightarrow$ **Z** (check $n\pi_0 \geq 5$ and $n(1-\pi_0) \geq 5$).
- 3 Compute the test statistic** = (estimate - null) / standard error; show the SE working.
- 4 Decide** by the critical value *or* $p < \alpha$ (they agree) and state **reject / fail to reject H_0** .
- 5 Write a one-sentence conclusion in context**, and where asked, comment on **practical** significance and assumptions.

★ Exam move — the one-sample test in five lines

(1) H_0 : **parameter = value**; H_1 carries the claim $\rightarrow$ sets the tail. (2) Choose **Z vs t** (mean: σ known/unknown) or the **proportion Z**; verify **assumptions / CLT conditions**. (3) **stat = (estimate - null)/SE**. (4) **Reject if $|stat| > critical$ or $p < \alpha$** . (5) **Conclude in context** and separate **statistical from practical** significance. The formula sheet gives the formulae — the **decision logic is what you memorise**.

Two-Sample Tests

Comparing two means (pooled, separate, paired) and two proportions

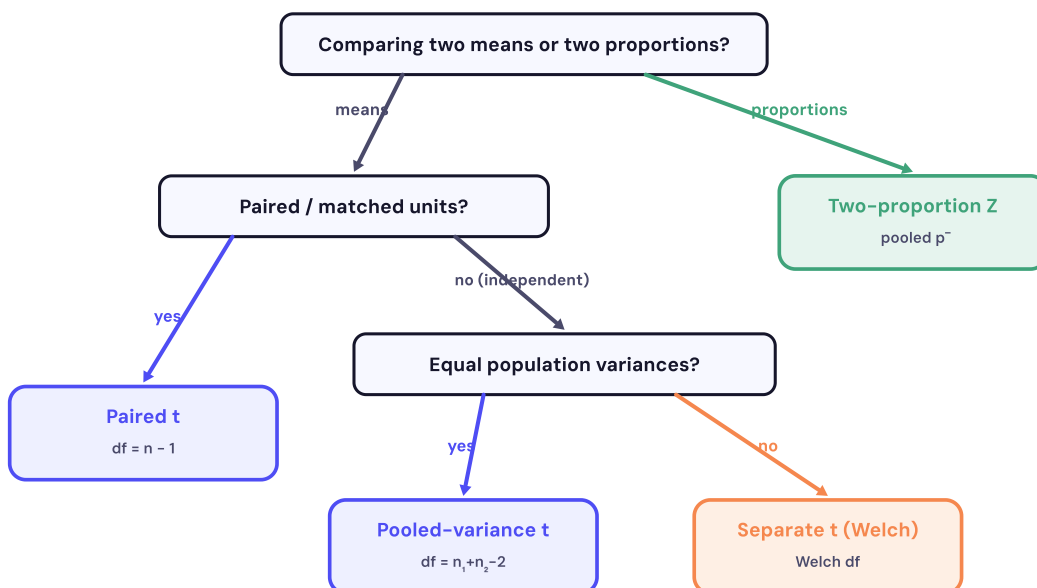
IN ONE LINE

Week 8 tested one sample against a fixed value; Week 9 compares two populations. Four routes, decided by two questions. Paired units → paired t on the differences ($df = n-1$). Independent means → pooled-variance t if variances are assumed equal ($df = n_1+n_2-2$), else separate-variance (Welch) t. Two proportions → a Z built on the pooled estimate $\bar{p}$.

Almost every business comparison is a two-sample question: does layout A keep visitors longer than layout B? Did a training course raise scores? Do two regions recommend the brand at the same rate? The arithmetic differs, but the **logic is unchanged** from Week 8 — set H_0 and H_1 about a **difference**, build a test statistic, compare it to a critical value (or read a p-value), and decide. The only new skill is **choosing the right statistic**, and that turns on two structural questions about the data.

9.1 Which two-sample test? Read the design

Ask first whether you are comparing **means** (numerical data) or **proportions** (success/failure data). For means, ask whether the two samples are **paired** — the *same* units measured twice (before/after) or deliberately matched — or **independent** (two separate, unrelated groups). Independent means then split on whether the two population **variances** are assumed equal.



A 95% CI for the difference that contains 0 $\Leftrightarrow$ fail to reject H_0 at $\alpha = 0.05$.

Two-sample test selector. First split: **means** vs **proportions**. For means, are the samples **paired/matched**? → **paired t** on the differences ($df = n-1$). If **independent**: equal population variances assumed → **pooled-variance t** ($df = n_1+n_2-2$); unequal → **separate-variance (Welch) t**. Two **proportions** → **two-proportion Z** with a pooled estimate $\bar{p}$. A 95% CI for the difference that contains 0 $\Leftrightarrow$ fail to reject H_0 at $\alpha = 0.05$.

CHAPTER 9 (CONT.)

Independent means, equal variances

Pool the two sample variances into one S_p^2 , then a single t with $df = n_1 + n_2 - 2$

9.2 Independent means, equal variances: pooled-variance t

When two **independent** samples come from populations whose variances we are willing to assume **equal**, we pool the two sample variances into one best estimate S_p^2 — a weighted average using degrees of freedom as the weights — and feed it into a single t statistic with **$df = n_1 + n_2 - 2$** .

POOLED-VARIANCE t (EQUAL VARIANCES ASSUMED)

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_p^2(1/n_1 + 1/n_2)}} \quad df = n_1 + n_2 - 2$$

Under H_0 the difference of population means $(\mu_1 - \mu_2)$ is usually **0**, so that term drops out and the statistic is just the observed mean gap divided by its standard error. The denominator $\sqrt{[S_p^2(1/n_1 + 1/n_2)]}$ is the **standard error of the difference**.

i Assumptions behind the pooled t

The pooled-variance t needs **both populations approximately normal** (or each $n \geq 30$ so the CLT applies to each mean) **and equal population variances**. The equal-variance assumption is what lets us pool S_1^2 and S_2^2 into one S_p^2 . A rough field check: if the larger sample variance is more than about **4×** the smaller, do not assume equality — use the separate-variance t instead (§9.3).

Why pool at all?

If both populations truly share one variance, the two samples each carry information about it. Pooling them into a single S_p^2 gives a **more precise** variance estimate and more degrees of freedom ($n_1 + n_2 - 2$) than either sample alone — which sharpens the test.

The df weighting

S_p^2 is a **df -weighted average** of S_1^2 and S_2^2 , not a plain average: the bigger sample (more df) pulls the pooled value toward its own variance. That is why you weight by $(n_1 - 1)$ and $(n_2 - 1)$, never simply average the two.

CHAPTER 9 (CONT.)

Worked: pooled-variance t

*Comparing time-on-task for two website layouts, step by step***WE-9A Time-on-task: layout A vs layout B (pooled-variance t)**WE-9a · POOLED-VARIANCE t, $df = n_1 + n_2 - 2$

Time-on-task (minutes) is recorded for two independent groups of visitors. **Layout A:** $n_1 = 12$, $\bar{X}_1 = 8.4$, $S_1 = 1.8$. **Layout B:** $n_2 = 14$, $\bar{X}_2 = 7.1$, $S_2 = 2.0$. Assuming equal variances, test at $\alpha = 0.05$ (two-tailed) whether the mean times differ.

- 1 **Hypotheses.** $H_0: \mu_1 - \mu_2 = 0$ (same mean time); $H_1: \mu_1 - \mu_2 \neq 0$ (two-tailed — we have no prior direction).
- 2 **Pool the variances.** $S_p^2 = [(12-1)(1.8^2) + (14-1)(2.0^2)] / (12+14-2) = [11(3.24) + 13(4.00)] / 24 = (35.64 + 52.00) / 24 = 87.64/24 = \mathbf{3.652}$.
- 3 **Standard error of the difference.** $SE = \sqrt{S_p^2(1/n_1 + 1/n_2)} = \sqrt{3.652(1/12 + 1/14)} = \sqrt{3.652 \times 0.1548} = \sqrt{0.5653} \approx \mathbf{0.752 \text{ min}}$.
- 4 **Test statistic.** $t = (\bar{X}_1 - \bar{X}_2) / SE = (8.4 - 7.1)/0.752 = 1.3/0.752 \approx \mathbf{1.73}$, with $df = 12 + 14 - 2 = \mathbf{24}$.
- 5 **Critical value & decision.** Two-tailed $t_{0.025, 24} \approx \pm 2.064$. Since $1.73 < 2.064$ (and $-1.73 > -2.064$), the statistic is *inside* the accept region: **fail to reject H_0** .
- 6 **Conclusion.** At $\alpha = 0.05$ there is **no significant difference** in mean time-on-task between the two layouts. Excel two-tailed p-value: **=T.DIST.2T(1.73, 24)**.

3.652POOLED S_p^2 **1.73**

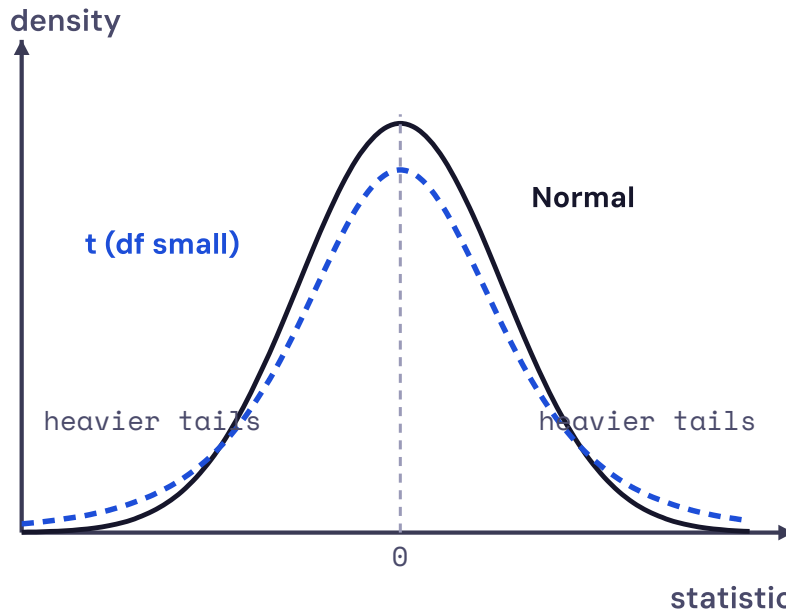
T STATISTIC

24DF = $N_1 + N_2 - 2$ **Fail**TO REJECT H_0

CHAPTER 9 (CONT.)

Why t, not Z — and the pooling traps

Heavier tails for small samples, and pooling variances the right way



Why t, not Z. With small samples and σ unknown, the standardised mean-difference follows a **Student t** (dashed) — a **lower peak and heavier tails** than the normal (solid). Heavier tails $\Rightarrow$ a **larger critical value** (± 2.064 at $df = 24$ vs ± 1.96 for Z), so t demands stronger evidence. As df grows the t collapses onto the normal.

! Pool the variances — do not pool the standard errors

A frequent slip is averaging S_1 and S_2 directly, or forgetting to weight by degrees of freedom. Pool the **variances** with the df weights $(n_1 - 1)$ and $(n_2 - 1)$: $S_p^2 = [(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2] / (n_1 + n_2 - 2)$. Then the standard error uses the bracket **$(1/n_1 + 1/n_2)$** , not $(1/(n_1 + n_2))$. Mixing these up shifts t enough to flip the decision.

i Sign and order are your choice — be consistent

You may label either group 1; just keep the **same order** in the numerator $(X_1 - X_2)$ and in the conclusion. For a two-tailed test the sign of t does not matter — compare **$|t|$** to the positive critical value.

CHAPTER 9 (CONT.)

Separate–variance t & the paired t

When variances are not equal, and when the same units are measured twice

IN ONE LINE

Drop the equal–variance assumption and you keep the variances apart: the **separate–variance (Welch) t** uses $S_1^2/n_1 + S_2^2/n_2$ under the root and an adjusted **Welch–Satterthwaite df**. When the two samples are the same units measured twice, collapse them to a single column of differences D and run a one–sample t on D — the **paired t**, $df = n-1$.

9.3 Independent means, unequal variances: separate–variance t

If the two populations clearly do **not** share a variance, pooling would be wrong. The **separate–variance (Welch) t** keeps each sample variance with its own sample size and uses a reduced, generally **non–integer df** (the Welch–Satterthwaite approximation, supplied by software or the formula sheet).

SEPARATE-VARIANCE (WELCH) T (VARIANCES NOT EQUAL)

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \quad \text{Welch–Satterthwaite df (usually non-integer)}$$

What changes from the pooled t

The numerator is identical — the observed mean gap minus the hypothesised difference. Only the **standard error** changes: each variance stays with its own n , so there is **no S_p^2** . The df is smaller than $n_1 + n_2 - 2$, which slightly **widens** the critical value.

When to reach for it

Use it whenever the equal–variance assumption is doubtful — very different sample SDs, or theory that one group should be more variable. In practice the Welch t is the **safe default** for independent means; it costs little when variances really are equal.

! Equal–variance assumption is a choice — state it

The pooled and separate t can give **different** decisions when sample SDs are far apart. In a written answer, **state which you assume** (and why): if the question says “assume equal variances” use the pooled t; if it flags unequal spread or gives wildly different S values, use the separate–variance t. Examiners award the marks for the **justified choice** as much as the arithmetic.

CHAPTER 9 (CONT.)

Paired-difference t

Same units measured twice: difference each pair, then a one-sample t on D

9.4 Paired-difference t

When each observation in sample 1 is naturally tied to one in sample 2 — the **same person before and after**, the same store in two periods, matched pairs — the samples are **not independent**. Compute the **difference D = (after – before)** for each pair, then run a **one-sample t-test on the single column D**: the pairing cancels the between-subject noise.

PAIRED-DIFFERENCE T (DEPENDENT SAMPLES)

$$D_i = X_{1i} - X_{2i} \quad \bar{D} = \sum D_i / n \quad S_D = \sqrt{[\sum (D_i - \bar{D})^2 / (n-1)]}$$

$$t = \frac{\bar{D} - \mu_D}{S_D / \sqrt{n}} \quad df = n - 1 \quad (n = \text{number of PAIRS})$$

✓ Pairing is a design win, not a complication

By differencing, the paired t removes the variation *between* subjects (some people are simply faster, some stores simply bigger) and tests only the **within-pair change**. This usually gives a **smaller standard error** and more power than an independent-samples test on the same totals. The cost is $df = n-1$ (n = number of pairs), smaller than $n_1 + n_2 - 2$.

How to spot a paired design

- **Before/after** on the same units (training, a diet, a redesign).
- **Matched pairs** deliberately paired on a confounder (twins, matched stores).
- Two columns with an obvious **row-by-row correspondence** — equal n by construction.

Then it is just Week 8

Once you have the single column D, the paired t is **exactly the one-sample t-test** from Chapter 8 with $\mu_D = 0$ as the null. Same statistic, same table, same decision rule — the only new step is the differencing that got you to D.

! Do not run an independent-samples t on paired data

Treating before/after columns as two **independent** groups throws away the pairing, inflates the standard error and usually **hides a real effect**. If the rows correspond one-to-one, the correct test is the **paired t** on the differences — every time.

CHAPTER 9 (CONT.)

Worked paired t; and two proportions

Before/after training scores, then comparing two success rates

WE-9B Productivity before vs after a training course (paired t)

WE-9b · PAIRED-DIFFERENCE t, $df = n-1$

Eight employees are scored **before** and **after** a training course. The differences $D = (\text{after} - \text{before})$ for the 8 staff are: **5, 3, 6, 2, 4, 7, 1, 4**. With $\bar{D} = 4.0$ and $S_D \approx 2.0$, test at $\alpha = 0.05$ whether training raised scores.

- 1 **Recognise the design.** The *same* 8 employees are measured twice → samples are **paired**, so work with the single column of differences D , not two independent groups.
- 2 **Hypotheses (one-tailed — we expect an increase).** $H_0: \mu_D = 0$; $H_1: \mu_D > 0$.
- 3 **Summaries.** $\bar{D} = (5+3+6+2+4+7+1+4)/8 = 32/8 = 4.0$; $S_D \approx 2.0$ (spread of the 8 differences); $n = 8$ pairs, so $df = n - 1 = 7$.
- 4 **Standard error.** $SE = S_D/\sqrt{n} = 2.0/\sqrt{8} = 2.0/2.828 \approx 0.707$.
- 5 **Test statistic.** $t = (\bar{D} - 0)/SE = 4.0/0.707 \approx 5.66$, $df = 7$.
- 6 **Critical value & decision.** One-tailed $t_{0.05, 7} \approx 1.895$ (two-tailed $t_{0.025, 7} = 2.365$). Either way 5.66 is far beyond the critical value → **reject H_0** ; training **significantly raised** scores. Excel: =T.DIST.RT(5.66, 7).

4.0

MEAN DIFF $\bar{D}$

5.66

T STATISTIC

7

DF = N-1

Reject

 H_0 — EFFECT

CHAPTER 9 (CONT.)

Comparing two proportions

A pooled-estimate Z for two success rates, worked end to end

9.5 Two proportions: the pooled-Z test

To compare two **population proportions** π_1 and π_2 (conversion rates, defect rates, recommend rates) we use a **Z** test. Under H_0 the two proportions are *equal*, so we estimate that single common proportion by pooling all successes over all trials: the **pooled estimate $\bar{p}$** . That $\bar{p}$ then builds the standard error.

TWO-PROPORTION Z (POOLED ESTIMATE)

$$\bar{p} = \frac{X_1 + X_2}{n_1 + n_2} \quad \text{pooled estimate (combine all successes)}$$

$$Z = \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1-\bar{p})(1/n_1 + 1/n_2)}}$$

WE-9C

Recommend rate: two customer regions (two-proportion Z)

WE-9C · TWO-PROPORTION Z, POOLED $\bar{p}$

In region 1, $X_1 = 84$ of $n_1 = 200$ customers would recommend the brand ($p_1 = 0.42$); in region 2, $X_2 = 78$ of $n_2 = 300$ ($p_2 = 0.26$). Test at $\alpha = 0.05$ (two-tailed) whether the recommend rates differ.

- 1 **Hypotheses.** $H_0: \pi_1 - \pi_2 = 0$; $H_1: \pi_1 - \pi_2 \neq 0$. Sample proportions $p_1 = 84/200 = \mathbf{0.42}$, $p_2 = 78/300 = \mathbf{0.26}$.
- 2 **Pooled estimate.** $\bar{p} = (X_1 + X_2)/(n_1 + n_2) = (84 + 78)/(200 + 300) = 162/500 = \mathbf{0.324}$.
- 3 **Standard error.** $SE = \sqrt{\bar{p}(1-\bar{p})(1/n_1 + 1/n_2)} = \sqrt{0.324 \times 0.676 \times (1/200 + 1/300)} = \sqrt{0.2190 \times 0.008333} = \sqrt{0.001825} \approx \mathbf{0.0427}$.
- 4 **Test statistic.** $Z = (p_1 - p_2)/SE = (0.42 - 0.26)/0.0427 = 0.16/0.0427 \approx \mathbf{3.74}$.
- 5 **Decision.** Two-tailed critical $Z = \pm 1.96$. Since $3.74 > 1.96 \rightarrow$ **reject H_0** : the two regions recommend the brand at **significantly different** rates. (CLT conditions hold: all of $X_1, n_1 - X_1, X_2, n_2 - X_2 \geq 5$.)

! Pool $\bar{p}$ for the TEST; do NOT pool for a confidence interval

The pooled $\bar{p}$'s are correct **only for the hypothesis test**, because H_0 assumes $\pi_1 = \pi_2$. For a **confidence interval** on $(\pi_1 - \pi_2)$ there is no such assumption — use the **separate** sample proportions in the SE: $\sqrt{p_1(1-p_1)/n_1 + p_2(1-p_2)/n_2}$. Using $\bar{p}$ in a CI is a classic error.

CHAPTER 9 (CONT.)

Confidence intervals & the contains-0 rule

A CI for every difference, and the bridge back to the test

IN ONE LINE

Every two-sample test has a matching confidence interval for the difference: point estimate $\pm$ (critical value)(standard error), with the same SE the test uses. The headline link: a 95% CI for a difference that contains 0 $\Leftrightarrow$ fail to reject H_0 (two-tailed) at $\alpha = 0.05$. A CI that excludes 0 $\Leftrightarrow$ reject.

9.6 Confidence intervals for each difference

Each statistic in this chapter has a parallel CI. The structure is always **(difference of estimates) $\pm$ (critical value) $\times$ (standard error)** — only the critical value (t or Z) and the SE change with the test.

CONFIDENCE INTERVALS FOR A DIFFERENCE

Pooled means: $(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2} \sqrt{[S_p^2(1/n_1 + 1/n_2)]}$ $df = n_1 + n_2 - 2$

Paired: $\bar{D} \pm t_{\alpha/2} (S_D / \sqrt{n})$ $df = n - 1$

Two prop.: $(p_1 - p_2) \pm Z_{\alpha/2} \sqrt{[p_1(1-p_1)/n_1 + p_2(1-p_2)/n_2]}$ **separate p's**

WE-9d

95% CI for the layout mean difference (from WE-9a)

WE-9d · CI FOR A DIFFERENCE + CONTAINS-0 CHECK

Reuse WE-9a: $\bar{X}_1 - \bar{X}_2 = 8.4 - 7.1 = 1.3$ min, $SE = 0.752$, $df = 24$, $t_{0.025, 24} = 2.064$. Build the 95% CI for $\mu_1 - \mu_2$ and link it to the test decision.

- Margin of error.** $ME = t_{0.025, 24} \times SE = 2.064 \times 0.752 \approx 1.55$ min.
- Interval.** $(\bar{X}_1 - \bar{X}_2) \pm ME = 1.3 \pm 1.55 \rightarrow (-0.25, +2.85)$ minutes.
- Read the interval.** The interval **includes 0**, so a zero difference ($\mu_1 = \mu_2$) is plausible at 95% confidence.
- Cross-check the test.** WE-9a failed to reject H_0 at $\alpha = 0.05$ — exactly what a contains-0 interval predicts. The two procedures **must** agree.

1.3

DIFF $\bar{X}_1 - \bar{X}_2$ (MIN)

 ± 1.55

MARGIN OF ERROR

(-0.25, 2.85)

95% CI

contains 0

 $\Leftrightarrow$ FAIL TO REJECT

CHAPTER 9 (CONT.)

The contains–0 bridge & the exam playbook

Why a CI and its test must agree – and the classify-compute-link routine

Design	Statistic	Standard error	df
Paired / matched	paired t on D	$S_D/\sqrt{n}$	$n - 1$
Indep., equal var.	pooled-variance t	$\sqrt{[S_p^2(1/n_1+1/n_2)]}$	n_1+n_2-2
Indep., unequal var.	separate-variance (Welch) t	$\sqrt{[S_1^2/n_1 + S_2^2/n_2]}$	Welch–Satt.
Two proportions	pooled-pZ	$\sqrt{[p(1-p)(1/n_1+1/n_2)]}$	—

The contains–0 ⇔ fail-to-reject bridge

A two-tailed test at α and the matching **(1– α)** CI for the difference are the **same decision** seen two ways. If the CI **contains 0**, a zero difference is plausible → **fail to reject** H_0 . If the CI **excludes 0**, zero is implausible → **reject** H_0 . Use this to sanity-check arithmetic: a CI and a test that disagree mean you slipped somewhere.

Exam move — classify, then compute, then link

(1) Name the design first: **paired** (same/matched units) → paired t, $df = n-1$; **independent means** → pooled t (equal variances, $df = n_1+n_2-2$) or separate/Welch t; **two proportions** → pooled-pZ. **(2)** Build the right standard error — pool variances (not SEs) for the pooled t; pool successes for the proportion *test* (but use separate p’s in a proportion *CI*). **(3)** State H_0/H_1 about the difference, compare to the critical value, and where asked, report the matching CI — a contains–0 interval must echo a fail-to-reject. Part B (Weeks 7–12) rewards the **justified choice of test** as much as the number.

Simple Linear Regression

Covariance, correlation, and turning a cloud of points into a straight-line model

IN ONE LINE

Two numerical variables can move together. Covariance measures the direction of that co-movement; the correlation coefficient r rescales it to $-1 \leq r \leq +1$ so its size is comparable across problems. r captures the strength and direction of a LINEAR relationship only — it is not causation and it is blind to curves.

Everything so far described *one* variable, or compared groups. The final block of BUSS1020 studies how **two numerical variables relate**: does monthly sales rise with training hours, does maintenance cost climb with machine age? We first **measure** the relationship (covariance, correlation), then **model** it with a straight line, then **test** whether the line is real and use it to **predict**.

10.1 Covariance — direction of co-movement

For each point, ask whether X and Y are *both* above their means, *both* below, or on opposite sides. The product $(X_i - \bar{X})(Y_i - \bar{Y})$ is **positive** when they agree and **negative** when they disagree. The sample **covariance** averages those products:

SAMPLE COVARIANCE

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1} \quad \text{sign} = \text{direction of the linear relationship}$$

- **Cov > 0**: X and Y tend to move *together* (up-up, down-down).
- **Cov < 0**: X and Y tend to move in *opposite* directions.
- **Cov $\approx$ 0**: no *linear* co-movement (a curve can still hide here).

! Covariance has units, so its size is meaningless on its own

$\text{Cov}(X, Y)$ is in the product of X 's and Y 's units (e.g. hours·\$000). Re-scale X from hours to minutes and the covariance changes by a factor of 60, though the relationship is identical. You can read its **sign**, but never its **strength** — that is exactly the problem the correlation coefficient fixes.



CHAPTER 10 (CONT.)

The correlation coefficient r

Rescale covariance to a unit-free $-1 \leq r \leq +1$ – strength and direction

10.2 The correlation coefficient r

Divide the covariance by both standard deviations and the units cancel, leaving a pure number between -1 and $+1$. This is the **correlation coefficient r** – the standard measure of the **strength and direction of a linear relationship**:

CORRELATION COEFFICIENT

$$r = \frac{\text{Cov}(X, Y)}{s_X \cdot s_Y} \quad -1 \leq r \leq +1, \text{ unit-free}$$

Value of r	Interpretation
$r = +1$	perfect positive linear relationship (all points on an upward line)
$0 < r < 1$	positive linear relationship; nearer 1 = stronger
$r = 0$	no linear relationship (but a curve may still exist)
$-1 < r < 0$	negative linear relationship; nearer -1 = stronger
$r = -1$	perfect negative linear relationship (all points on a downward line)

i Strength bands (rule of thumb, not a law)

$|r|$ around 0.1–0.3 is weak, 0.3–0.7 moderate, 0.7–0.9 strong, > 0.9 very strong. These are **guides** for words, not thresholds for significance – whether r is *statistically* different from 0 is the t-test on page 4, and it depends on the sample size n as much as on the size of r .

CHAPTER 10 (CONT.)

From correlation to a fitted line

The two great cautions about r , then the least-squares model $\hat{Y} = b_0 + b_1X$

! Correlation is NOT causation, and r only sees straight lines

(1) Not causation. A strong r means X and Y move together, not that X *causes* Y . Ice-cream sales and drownings correlate because a **lurking variable** (hot weather) drives both. **(2) Linear only.** r measures *straight-line* association: a perfect U-shaped relationship (e.g. $Y = X^2$) can give $r \approx 0$ even though X completely determines Y . **Always plot the scatter first** — a single number can lie about the shape.

"A scatterplot is worth a thousand correlation coefficients. Look at the cloud before you trust any r — r is honest only about lines."

EXAM-CRAFT · ALWAYS PLOT FIRST

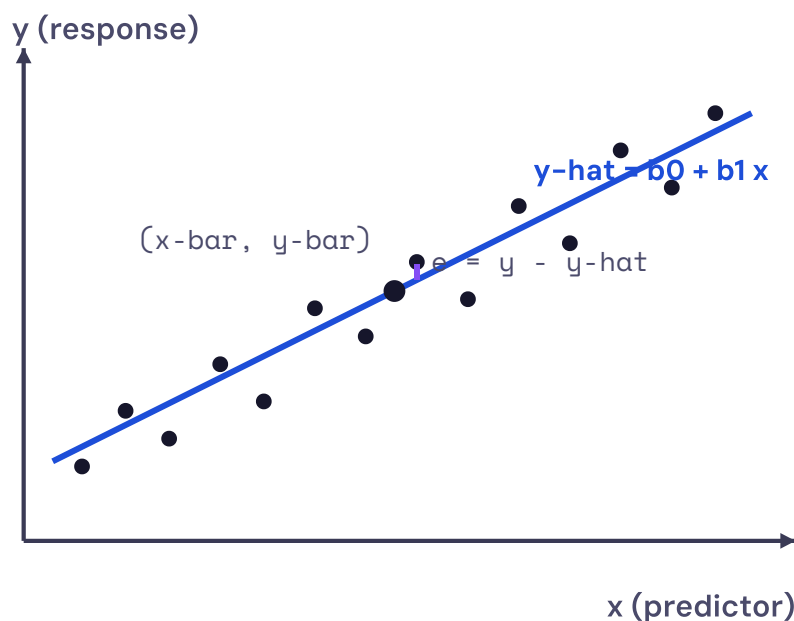
10.3 The simple linear regression model

Once a scatterplot shows a roughly linear, useful relationship, we **fit a line** that lets us predict Y from X . The fitted (estimated) model writes the predicted value $\hat{Y}$ as an intercept plus a slope times X :

THE FITTED SLR MODEL

$$\hat{Y} = b_0 + b_1X$$

b_0 = sample intercept · b_1 = sample slope · $\hat{Y}$ = predicted (fitted) Y



The least-squares line through the cloud. Of all possible straight lines, least squares picks the one that **minimises** $\sum e^2 = \sum (Y_i - \hat{Y}_i)^2$, the sum of squared **vertical residuals** $e = Y - \hat{Y}$ (purple drop). The line always passes through the **centroid** $(\bar{X}, \bar{Y})$ (violet dot), so $b_0 = \bar{Y} - b_1\bar{X}$ falls straight out once the slope is known.

CHAPTER 10 (CONT.)

Estimating the slope & intercept

Least squares: $b_1 = SSXY/SSXX$, then b_0 locks the line onto the centroid

Estimating the slope and intercept

The least-squares slope is the covariation of X and Y divided by the variation in X; the intercept then locks the line onto the centroid:

LEAST-SQUARES SLOPE & INTERCEPT

$$b_1 = \frac{SSXY}{SSXX} \quad \text{where } SSXY = \sum (X_i - \bar{X})(Y_i - \bar{Y}), \quad SSXX = \sum (X_i - \bar{X})^2$$

$$b_0 = \bar{Y} - b_1\bar{X} \quad \text{the line passes through } (\bar{X}, \bar{Y})$$

✓ The slope carries the sign of r

b_1 and r always share the same sign — both come from $SSXY$ (and $\text{Cov}(X, Y)$). In fact $b_1 = r \cdot (S_Y / S_X)$. So a positive correlation gives an upward line, and you can sanity-check a computed slope against the sign of the scatter.

Why “least squares”?

Among all candidate lines, least squares minimises $\sum (Y_i - \hat{Y}_i)^2$ — the sum of squared *vertical* gaps. Squaring stops positive and negative residuals from cancelling and penalises big misses harder, which is what pins down a single best-fit line.

The centroid anchor

Because $b_0 = \bar{Y} - b_1\bar{X}$, the fitted line is **forced through** $(\bar{X}, \bar{Y})$. That gives a quick check: plug $\bar{X}$ into the equation and you must get $\bar{Y}$ back. It also means errors in the slope pivot the line about the centroid rather than shifting it bodily.

i SSXY and SSXX are just Week-2 building blocks

$SSXX = \sum (X_i - \bar{X})^2$ is the same numerator you used for the variance of X, and $SSXY = \sum (X_i - \bar{X})(Y_i - \bar{Y})$ is the covariance numerator. So the slope $b_1 = SSXY/SSXX$ reuses sums you already know how to build — no new machinery, just a ratio of two of them.

CHAPTER 10 (CONT.)

Reading the line, then using it

Interpreting b_0 and b_1 in business words, then a full prediction and residual

Interpreting the coefficients

Slope b_1

The **change in predicted Y for a one-unit increase in X**. Always quote it with units: "each extra hour of training is associated with a \$1,800 rise in predicted monthly sales". It is an *association within the data range*, not a guaranteed causal effect.

Intercept b_0

The **predicted Y when X = 0**. It is only meaningful if X = 0 is realistic and inside the data range; otherwise it is just where the maths puts the line and should **not** be interpreted as a real prediction.

! Do not extrapolate beyond the range of X

The line is only evidence over the span of X you actually observed. Predicting sales for **40 training hours** when the data only ran 0–15 hours is **extrapolation** — the linear pattern may break down entirely outside the observed range. Predict *inside* the cloud, and treat b_0 with the same suspicion if X = 0 lies outside it.

WE-
10A

Training hours → monthly sales: predict, interpret, residual

WE-10a · PREDICT · INTERPRET · r · RESIDUAL

For a chain of stores, monthly sales (Y, in \$000) are regressed on hours of staff training (X). The fitted line is $\hat{Y} = 22 + 1.8X$ with $r^2 = 0.49$. (a) Predict sales at X = 10 hours. (b) Interpret the slope and intercept. (c) Recover r. (d) A store with X = 6 actually recorded \$35,000 in sales — find its residual.

- 1 Predict at X = 10:** $\hat{Y} = 22 + 1.8(10) = 22 + 18 = 40$ → predicted sales of **\$40,000**.
- 2 Slope $b_1 = 1.8$:** each **extra hour of training** is associated with a **\$1,800 rise** in predicted monthly sales ($1.8 \times \$1,000$).
- 3 Intercept $b_0 = 22$:** a store with **zero** training hours has predicted sales of \$22,000 — meaningful only if X = 0 is realistic and within the data range.
- 4 Recover r:** $r = \pm\sqrt{r^2} = \pm\sqrt{0.49} = +0.70$ — **positive** because the slope is positive. A moderate-to-strong positive linear relationship.
- 5 Residual at X = 6:** fitted $\hat{Y} = 22 + 1.8(6) = 22 + 10.8 = 32.8$ (\$32,800). Residual $e = Y - \hat{Y} = 35 - 32.8 = +2.2$ → the store sold **\$2,200 above** the line.

40

$\hat{Y}$ AT X = 10 (\$000)

+1.8

SLOPE b_1 (\$000/HR)

+0.70

R = $\pm\sqrt{R^2}$

+2.2

RESIDUAL (\$000)

! Take the square root of r^2 , then fix the sign from the slope

r^2 is always between 0 and 1 and is **always positive**, so $\sqrt{r^2}$ alone cannot tell you direction. $r = +0.70$ here only because $b_1 > 0$; if the slope were negative the same $r^2 = 0.49$ would give $r = -0.70$. Read the sign off the slope (or the scatter), never off r^2 .

CHAPTER 10 (CONT.)

How good is the fit?

Decomposing the variation: $SST = SSR + SSE$, then r^2 and the standard error of the estimate

IN ONE LINE

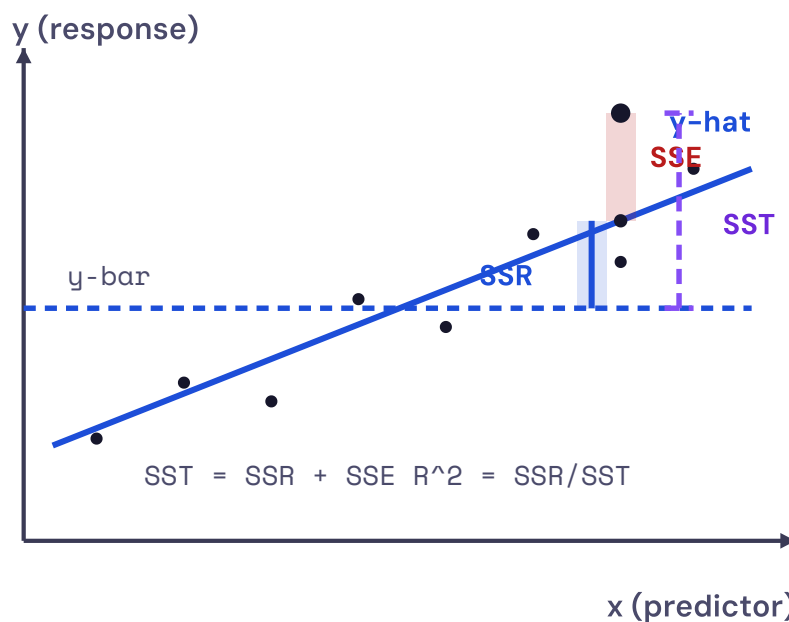
Total spread in Y (SST) splits into a part the line explains (SSR) plus a part it cannot (SSE). The fraction explained is $r^2 = SSR/SST$; the typical size of a residual is the standard error of the estimate S_{ER} .

10.4 Measures of variation

Why does Y vary at all? Some of the spread is tracked by the line (as X changes, $\hat{Y}$ changes); the rest is scatter around the line. The total sum of squares splits cleanly into these two pieces — and the cross term vanishes *exactly* for the least-squares line:

DECOMPOSITION OF VARIATION

$SST = SSR + SSE$
 $SST = \sum(Y_i - \bar{Y})^2 \cdot SSR = \sum(\hat{Y}_i - \bar{Y})^2 \cdot SSE = \sum(Y_i - \hat{Y}_i)^2$ total = explained + unexplained



The split for one point. Its total deviation from $\bar{Y}$ (violet, **SST** piece) is exactly the part the line explains — $\hat{Y} - \bar{Y}$ (blue, **SSR**) — plus the residual it leaves behind — $Y - \hat{Y}$ (red, **SSE**). Square and sum over all points and the identity $SST = SSR + SSE$ holds. $r^2 = SSR/SST$ is the share of total spread the line accounts for.

CHAPTER 10 (CONT.)

Scoring the fit

r^2 = the share of Y explained, S_{ER} = the typical residual, and regression to the mean

Coefficient of determination r^2

The proportion of total variation in Y **explained** by the regression on X :

R-SQUARED

$$r^2 = \frac{SSR}{SST} \quad 0 \leq r^2 \leq 1$$

In SLR only, r^2 is literally the square of the correlation r . $r^2 = 0.49$ means **49% of the variation in sales** is explained by training hours; the other 51% is other factors and noise.

Standard error of the estimate S_{ER}

The typical vertical distance of points from the line — the SD of the residuals, with $df = n - 2$ (two estimated coefficients):

STANDARD ERROR OF THE ESTIMATE

$$S_{ER} = \sqrt{\frac{SSE}{n - 2}}$$

Same units as Y . A small S_{ER} relative to $\bar{Y}$ means tight, accurate predictions. It feeds directly into every interval on page 5.

Regression to the mean

Extreme observations tend to be followed by less extreme ones. A store with a record sales month is, on average, predicted closer to the mean next period — not because anything *caused* a fall, but because the extreme was partly luck that does not repeat. Mistaking this statistical pull for a real effect (e.g. crediting an intervention applied right after a bad month) is the **regression fallacy**.

Read r^2 as a percentage, in context

Always state r^2 as “**__% of the variation in $[Y]$ is explained by $[X]$** ”, and name the leftover too. $r^2 = 0.49$ is not “the model is 49% accurate”; it is “49% of the spread in sales is tracked by training hours, 51% is everything else.” Pair it with S_{ER} for the typical error in Y ’s own units — together they tell the full goodness-of-fit story.



Inference on the slope

Is the relationship real? The t-test for β_1 ($df = n - 2$) and a confidence interval

IN ONE LINE

A non-zero sample slope b_1 could be sampling noise. The t-test for the slope ($df = n - 2$) asks whether the *population* slope β_1 differs from 0. Rejecting $H_0: \beta_1 = 0$ means there is a **significant linear relationship**. A CI for β_1 then says how big the slope plausibly is.

11.1 Testing whether the slope is real

If the true slope β_1 were 0, X would carry **no linear information** about Y. We test that null using the estimated slope b_1 , its standard error S_{b_1} , and a t-distribution with $n - 2$ degrees of freedom (two coefficients estimated):

T-TEST FOR THE SLOPE

$$t = \frac{b_1 - \beta_1}{S_{b_1}} \quad (\text{under } H_0: \beta_1 = 0, \quad t = b_1 / S_{b_1})$$
$$S_{b_1} = \frac{S_{ER}}{\sqrt{SS_{XX}}} \quad df = n - 2$$

The equivalent test for r

Testing $\beta_1 = 0$ is **algebraically the same test** as testing whether the population correlation $\rho = 0$. Both give an identical t-statistic and the same $df = n - 2$:

EQUIVALENT T-TEST FOR R

$$t = \frac{r \sqrt{(n-2)}}{\sqrt{(1-r^2)}} \quad df = n - 2$$

Confidence interval for β_1

Invert the test to bracket the true slope. If the interval **excludes 0**, the slope is significant at that level — the CI and the two-tailed test always agree:

CONFIDENCE INTERVAL FOR B_1

$$b_1 \pm t_{\alpha/2, n-2} \cdot S_{b_1}$$

CHAPTER 11 (CONT.)

Worked: slope t-test & a 95% CI for β_1

Machine age $\rightarrow$ maintenance cost, $df = n - 2$, and the CI-excludes-0 link

WE-10B

Machine age $\rightarrow$ maintenance cost: slope t-test and 95% CIWE-10B · SLOPE t-TEST + CI FOR β_1

Monthly maintenance cost (Y) is regressed on machine age (X) for $n = 30$ machines. The fitted slope is $b_1 = 4.2$ with standard error $S_{b_1} = 1.1$. At $\alpha = 0.05$: (a) test whether age has a significant linear relationship with cost; (b) build a 95% CI for β_1 .

- 1 **Hypotheses:** $H_0: \beta_1 = 0$ (no linear relationship) vs $H_1: \beta_1 \neq 0$ (two-tailed).
- 2 **Degrees of freedom:** $df = n - 2 = 30 - 2 = 28$.
- 3 **Test statistic:** $t = b_1 / S_{b_1} = 4.2 / 1.1 = 3.82$.
- 4 **Critical value:** $t_{0.025, 28} = 2.048$. Since $|3.82| > 2.048$, **reject H_0** — there is a significant linear relationship between age and cost.
- 5 **95% CI for β_1 :** $4.2 \pm 2.048(1.1) = 4.2 \pm 2.25 \rightarrow (1.95, 6.45)$. The interval **excludes 0**, agreeing with the test: each extra year of age plausibly adds between \$1.95 and \$6.45 of monthly maintenance cost.

28

DF = N - 2

3.82

T = B1/S_B1

2.048

T CRIT (0.025, 28)

(1.95, 6.45)

95% CI FOR B1

! Slope inference uses $df = n - 2$, not $n - 1$

Two coefficients (b_0 and b_1) are estimated from the data, so a regression t-test loses **two** degrees of freedom: $df = n - 2$. Using $n - 1$ (the one-sample df) is a classic slip. Also remember: a **two-tailed** CI that *excludes 0* is exactly equivalent to rejecting $H_0: \beta_1 = 0$ at the matching α .

CHAPTER 11 (CONT.)

Confidence vs prediction intervals

Estimating the average Y at a given X (narrow) vs predicting one new Y (wide)

IN ONE LINE

At a chosen X there are two intervals. The **confidence interval** brackets the mean response $E[Y|X]$ — narrow. The **prediction interval** brackets a single new observation Y — always wider, because it carries the extra scatter of one individual point. Both pinch at $\bar{X}$ and flare toward the extremes.

11.2 Two questions, two intervals

Both intervals are centred on the same $\hat{Y}$ from the fitted line, but they answer different questions and so have different widths. The leverage term $h_i = 1/n + (X_i - \bar{X})^2 / SSXX$ measures how far the chosen X is from $\bar{X}$.

CI for the MEAN of Y given X — narrow

"What is the **average** Y for *all* units with this X?" Only the uncertainty in where the line sits:

CI FOR THE MEAN OF Y | X

$$\hat{Y}_i \pm t_{\alpha/2} \cdot s_{ER} \cdot \sqrt{h_i}$$

$$h_i = 1/n + (X_i - \bar{X})^2 / SSXX$$

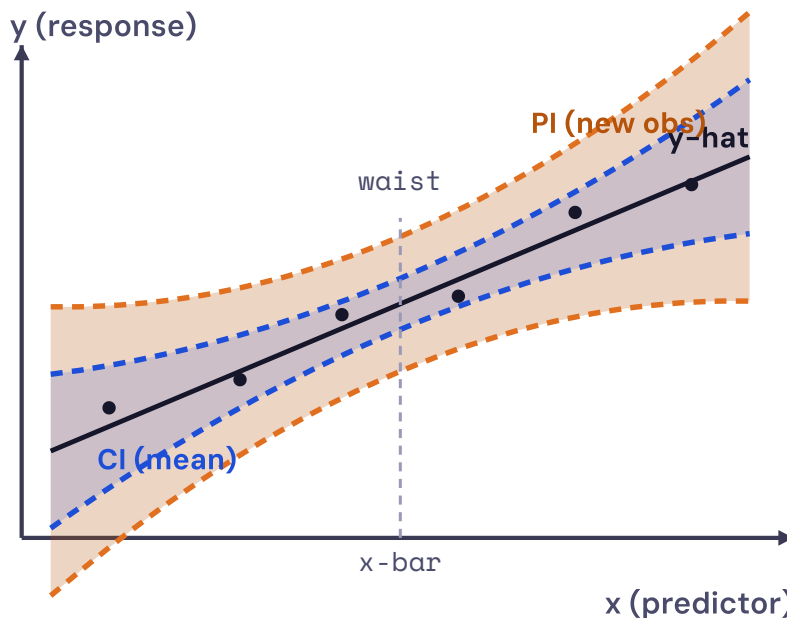
PI for an INDIVIDUAL Y — wide

"What value of Y will **one new** unit with this X take?" Adds the point's own scatter — the extra **+1** under the root:

PREDICTION INTERVAL FOR ONE Y

$$\hat{Y}_i \pm t_{\alpha/2} \cdot s_{ER} \cdot \sqrt{1 + h_i}$$

always wider — the +1 is one individual's noise



Confidence band inside prediction band. The narrow **CI band** (blue) brackets the mean response; the wide **PI band** (amber) brackets a single new observation and sits **outside it everywhere** because of the extra +1. Both are **narrowest at $\bar{X}$** (the waist) and **flare** as X moves away — predictions are most precise near the centre of the data and least precise at the edges.

■ confidence band ■ prediction band (wider)

CHAPTER 11 (CONT.)

Picking the right interval & the chapter playbook

Mean-response CI vs single-point PI, and the whole regression arc in one breath

! Mean-of-Y CI vs individual-Y PI — pick the right one

If the question asks for the **average** outcome at a given X (e.g. mean sales for all stores training 10 hours), use the **narrow CI** ($\sqrt{h_i}$). If it asks to **predict one** specific new outcome (e.g. next month's sales for *this* store), use the **wide PI** ($\sqrt{1 + h_i}$). The PI is **always wider** at the same X ; mixing them up loses the mark.

★ Exam move — the whole regression chapter in one breath

Measure the link: r (sign + strength, linear only, never causation). **Model** it: $\hat{Y} = b_0 + b_1X$, $b_1 = SSXY/SSXX$, interpret per unit and don't extrapolate. **Score** the fit: $r^2 = SSR/SST$ explained, $S_{ER} = \sqrt{[SSE/(n-2)]}$ typical error. **Test** it: $t = b_1/S_{b_1}$, **df** = $n - 2$, CI for β_1 excludes 0 $\Leftrightarrow$ significant. **Predict**: narrow CI for the mean, wide PI for one new Y . State the assumption or df you used — that is the examined sentence.

Stage	Tool	Key number
Measure	correlation r	$-1 \leq r \leq +1$ (linear only)
Model	least-squares line	$b_1 = SSXY/SSXX$, $b_0 = \bar{Y} - b_1\bar{X}$
Score fit	r^2 and S_{ER}	$r^2 = SSR/SST$; $S_{ER} = \sqrt{[SSE/(n-2)]}$
Test slope	t-test on β_1	$t = b_1/S_{b_1}$, $df = n - 2$
Predict	CI (mean) vs PI (one Y)	$\sqrt{h_i}$ vs $\sqrt{1 + h_i}$ — PI always wider

Multiple Linear Regression

Several predictors at once, each slope read holding the others constant

IN ONE LINE

Extend the straight line to several X s: $\hat{Y} = b_0 + b_1X_1 + \dots + b_kX_k$. Each b_j is the effect of that X on Y holding all the other predictors constant. Fit is read off r^2 and the penalised adjusted r^2 ; the F -test judges the model as a whole, the t -tests judge one coefficient at a time.

Simple regression used a single driver. In business almost nothing has a single driver — revenue depends on advertising *and* staffing *and* location at once. **Multiple linear regression (MLR)** fits one equation in which each predictor gets its own slope, so we can ask "what does **this** variable do, once the others are already accounted for?"

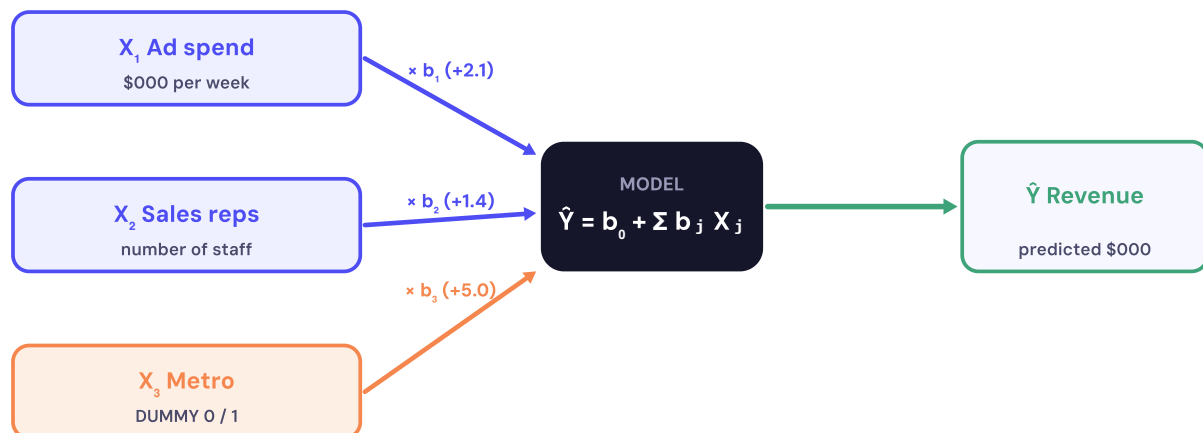
11.1 The model

For k predictors the least-squares model estimates a sample equation that we use to **predict** Y and to **interpret** each driver:

MULTIPLE REGRESSION MODEL

$\hat{Y} = b_0 + b_1X_1 + b_2X_2 + \dots + b_kX_k$ $b_0 = \text{intercept (all } X = 0)$ · $b_j = \text{slope of } X_j$ · $k = \text{number of predictors}$

MLR + DUMMY + F-TEST



F-test (overall): is the WHOLE set of slopes useful?

$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ (model explains nothing)

$H_1 : \text{at least one } \beta_j \neq 0$ (model has real explanatory power)

$F = \text{MSR} / \text{MSE} \rightarrow \text{large } F \text{ (small } p) \text{ rejects } H_0$. Then read each t -test to see WHICH X matters.

The MLR machine. Several predictors each enter the model with their own slope b_j (here +2.1, +1.4, +5.0). The **Metro** driver is a **0/1 dummy** — a switch that adds b_3 only for metro stores. The model fuses them into a single prediction $\hat{Y}$. The **F-test** asks the one big question first: are the slopes **jointly** worth anything (reject H_0 : all $\beta_j = 0$)? Only then do the individual t -tests say *which* X is pulling its weight.



CHAPTER 11 (CONT.)

“Holding the others constant”

Each slope is a partial effect — the heart of multiple regression

11.2 "Holding the others constant"

This phrase is the heart of MLR and a guaranteed exam marker. Each slope b_j is a **partial effect**: the predicted change in Y for a **one-unit** rise in X_j **while every other predictor is kept fixed**. It is the contribution of X_j *after* the other variables have already had their say — not its raw, alone-in-the-room effect.

Term	Reads as	Watch
b_0 (intercept)	predicted Y when every $X = 0$	often not meaningful (no X is ever 0)
b_j (slope)	$\Delta \hat{Y}$ per +1 in X_j , others held constant	always quote the units of X_j and Y
sign of b_j	direction of the partial effect	can flip vs simple regression if predictors are correlated

i Why the word “partial” matters

A coefficient can change — even change sign — when you add or drop another predictor. That is not a mistake: each b_j is measured **net of the other variables in the model**. So always interpret a slope **relative to the specific model it sits in**, and use the words **“holding the other predictors constant”** every time.

Simple vs multiple slope

In simple regression a slope absorbs the effect of *everything* correlated with X . In MLR the slope is **cleaned** of the other predictors in the model, so it isolates X_j 's own contribution. That is why the two can differ — and why a sign flip is a signal that a lurking predictor was doing the work all along.

The phrase that earns the mark

Every coefficient interpretation in Part B should end with **“holding the other predictors constant”** (or “all else equal”). Omitting it turns a partial effect into a false absolute claim — the single most common MLR interpretation error, and an easy half-mark to lose.

✓ Always carry the units

A slope is meaningless without units on both sides: “each extra **\$1,000 of ad spend** raises predicted revenue by **\$2,100**”, not “2.1”. State what a one-unit change in X_j *is*, and what the change in Y *is*, in their real-world units.

CHAPTER 11 (CONT.)

Model fit: r^2 , adj r^2 , and the F-test

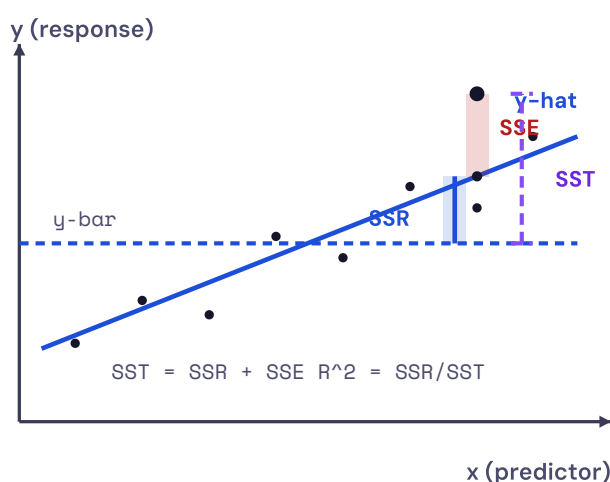
How well does the whole model explain Y – and is that explanation real?

11.3 Coefficient of multiple determination r^2

Exactly as in simple regression, total spread splits into explained and unexplained: $SST = SSR + SSE$. The **coefficient of multiple determination** is the share of the total variation in Y that **all the predictors together** explain.

COEFFICIENT OF MULTIPLE DETERMINATION

$r^2 = \frac{SSR}{SST}$ $SST = SSR + SSE$ · $0 \leq r^2 \leq 1$ · "fraction of variation in Y explained by the model"



$SST = SSR + SSE$, the same decomposition as simple regression but now SSR is the spread captured by the **whole set** of predictors. $r^2 = SSR/SST$ is the blue (explained) share of the total violet bracket. Adding any predictor can only *raise* SSR – which is exactly why plain r^2 needs a penalty.

11.4 Adjusted r^2 – the penalty for extra X s

Plain r^2 **never falls** when you add a predictor, even a useless random one – so it rewards stuffing the model.

Adjusted r^2 docks a penalty for each extra predictor (each lost degree of freedom), so it **only rises if the new X earns its keep**. Use it to compare models with different numbers of predictors.

ADJUSTED R^2

$r^2_{adj} = 1 - (1 - r^2) \cdot \frac{(n - 1)}{(n - k - 1)}$ $n = \text{sample size}$ · $k = \text{number of predictors}$ ·
 $r^2_{adj} < r^2$ always

✓ How to read the r^2 / adj- r^2 gap

A **small gap** (e.g. 0.58 vs 0.55) means the predictors are mostly pulling their weight. A **large gap**, or an adjusted r^2 that **drops** when you add a variable, is the signal that the extra predictor is just fitting noise – **drop it**. Compare competing models on **adjusted r^2** , never on raw r^2 .

CHAPTER 11 (CONT.)

The F-test vs the individual t-tests

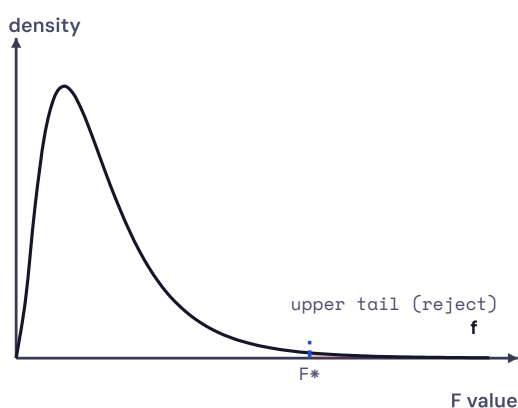
One test for the whole model, then one test per coefficient

11.5 F-test for overall significance

The **F-test** asks a single yes/no question about the entire model: **do the predictors jointly explain anything?** It compares the variance the model explains (MSR) against the variance it leaves behind (MSE). If the explained part is large relative to the leftover noise, F is large and we reject the "all slopes zero" null.

F-TEST – OVERALL SIGNIFICANCE

$$F = \frac{MSR}{MSE} = \frac{SSR / k}{SSE / (n - k - 1)} \quad df = (k, n - k - 1) \quad H_0: \beta_1 = \dots = \beta_k = 0 \quad \text{Excel: } =F.INV(1-\alpha, k, n-k-1)$$



The **F-distribution** lives on $x \geq 0$ and is right-skewed. The test is **upper-tailed**: a large observed F lands in the shaded tail beyond **F***, so we **reject H_0** – the model explains real variance. Equivalently, reject when the **ANOVA p-value** $< \alpha$.

i F-test = the ANOVA line of the Excel output

In **Data → Regression**, the ANOVA table gives SSR (Regression), SSE (Residual), their df (k and n-k-1), the mean squares MSR and MSE, the **F** statistic and its **Significance F** (the p-value). Reject the overall null when Significance F $< \alpha$.

CHAPTER 11 (CONT.)

Individual t-tests on each coefficient

After the F gate, one test per slope to see which predictor pulls its weight

11.6 Individual t-tests on each coefficient

Once F says the model as a whole works, the **t-test on each slope** says **which** predictors are doing the work. For predictor j it tests $H_0: \beta_j = 0$ against $H_1: \beta_j \neq 0$, holding the others in the model.

INDIVIDUAL T-TEST ON B_j

$$t = \frac{b_j}{s_{b_j}} = \frac{\text{Coefficient}}{\text{Standard Error}} \quad df = n - k - 1 \cdot \text{reject if } |t| > t_{\alpha/2} \text{ or } p\text{-value} < \alpha$$

Two layers, one logic

- **F-test (global)**: is the model useful at all? *Run first.*
- **t-tests (local)**: which specific X_j contributes, holding others constant?
- Each Excel coefficient row already gives Coef, SE, t-Stat and P-value — just compare the P-value to α .

When the two disagree

A significant **F** with **no** significant individual t-tests is the classic **multicollinearity** footprint: the predictors overlap so heavily that the model clearly works, yet no single X can be credited. Cross-check the predictor correlations before trusting any one coefficient.

! Don't skip the F-test and t-shop the coefficients

Running lots of individual t-tests and keeping whichever look significant inflates your Type I error. The **F-test gates the whole model first**; only after it rejects do the individual t-tests earn their meaning. Reporting just the “nice” coefficients is the p-hacking the ethics section warns against.

CHAPTER 11 (CONT.)

Dummy variables & residual diagnostics

Putting categories into the model, then checking the LINE assumptions

11.7 Dummy (0/1) variables

Regression needs numbers, but drivers like **Metro / Regional** or **Member / Non-member** are categories. Code a two-level category as a **dummy variable**: 1 for one level, 0 for the other. The level coded **0** is the **reference (baseline)** category.

DUMMY VARIABLE CODING

$D = 1$ (category present) · $D = 0$ (reference category)
coefficient b_D = the **difference in mean Y** between the $D = 1$ group and the reference group, others held constant

Reading the dummy coefficient

In $\hat{Y} = \dots + 5.0 \cdot \text{Metro}$, with $\text{Metro} = 1$ for metro stores and 0 for regional (the reference), the coefficient **5.0** means a metro store is predicted to earn **\$5,000 more** than an otherwise-identical regional store — same ad spend, same reps. The dummy **shifts the whole line up by b_D** ; it does not change the slopes of the numeric predictors.

The c – 1 rule

A category with **c levels** needs **c – 1 dummies** (one level is the reference). Coding all c levels is the **dummy-variable trap** — perfect multicollinearity that breaks the fit. Three regions → two dummies; the third region is the baseline every coefficient is measured against.

! The dummy coefficient is ALWAYS “vs the reference”

A dummy slope is never an absolute level — it is the gap between that category and the **omitted reference category**, holding the numeric predictors constant. Always name the baseline when you interpret it (“\$5,000 more than a **regional** store”), and remember the intercept already bakes in the reference level.

11.8 Residual analysis — the LINE assumptions

The inference (F, t, intervals) is only valid if the model’s errors behave. Check the **residuals** ($e = Y - \hat{Y}$) against four assumptions, remembered as **L-I-N-E**:

Assumption	What it means	How you check it
Linearity	relationship really is linear in the Xs	residual-vs-fitted shows no bend
Independence	errors are not correlated (esp. over time)	Durbin-Watson ≈ 2 ; plot residuals in order
Normality	errors are roughly normal	histogram / normal probability plot of residuals
Equal variance	constant spread (homoscedasticity)	residual-vs-fitted is a flat band , not a funnel

★ Half a mark each — name the plot AND the verdict

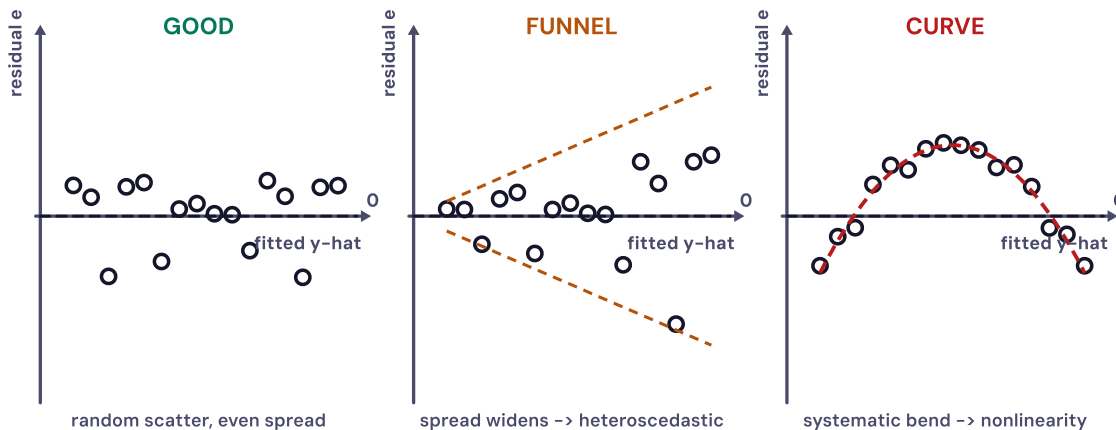
Diagnostics questions want the **plot** and the **conclusion**: “the residual-vs-fitted plot **fans out** → **unequal variance / heteroscedasticity** → the standard errors are wrong → try a **log or $\sqrt{}$ transform of Y**.” State the fault, its consequence, and the fix.

CHAPTER 11 (CONT.)

Reading the residual plot

Good scatter vs the funnel of heteroscedasticity, and autocorrelation

Almost every diagnostics mark comes from **one picture**: residuals on the vertical axis, fitted $\hat{Y}$ on the horizontal, centred on the $e = 0$ line. A healthy model leaves a **formless, even-width band**. Any **shape** in the cloud is the model telling you an assumption has failed.



GOOD (left): a flat, formless band of constant spread – assumptions hold, inference is valid.

FUNNEL (middle): the spread **widens** as $\hat{Y}$ grows = **heteroscedasticity** (unequal variance) – the standard errors, and so the t- and F-tests, are unreliable. **CURVE** (right): a systematic **bend** = a missed **nonlinearity** – add a term or transform.

Heteroscedasticity (the funnel)

When error spread grows with the size of $\hat{Y}$ – common with revenue, sales or income data – the band **fans out**. The coefficients are still unbiased, but their **standard errors are wrong**, so p-values and intervals can't be trusted. Usual fix: model $\log(Y)$ or $\sqrt{Y}$ to stabilise the variance.

Autocorrelation (Durbin–Watson)

With **time-ordered** data, errors can be correlated with their neighbours, breaking **independence**. The **Durbin–Watson** statistic detects it: $DW \approx 2$ means no autocorrelation, $DW \rightarrow 0$ signals positive autocorrelation, $DW \rightarrow 4$ negative. Always plot residuals **in time order** for time-series models.

§ Multicollinearity – correlated predictors

When predictors are themselves highly correlated, individual coefficients become **unstable** (big standard errors, signs that flip, t-tests insignificant even with a significant F). It does not bias prediction, but it wrecks **interpretation** of single slopes. Spot it via a high pairwise correlation among the Xs; the cure is usually to **drop one** of the overlapping predictors.

i Assumptions decide whether the numbers mean anything

r^2 , F and the t-tests are only trustworthy if the LINE assumptions hold. A model with a gorgeous r^2 but a clear funnel or curve in its residuals is **not** a good model – the fit statistics are built on broken foundations.



CHAPTER 11 (CONT.)

The KEY example: a revenue model

Interpret every coefficient, predict, and judge the fit – final Part B style

This is the flagship MLR question. A store chain regresses **weekly revenue Y (in \$000)** on weekly **ad spend X_1 (\$000)**, number of **sales reps X_2** , and a **Metro dummy X_3** (1 = metro, 0 = regional). On $n = 80$ stores the fitted model and fit statistics are:

WE-11A · THE FITTED REVENUE MODEL

$$\hat{Y} = 30 + 2.1 \cdot \text{AdSpend} + 1.4 \cdot \text{Reps} + 5.0 \cdot \text{Metro} \quad R^2 = 0.58 \cdot r^2_{\text{adj}} = 0.55 \cdot n = 80 \cdot k = 3$$

· Metro = 1 (metro) / 0 (regional)

WE-11A Interpret, predict, and judge the revenue MLR

WE-11a · INTERPRET + PREDICT + FIT

Part 1 — interpret each coefficient (all in \$000, others held constant).

- 1 **$b_0 = 30$:** a regional store (Metro = 0) with zero ad spend and zero reps is predicted to earn **\$30,000** — a baseline, not literally meaningful since no store runs at all-zero.
- 2 **$b_1 = 2.1$:** each extra **\$1,000 of ad spend** is predicted to raise revenue by **\$2,100**, holding reps and location fixed.
- 3 **$b_2 = 1.4$:** each extra **sales rep** adds a predicted **\$1,400**, holding ad spend and location fixed.
- 4 **$b_3 = 5.0$ (the Metro dummy):** a **metro** store earns a predicted **\$5,000 more** than an otherwise-identical **regional** store (the reference), same ad spend and reps.

Part 2 — predict revenue for a **metro** store with AdSpend = 10 and Reps = 4 (so $X_1 = 10$, $X_2 = 4$, $X_3 = 1$).

- 1 Substitute: $\hat{Y} = 30 + 2.1(10) + 1.4(4) + 5.0(1)$.
- 2 Evaluate: $\hat{Y} = 30 + 21 + 5.6 + 5.0 = 61.6$.
- 3 Units: revenue is in \$000, so the prediction is **\$61,600**.

Part 3 — judge the fit (r^2 , $\text{adj } r^2$, and the F-test).

- 1 **$r^2 = 0.58$:** the three predictors together explain **58%** of the variation in weekly revenue.
- 2 **$\text{adj } r^2 = 0.55$:** after the penalty for $k = 3$ predictors, **55%** — the small 0.03 gap means the predictors are mostly earning their keep (no obvious dead weight).
- 3 **F-test:** $H_0: \beta_1 = \beta_2 = \beta_3 = 0$. With a high r^2 on $n = 80$, $F = \text{MSR}/\text{MSE}$ is large and Significance $F < 0.05 \rightarrow$ **reject H_0** : the model is jointly useful. Then read the individual t-tests to see which of ad spend, reps and location are significant on their own.

\$61.6k

PREDICTED METRO-STORE REVENUE

58% R^2 EXPLAINED BY THE MODEL**55%**ADJ R^2 AFTER THE K-PENALTY

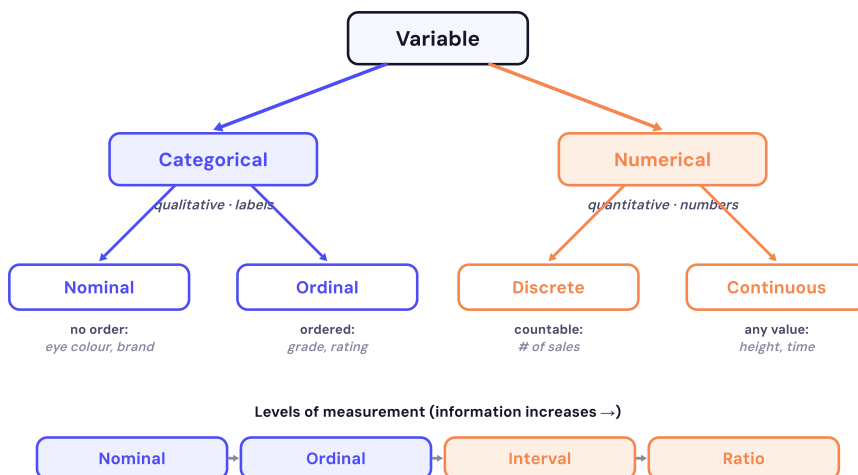
Mini practice exam — Part 1

In-semester-test style MCQ (Weeks 1–6, closed-book): classify, describe, probability

IN ONE LINE

The real in-semester test is closed-book, in person, 90 minutes, ~18 MCQ + 12 short-answer, covering Weeks 1–6 only (data, descriptive measures, probability, distributions, sampling). A formula sheet and an Excel-formula page are supplied; bring a non-programmable calculator. Treat every MCQ below as 2 marks and work it to the answer *before* reading the solution.

These questions are re-authored with fresh numbers and business contexts — they teach the same examinable **skills** as the real paper without reproducing any stem. Time yourself at roughly **2 minutes per MCQ**: the test rewards quick, correct classification as much as arithmetic.



Q1–Q2 lean on this. First split a variable into **categorical** (labels) vs **numerical** (counts/measurements); then categorical → nominal/ordinal, numerical → discrete/continuous. The four **levels of measurement** rank the information carried: nominal < ordinal < interval < ratio.

Q1 MCQ · WK 1

2 marks

A logistics firm records, for each delivery: the **courier's star rating** (1–5 stars), the **postcode** of the drop-off, and the **parcel weight in kg**. Which classification is correct?

- (A) star rating = ratio; postcode = numerical; weight = continuous
- (B) star rating = ordinal; postcode = nominal; weight = continuous
- (C) star rating = interval; postcode = ordinal; weight = discrete
- (D) all three are numerical

✓ Worked solution

Answer (B). A 1–5 **star rating** is **ordinal** categorical — ordered labels with no meaningful equal spacing. A **postcode** is a number used purely as a label, so it is **nominal**, not numerical. **Weight in kg** can take any value in a range → **continuous numerical**. (A)/(C)/(D) misread one or more of these.

PRACTICE EXAM (CONT.)

Mini practice exam — Part 1 (cont.)

In-semester-test style MCQ: sampling methods and the geometric mean

Q2 MCQ · WK 1

2 marks

A market-research team surveys every 50th shopper entering a mall from a random start. This sampling method is best described as:

- (A) simple random sampling
- (B) stratified sampling
- (C) systematic sampling
- (D) convenience (non-probability) sampling

✓ **Worked solution**

Answer (C). Choosing every k-th unit (here every 50th) after a random start is **systematic** sampling — a probability method. It is *not* simple random (not every subset is equally likely), not stratified (no grouping into strata then sampling within), and not convenience (the selection rule is defined and random-started).

Q3 MCQ · WK 2

2 marks

Monthly returns on a fund were +30%, -10%, +5% over three years. The **geometric mean return per year** is closest to:

- (A) 8.33% (the arithmetic mean)
- (B) 7.10%
- (C) 6.00%
- (D) 22.85%

✓ **Worked solution**

Answer (B). For multiplicative returns use the **geometric** mean: $\bar{R}_G = [(1.30)(0.90)(1.05)]^{1/3} - 1 = (1.2285)^{1/3} - 1 \approx 1.0710 - 1 = \mathbf{7.10\% \text{ p.a.}}$ (A) is the arithmetic mean, which overstates compounded growth; (D) is the product factor minus 1, not annualised.

PRACTICE EXAM (CONT.)

Mini practice exam — Part 2

In-semester short-answer & calculation (Weeks 2–6): describe, probability, distributions

IN ONE LINE

Short-answer items want the **working shown**: a formula, the substitution, the number, and a one-line interpretation. In Weeks 2–6 the high-frequency skills are the **five-number summary / boxplot**, **conditional probability from a table**, and **choosing the right discrete or continuous distribution**. Each carries part-marks, so always write the method even if the arithmetic stalls.

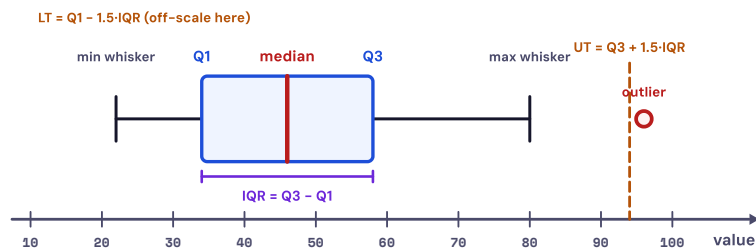
Q4 SHORT ANSWER · WK 2

6 marks

Nine weekly delivery times (minutes) are: **22, 25, 28, 30, 31, 33, 35, 40, 52**. (a) Find the five-number summary and the IQR. (b) State, with reason, the shape of the distribution.

✓ Worked solution

(a) Ranked already. $n = 9$. Median is the 5th value = **31**. Q_1 position = $(n+1)/4 = 2.5 \rightarrow$ halfway between the 2nd (25) and 3rd (28) = **26.5**. Q_3 position = $3(n+1)/4 = 7.5 \rightarrow$ halfway between the 7th (35) and 8th (40) = **37.5**. Five-number summary = **22 / 26.5 / 31 / 37.5 / 52**; $IQR = Q_3 - Q_1 = 37.5 - 26.5 =$ **11 min**. (b) The upper whisker (40, 52) is far longer than the lower whisker, and the max sits 14.5 above Q_3 — a long right tail $\rightarrow$ **right-skewed**.



Q4 visual. The box spans Q_1 to Q_3 (the IQR) with the median line inside; whiskers reach the most extreme points inside the fences $LT = Q_1 - 1.5 \cdot IQR$ and $UT = Q_3 + 1.5 \cdot IQR$. A long **right** whisker (and any high outlier) signals right-skew — exactly the pattern in this delivery-time data.

PRACTICE EXAM (CONT.)

Mini practice exam — Part 2 (cont.)

In-semester short-answer: conditional probability and the Poisson model

Q5 SHORT ANSWER · WK 3

6 marks

Of 500 loan applications, 300 were approved. 240 applicants used a broker; of those, 180 were approved. (a) Find $P(\text{Approved} | \text{Broker})$. (b) Are “broker use” and “approval” independent? Justify numerically.

✓ Worked solution

(a) $P(\text{Approved} | \text{Broker}) = (\text{approved brokered})/(\text{all brokered}) = 180/240 = \mathbf{0.75}$. (b) Marginal $P(\text{Approved}) = 300/500 = \mathbf{0.60}$. Independence requires $P(\text{Approved} | \text{Broker}) = P(\text{Approved})$. Since $0.75 \neq 0.60$, the events are **NOT independent** — using a broker is associated with a higher approval rate.

Q6 SHORT ANSWER · WK 4

5 marks

A call centre receives on average $\lambda = 4$ calls per 5-minute window. (a) Which distribution models the number of calls in a window, and why? (b) Find $P(\text{exactly 2 calls})$ and $P(\text{at least 1 call})$.

✓ Worked solution

(a) **Poisson**: we count events in a continuous interval (a 5-minute window) with no fixed upper limit and a constant mean rate λ . (b) $P(X=2) = e^{-4} \cdot 4^2/2! = e^{-4} \cdot 16/2 \approx 0.01832 \cdot 8 = \mathbf{0.1465}$. $P(X \geq 1) = 1 - P(X=0) = 1 - e^{-4} \approx 1 - 0.0183 = \mathbf{0.9817}$ (complement). Excel: =POISSON.DIST(2,4,0) and =1-POISSON.DIST(0,4,0).

Mini practice exam — Part 3

Final-exam Part A style MCQ (Weeks 1–12 spread, 2 marks each): the back-half topics

IN ONE LINE

The final is closed-book, 120 min + 10 min reading. Part A = 18 MCQ × 2 = 36 (Weeks 1–12); Part B = 5 written = 64 (Weeks 7–12). Formula sheet in the paper; the final is **not a hurdle** (pass on a weighted average $\geq 50\%$). For Part A the trap is **concept choice** — z vs t, which two-sample test, what a confidence level means — so read each stem for the structural cue (σ known? paired? proportion?) before the calculator.

Q7 MCQ · WK 7

2 marks

A 95% confidence interval for a population mean is (56.2, 67.8). Which statement is the correct interpretation?

- (A) There is a 95% probability the true mean lies in (56.2, 67.8).
- (B) 95% of the data lie between 56.2 and 67.8.
- (C) If we repeated the sampling many times, about 95% of such intervals would contain the true mean.
- (D) The sample mean has a 95% chance of being 62.0.

✓ Worked solution

Answer (C). The confidence level is a property of the **procedure**: 95% of intervals built this way capture the fixed (but unknown) parameter. Once computed, a single interval either does or does not contain μ — so (A) is the classic misstatement. (B) confuses a CI with a data range; (D) is meaningless for a fixed sample mean.

Q8 MCQ · WK 7–8

2 marks

You want a confidence interval for a population mean. The population SD σ is **unknown** and the sample size is $n = 25$. The correct critical value comes from:

- (A) the standard normal Z, because n is fixed
- (B) the t-distribution with 25 degrees of freedom
- (C) the t-distribution with 24 degrees of freedom
- (D) the t-distribution with 26 degrees of freedom

✓ Worked solution

Answer (C). σ unknown $\Rightarrow$ estimate it by S and use the **t-distribution**; for one sample the degrees of freedom are $n - 1 = 24$. Z (A) is only for σ known (or very large n); df is $n-1$, not n (B) or $n+1$ (D).

PRACTICE EXAM (CONT.)

Mini practice exam — Part 3 (cont.)

Final-exam Part A style MCQ: the two error types in hypothesis testing

IN ONE LINE

Part A leans hard on the vocabulary of testing: Type I vs Type II, the meaning of α and β , and what “significant” does and does not claim. These are pure-concept marks — get the definitions exact and you bank them in seconds.

Q9

MCQ · WK 8

2 marks

In a hypothesis test, a **Type I error** is:

- (A) failing to reject H_0 when H_0 is actually false
- (B) rejecting H_0 when H_0 is actually true
- (C) choosing the wrong test statistic
- (D) using a one-tailed test when a two-tailed test was needed



Worked solution

Answer (B). A **Type I error** = rejecting a *true* H_0 (a false positive); its probability is α . (A) describes a **Type II error** (probability β ; power = $1 - \beta$). (C)/(D) are modelling mistakes, not the formal error types.

Reality →	H_0 true	H_0 false
Fail to reject H_0	correct ($1 - \alpha$)	Type II error (β)
Reject H_0	Type I error (α)	correct — power ($1 - \beta$)



Read the 2×2 the way the exam asks it

Match the cell to the words: a **false positive** (reject a true H_0) is **Type I**, rate α ; a **miss** (fail to reject a false H_0) is **Type II**, rate β ; and **power** = $1 - \beta$ is correctly catching a real effect. Shrinking α for fixed n raises β — only a bigger sample lowers both.

PRACTICE EXAM (CONT.)

Mini practice exam — Part 3 (cont.)

Final-exam Part A style MCQ: two-sample choice, correlation, and model fit

IN ONE LINE

The back-half MCQs reward a fast which-test read. Paired design → paired t; $r = \pm\sqrt{r^2}$ with the slope's sign; r^2 never falls when predictors are added, so compare models on adjusted r^2 . Decide the concept first, then confirm with a one-line calculation.

Q10

MCQ · WK 9

2 marks

The same 30 employees are scored before and after a training course; you compare the mean scores. The appropriate test is:

- (A) a pooled-variance (independent) two-sample t-test
- (B) a separate-variance (Welch) t-test
- (C) a paired-difference t-test, $df = 29$
- (D) a two-proportion Z-test

✓ Worked solution

Answer (C). The *same* units measured twice are **paired**: reduce each pair to a difference $D = \text{after} - \text{before}$ and run a one-sample t on D with $df = n - 1 = 29$. The independent-samples t-tests (A)/(B) ignore the pairing and lose power; (D) is for success/failure proportions.

Q11

MCQ · WK 10

2 marks

A simple linear regression of sales on training hours gives $r^2 = 0.49$ with a **positive** slope. The correlation coefficient r is:

- (A) +0.70
- (B) -0.70
- (C) +0.24
- (D) +0.49

✓ Worked solution

Answer (A). $r = \pm\sqrt{r^2} = \pm\sqrt{0.49} = \pm 0.70$; the sign follows the slope, which is positive, so $r = +0.70$. (B) has the wrong sign; r^2 itself is not r (D); $0.24 \approx r^{22}$ is a distractor.

PRACTICE EXAM (CONT.)

Mini practice exam — Part 3 (cont.)

Final-exam Part A style MCQ: r^2 vs adjusted r^2 when a predictor is added

IN ONE LINE

A favourite Part A trap: what happens to r^2 and adjusted r^2 when you add a predictor. r^2 can only rise (or hold); adjusted r^2 can fall if the new variable does not earn its keep — which is exactly why models are compared on adjusted r^2 .

Q12

MCQ · WK 12

2 marks

In a multiple regression, adding an extra predictor that contributes almost nothing will:

- (A) always increase both r^2 and adjusted r^2
- (B) increase (or not decrease) r^2 , but can decrease adjusted r^2
- (C) decrease r^2 but increase adjusted r^2
- (D) leave both r^2 and adjusted r^2 unchanged

 Worked solution

Answer (B). r^2 **never falls** when a predictor is added (it can only explain more or the same variance). **Adjusted r^2** penalises extra predictors — $r^2_{\text{adj}} = 1 - (1-r^2)(n-1)/(n-k-1)$ — so a useless predictor can **lower** it. That is exactly why adjusted r^2 is used to compare models with different numbers of predictors.

 Compare models on adjusted r^2 , never raw r^2

Because raw r^2 can only climb as you bolt on predictors, picking the model with the highest r^2 always rewards the most bloated one. **Adjusted r^2** is the fair referee: it rises only when a new variable explains more than its degree-of-freedom cost, so the model with the **highest adjusted r^2** is the better-justified one.

PRACTICE EXAM (CONT.)

Mini practice exam — Part 4

Final Part B style written answers: confidence interval & two-sample test (Weeks 7–9)

IN ONE LINE

Part B (Weeks 7–12, 64 marks) wants full written solutions: state H_0/H_1 , justify z vs t and the test choice, substitute into the formula sheet equation, decide, and interpret in context. Part-marks reward the method, so show every line. The two staples below are a confidence interval with a sample-size follow-up and a two-sample comparison.

Q13 WRITTEN · WK 7

12 marks

A random sample of **40 receipts** at a bookshop has mean spend $\bar{X} = \$62$ and sample SD $S = \$18$. (a) Construct a 95% confidence interval for the mean spend. (b) The manager wants the margin of error reduced to **\$4**; assuming $\sigma \approx \$18$, how many receipts are needed? (c) Interpret the interval in one sentence.

✓ Worked solution

(a) σ is **unknown** $\Rightarrow$ use t with $df = n - 1 = 39$; $t_{0.025,39} \approx 2.02$. $CI = \bar{X} \pm t(S/\sqrt{n}) = 62 \pm 2.02(18/\sqrt{40}) = 62 \pm 2.02(2.846) = 62 \pm 5.75 \rightarrow (\$56.25, \$67.75)$. (b) Sample size for a mean: $n = Z^2\sigma^2/e^2$ with $Z = 1.96$, $\sigma = 18$, $e = 4 \rightarrow n = (1.96^2 \cdot 18^2)/4^2 = (3.8416 \cdot 324)/16 = 1244.7/16 \approx 77.8 \rightarrow$ round **up** to **$n = 78$ receipts**. (c) We are 95% confident that the *true* mean spend per receipt lies between \$56.25 and \$67.75 — i.e. 95% of intervals built this way would capture the population mean.

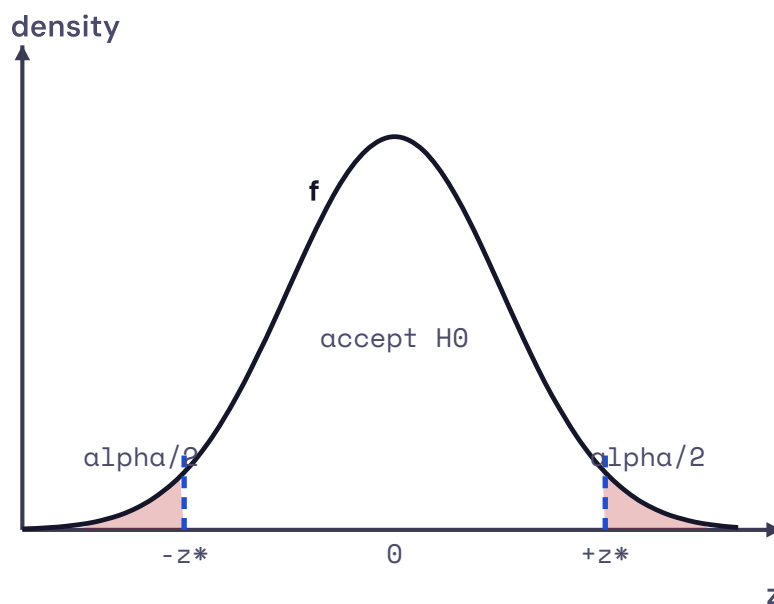
(\$56.25, \$67.75)

95% CI FOR μ

\$5.75

MARGIN OF ERROR

78

RECEIPTS FOR $E = \$4$


The two-sample test (Q14) uses this picture. A two-tailed test at $\alpha = 0.05$ splits α into two tails of 0.025 beyond the critical values (here $\pm t^*$). If the observed statistic lands in a shaded tail $\rightarrow$ **reject H_0** ; if it sits in the unshaded middle $\rightarrow$ **fail to reject**. The same logic drives the one-tailed test, with the whole α in a single tail.

PRACTICE EXAM (CONT.)

Mini practice exam — Part 4 (cont.)

Final Part B style: a full two-sample hypothesis test, line by line

IN ONE LINE

A Part-B two-sample answer is a seven-line drill: hypotheses → name and justify the test → pool the variances → standard error → statistic and df → decision against the critical value → a plain-English conclusion. The last mark is the conclusion in context, so never stop at the number.

Q14

WRITTEN · WK 9

14 marks

Time-on-task (minutes) is measured for two independent groups of website visitors. **Layout A:** $n_1 = 12$, $\bar{X}_1 = 8.4$, $S_1 = 1.8$. **Layout B:** $n_2 = 14$, $\bar{X}_2 = 7.1$, $S_2 = 2.0$. Assuming equal variances, test at $\alpha = 0.05$ (two-tailed) whether the mean times differ.

✓ Worked solution

1. Hypotheses. $H_0: \mu_1 - \mu_2 = 0$; $H_1: \mu_1 - \mu_2 \neq 0$ (two-tailed; no prior direction). **2. Choose the test.** Two independent samples, σ unknown, equal variances assumed → **pooled-variance t**, $df = n_1 + n_2 - 2$. **3. Pool.** $S_p^2 = [(11)(1.8^2) + (13)(2.0^2)] / (12 + 14 - 2) = (35.64 + 52.00) / 24 = 3.652$. **4. Standard error.** $SE = \sqrt{S_p^2(1/n_1 + 1/n_2)} = \sqrt{3.652 \cdot (1/12 + 1/14)} = \sqrt{3.652 \cdot 0.1548} \approx 0.752$. **5. Statistic.** $t = (8.4 - 7.1) / 0.752 = 1.3 / 0.752 \approx 1.73$, $df = 24$. **6. Decision.** Critical $t_{0.025, 24} \approx \pm 2.064$. Since $|1.73| < 2.064 \rightarrow$ **fail to reject H_0** . **7. Conclusion.** At $\alpha = 0.05$ there is no significant difference in mean time-on-task between the layouts. Excel two-tailed p-value: $=T.DIST.2T(1.73, 24)$.

! Pool the variances, never the standard errors — and keep V/order consistent

The most expensive Part-B slips are: (i) averaging S_1 and S_2 directly instead of pooling the **variances** with df weights; (ii) using $(1/(n_1 + n_2))$ instead of $(1/n_1 + 1/n_2)$ in the SE; (iii) reading the wrong df (it is $n_1 + n_2 - 2$ for the pooled t, not $n - 1$). Get the standard error right and the decision usually follows.

i Sample-size formulas always round UP

Back on the CI question: whether for a mean ($n = Z^2 \sigma^2 / e^2$) or a proportion ($n = Z^2 \pi(1 - \pi) / e^2$), a fractional n must be **rounded up** to the next whole unit — rounding down would leave the margin of error larger than required. If π is unknown for a proportion, use $\pi = 0.5$ for the most conservative (largest) sample size.

PRACTICE EXAM (CONT.)

Mini practice exam — Part 5

Final Part B style: regression (Weeks 10–12) and how to revise the whole unit

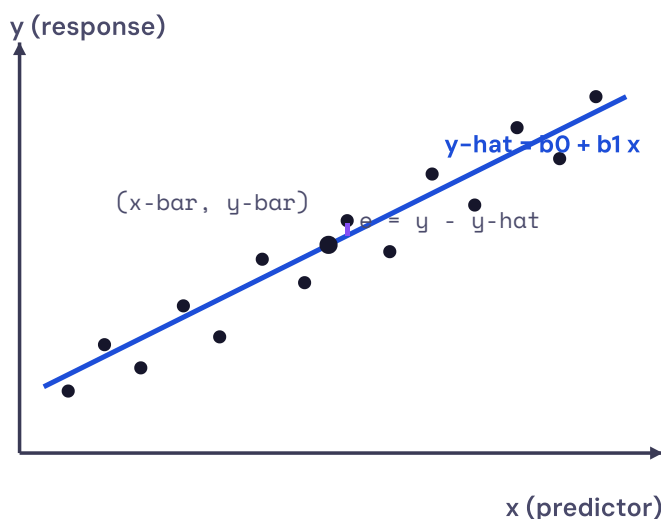
Q15 WRITTEN · WK 10–11

16 marks

A regression of monthly maintenance cost Y (\$) on machine age X (years) gives $\hat{Y} = 120 + 4.2X$, with slope standard error $S_{b_1} = 1.1$, $n = 30$, $r^2 = 0.46$. (a) Interpret the slope and intercept. (b) Test $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$ at $\alpha = 0.05$. (c) Build a 95% CI for the slope. (d) Interpret r^2 .

Worked solution

(a) Slope $b_1 = 4.2$: each additional **year** of machine age is associated with an extra **\$4.20** of monthly maintenance cost, on average. Intercept $b_0 = 120$: the predicted cost at $X = 0$ (a brand-new machine) — often an extrapolation, so interpret with care. (b) $t = (b_1 - 0)/S_{b_1} = 4.2/1.1 \approx 3.82$, $df = n - 2 = 28$. Critical $t_{0.025, 28} \approx \pm 2.048$. Since $3.82 > 2.048 \rightarrow$ **reject H_0** : there is a significant linear relationship between age and cost. (c) 95% CI for β_1 : $b_1 \pm t_{0.025, 28} \cdot S_{b_1} = 4.2 \pm 2.048 \cdot 1.1 = 4.2 \pm 2.25 \rightarrow$ **(1.95, 6.45)**. It excludes 0, echoing the rejection in (b). (d) $r^2 = 0.46$: **46%** of the variation in maintenance cost is explained by the linear relationship with machine age; the remaining 54% is unexplained.



Q15 picture. The least-squares line $\hat{Y} = b_0 + b_1X$ passes through the centroid $(\bar{X}, \bar{Y})$ and minimises $\sum$ of the squared vertical residuals $e = Y - \hat{Y}$. A significant positive slope (b) and a moderate r^2 (d) mean the cloud trends up but still scatters around the line.

PRACTICE EXAM (CONT.)

Mini practice exam — Part 5 (cont.)

Final Part B style: a multiple-regression written answer, interpret and predict

Q16

WRITTEN · WK 12

10 marks

A multiple regression of weekly revenue Y (\$000) gives $\hat{Y} = 30 + 2.1 \cdot \text{AdSpend} + 1.4 \cdot \text{Reps} + 5.0 \cdot \text{Metro}$ (Metro = 1 if a metro store, else 0), with $R^2 = 0.58$, adjusted $R^2 = 0.55$, $n = 80$. (a) Interpret the AdSpend and Metro coefficients. (b) Predict revenue for a metro store with AdSpend = 10 and Reps = 4. (c) Why report adjusted R^2 alongside R^2 ?

✓ Worked solution

(a) AdSpend coefficient 2.1: holding Reps and Metro constant, each extra **\$1,000** of ad spend is associated with **+\$2,100** revenue. Metro coefficient 5.0 (a dummy): a metro store averages **\$5,000** more per week than a regional store (the reference category), all else equal. (b) $\hat{Y} = 30 + 2.1 \cdot 10 + 1.4 \cdot 4 + 5.0 \cdot 1 = 30 + 21 + 5.6 + 5.0 = 61.6 \rightarrow \text{\$61,600}$. (c) R^2 never falls as predictors are added, so it cannot fairly compare models of different size; **adjusted R^2** penalises extra predictors and only rises if a new variable earns its place — making it the honest model-comparison measure.

★ Exam move — the MLR written-answer drill

For any multiple-regression Part B item: (1) interpret each slope with **units** and the phrase “**holding the other predictors constant**”; (2) read a **dummy** as the gap **vs the reference category**; (3) **predict** by substituting every X (dummy = 0 or 1) and keeping the units of Y ; (4) judge the model with **r^2** (explained share) and **adjusted r^2** (the fair cross-model comparison). The equation is on the sheet — the marks are in the words.

! Two faults that cost MLR marks every year

(i) Dropping the **units** — write **\$2,100**, not “2.1”, when Y is in \$000. (ii) Reading a **dummy as an absolute level** — it is “\$5,000 more than a regional store”, never “metro stores earn \$5,000”. Name the baseline every time.

PRACTICE EXAM (CONT.)

Mini practice exam — Part 5 (cont.)

The Part B answer template and a whole-unit revision plan

IN ONE LINE

Two takeaways close out the practice exam: a **reusable five-line template** for any written answer, and a **week-by-week revision map**. Across the whole unit the marks reward the same thing — **choosing the right method and interpreting the result** — because the formula sheet hands you the equations.

i Part B answer template (use it on every written question)

(1) State H_0/H_1 or the quantity sought. **(2)** Name and justify the procedure — z vs t, which two-sample test, SLR vs MLR. **(3)** Write the formula-sheet equation, substitute, and compute. **(4)** Decide against the critical value (or p-value vs α) / report the interval. **(5)** Interpret in the **business context** of the question — the last mark is almost always for the plain-English conclusion.

★ How to revise BUSS1020 — the whole-unit plan

Weeks 1–6 (test + Part A): drill classification (data types, levels), the five-number summary/boxplot, and the **distribution decoder** — binomial (fixed n trials), Poisson (events per interval), hypergeometric (without replacement), normal/uniform/exponential. **Weeks 7–12 (Part B):** master the **inference spine** — $CI = \text{estimate} \pm (\text{crit})(SE)$; the z-vs-t and which-two-sample-test decisions; and regression interpretation (slope, r^2 , adjusted R^2 , dummies). Learn the **method**, not the numbers: the formula sheet gives the equations, so marks come from **choosing the right tool and interpreting the answer**.

Closed-book, on a provided formula sheet, with a non-programmable calculator — so the exam tests whether you can **pick the right method and read the result**, not whether you memorised equations.

BUSS1020 · EXAM STRATEGY



The 30-minute recap

Map the paper, then let the decision trees pick your tool - the constants are on the sheet

IN ONE LINE

This is a closed-book exam with a formula sheet and an Excel-function page supplied in the paper. So you are not marked on recalling formulas – you are marked on picking the right tool (which distribution, z or t, which test) and interpreting the answer in plain business English. Bring a non-programmable calculator; no Excel in the room.

1 Know the shape of the paper

The **final (40%)** is closed-book, in-person, 120 min + 10 min reading. **Part A = 18 MCQ × 2 marks = 36** and covers **Weeks 1-12**; **Part B = 5 written-answer questions = 64 marks** and covers **Weeks 7-12** (CIs, hypothesis tests, regression). The **in-semester test (20%)** is closed-book, 90 min, and covers up to and including **Week 6**.

120 min

FINAL + 10 MIN READING

36 mk

PART A - 18 MCQ X2 - WK 1-12

64 mk

PART B - 5 WRITTEN - WK 7-12

i The final is NOT a hurdle

You pass on a **weighted average $\geq 50\%$** across all components; there is no single-component hurdle. So Part B written marks (CIs, tests, regression) are where the big, recoverable marks live – **show every step**, because method marks survive an arithmetic slip.

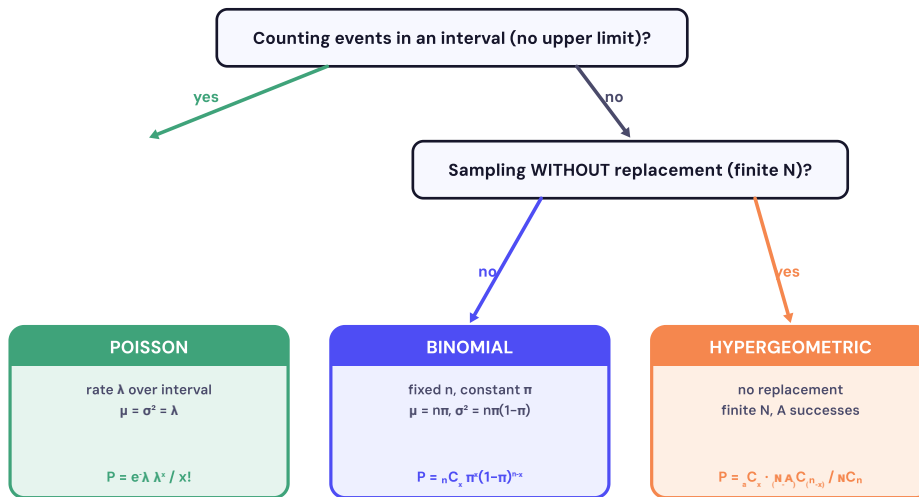
EXAM MORNING (CONT.)

First decision: which distribution?

Read the stem for the cue words, then commit before you compute

2 First decision: which distribution?

Half the MCQ marks turn on naming the model before you compute. Read the stem for the **cue words**, then commit.



Binomial sampling WITH replacement $\rightarrow \pi$ constant; without $\rightarrow$ hypergeometric.

The discrete fork. **BINOMIAL** - a fixed number n of independent trials, each a success/failure with constant π ; count the successes ($\mu = n\pi, \sigma^2 = n\pi(1-\pi)$). **POISSON** - count events in a continuous interval (time/area/volume) at mean rate λ , no fixed upper limit ($\mu = \sigma^2 = \lambda$). **HYPERGEOMETRIC** - sample *without replacement* from a finite population, so π changes draw to draw. Continuous cousins: **uniform** (flat), **normal** (bell), **exponential** (time *between* Poisson events).

✓ Cue-word shortcuts

"out of n ", "% defective", "probability of exactly x successes" $\rightarrow$ binomial. "per hour / per page / on average λ ", "window" $\rightarrow$ Poisson. "without replacement", small finite N $\rightarrow$ hypergeometric. "between events", "time until" $\rightarrow$ exponential (mean $1/\lambda$). "bell-shaped", a mean and SD $\rightarrow$ normal, standardise with $Z = (X-\mu)/\sigma$.

EXAM MORNING (CONT.)

The inference forks: z-or-t, and which two-sample test

Two trees that decide most of Part B

3 Second decision: Z or t?

For any inference **about a mean**, the only question is whether you know the **population SD σ** . **Known $\sigma \rightarrow$ Z. Unknown σ (you only have the sample S) $\rightarrow$ t with $n-1$ degrees of freedom.** For a **proportion**, always Z (with the CLT conditions checked). In practice almost every real Part-B mean question gives you S, not σ – so it is usually **t**.

You want inference about...	What you know	Tool	df
A mean μ	population σ known	Z	–
A mean μ	only sample S (σ unknown)	t	$n - 1$
A proportion π	n large ($n\pi \geq 5, n(1-\pi) \geq 5$)	Z	–
Two means , equal var	pooled S_p^2	pooled t	$n_1 + n_2 - 2$
Two means , unequal var	separate S_1^2, S_2^2	Welch t	Welch
Paired differences	differences D	paired t	$n - 1$
A slope β_1	regression output	t	$n - 2$

Z CRITICAL VALUES (ON THE SHEET - KNOW WHICH TO PICK)

Common critical $Z_{\alpha/2}$: 90% $\rightarrow$ 1.645 95% $\rightarrow$ 1.96 98% $\rightarrow$ 2.33 99% $\rightarrow$ 2.58 **one-tail 5% $\rightarrow$ 1.645**

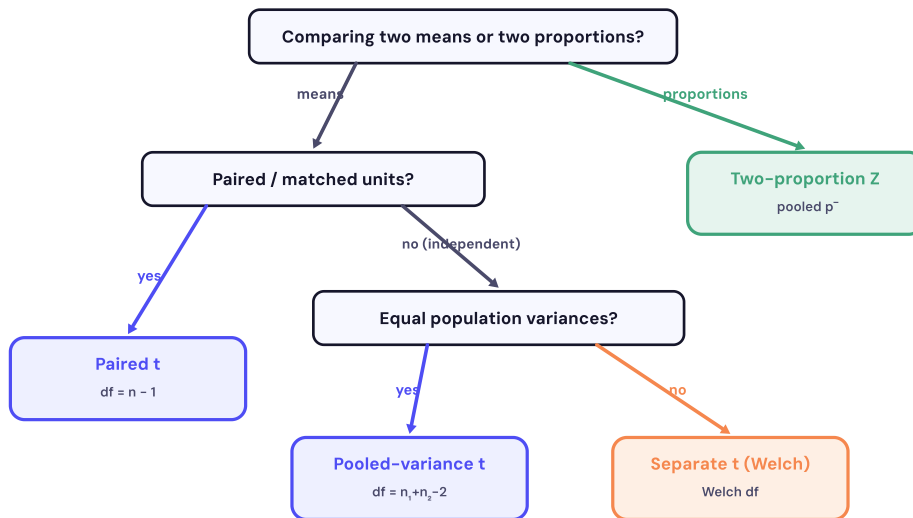
! The classic Z-vs-t slip

Seeing a sample standard deviation S and reaching for Z because "n is large" loses marks. The rule is about **what parameter you know**, not n: if σ is unknown you use **t**, full stop. Yes, for large n the t critical value is very close to Z – but the exam expects you to **name t and use df = n-1**.

Third decision: which two-sample test?

Paired or independent, equal or unequal variance, means or proportions

4 Third decision: which two-sample test?



A 95% CI for the difference that contains 0 $\Leftrightarrow$ fail to reject H_0 at $\alpha = 0.05$.

Picking a two-sample test. First ask: are the two columns **PAIRED** (the same units measured twice, or matched pairs)? $\rightarrow$ run a **paired t on the differences**, $df = n-1$. If the samples are **INDEPENDENT** and you are comparing **means**: equal population variances assumed $\rightarrow$ **pooled-variance t** ($df = n_1+n_2-2$); unequal $\rightarrow$ **separate-variance (Welch) t**.

Comparing two **proportions** $\rightarrow$ **two-proportion Z** with a pooled estimate $\hat{p} = (X_1+X_2)/(n_1+n_2)$.

! Paired data run as independent

Before/after, matched-pair, or same-subject designs are **paired** – collapse them to one column of differences D and run a one-sample t on D . Treating them as two independent samples throws away the pairing and is the single most common two-sample error.

EXAM MORNING (CONT.)

The universal templates + the regression read

One CI shape, one test shape - fill in the standard error and go

5 Every CI and every test is one template

Confidence interval

UNIVERSAL CI

estimate $\pm$ (critical value) $\times$ (standard error)

Pick the estimate ($\bar{X}$ or p or b_1), the critical value (Z or t with the right df), and the standard error ($\sigma/\sqrt{n}$, $S/\sqrt{n}$, $\sqrt{[p(1-p)/n]}$, ...). Interpretation: "**we are 95% confident the true parameter lies in this interval**" - a statement about the *method*, not a probability about a fixed parameter.

Hypothesis test

UNIVERSAL TEST

test stat = (estimate - null value) / standard

error

reject H_0 if $|stat|$ in rejection region, or if $p < \alpha$

Same standard errors as the CI. **Five steps:** state H_0/H_1 , pick α and the tail, compute the stat, find the critical value (or p-value), **decide and write a sentence in context**.

✓ The CI $\leftrightarrow$ test bridge (free marks)

A **two-tailed test at α** and a **(1- α) CI** are the same decision. If the 95% CI for $\mu_1 - \mu_2$ **contains 0**, you **fail to reject H_0** at $\alpha = 0.05$; if it excludes 0, you reject. Use this to sanity-check, or to answer a test from a CI you already built.

RECAP

The five-step test, end to end

RECAP - ONE-SAMPLE t

A vending machine should pour 250 ml. A sample of 36 cups gives $\bar{X} = 246$ ml, $S = 9$ ml; test for under-filling at $\alpha = 0.05$. (σ unknown $\rightarrow t$.)

- 1 **Hypotheses + tail:** $H_0: \mu = 250$ vs $H_1: \mu < 250$ - a left-tailed test.
- 2 **Test stat:** $t = (246 - 250)/(9/\sqrt{36}) = -4/1.5 = -2.67$, $df = 35$.
- 3 **Critical value:** $t_{0.05,35} \approx -1.69$ (one tail).
- 4 **Decide:** $-2.67 < -1.69 \rightarrow$ **reject H_0** . (Excel p-value = $T.DIST(-2.67, 35, 1)$.)
- 5 **Sentence:** there is significant evidence at the 5% level that the machine is **under-filling**.

EXAM MORNING (CONT.)

Reading regression output

Know what each Excel number means before you touch the calculator

6 Reading regression output

Part B almost always shows Excel regression output. Know what each number means before you touch a calculator.

Output	Reads as	Exam move
Slope b_1	ΔY per one-unit ΔX	state units; in MLR add "holding others constant"
Intercept b_0	predicted Y when $X = 0$	often not meaningful – say so
r^2	% of variation in Y explained	$r^2 = SSR/SST$; $r = \pm\sqrt{r^2}$ (sign of slope)
t-Stat / P-value	is this slope $\neq 0$?	$p < \alpha \rightarrow$ predictor is significant
Adjusted r^2	r^2 penalised for extra X's	compare models; use over r^2 in MLR
F (ANOVA)	are ALL slopes 0?	overall-model test in MLR

! **r is not causation, and r^2 is not r**

A strong r measures a **linear** association only – never claim X *causes* Y, and watch for nonlinear patterns r misses. And do not confuse them: $r^2 = 0.49$ means $r = 0.70$ (take the square root), with r 's sign set by the slope.

SLR vs MLR — the two t-questions

In **simple** regression the slope t-test ($df = n - 2$) and the F-test say the **same** thing. In **multiple** regression they split: the **F-test** asks whether the model as a whole works (any $\beta \neq 0$), then each **coefficient t-test** asks which predictor pulls its weight, holding the others constant.

The interpretation, every time

Whatever the output, the marks are in the **sentence**: state the slope with **units**, add "holding the other predictors constant" in MLR, read a **dummy vs its reference category**, and quote r^2 as a **percentage of variation explained**. The numbers are printed — the reading is the exam.

i **The ANOVA line and the coefficient rows**

Excel regression output has two parts you must read: the **ANOVA table** (SSR, SSE, the F statistic and Significance F = the overall p-value) and the **coefficient rows** (each Coef, its Standard Error, t-Stat and P-value). Overall model: Significance F $< \alpha$. Each predictor: its own P-value $< \alpha$.



EXAM MORNING (CONT.)

Top traps + the calm closing checklist

The handful of slips that cost the most marks - then walk in

7 The traps that cost the most marks

Trap	Fix
Mixing population vs sample notation	μ, σ, π = population (Greek); X, S, p = sample (Latin)
Forgetting CLT conditions	means: $n \geq 30$ (or symmetric); proportions: $n\pi \geq 5$ and $n(1-\pi) \geq 5$
Using Z when σ is unknown	unknown $\sigma \rightarrow t$ with $df = n-1$, every time
Reading the wrong tail	let H_1 set the tail; halve α for two-tailed critical values
Saying " accept H_0 "	you fail to reject H_0 - never prove it true
Standard deviation vs standard error	$SE = \sigma/\sqrt{n}$ - it shrinks with n ; SD does not
" 95% probability " about a CI	the parameter is fixed; 95% of such intervals capture it
r^2 vs r , and r = causation	$r = \pm\sqrt{r^2}$ with slope's sign; association $\neq$ causation
No units / context sentence	every Part-B answer ends in a plain-English claim

8 Reach-for-it formulas (verify on the sheet)

THE ONES YOU'LL ACTUALLY USE

$$Z = \frac{(\bar{X} - \mu)_{\sigma}}{\sigma} \cdot SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} \cdot CI = \bar{X} \pm t_{\alpha/2, n-1} \cdot \frac{S}{\sqrt{n}} \cdot r^2 = \frac{SSR}{SST}$$

$$\text{Binomial } \mu = n\pi, \sigma^2 = n\pi(1-\pi) \cdot \text{Poisson } \mu = \sigma^2 = \lambda \cdot \text{Exponential mean} = 1/\lambda$$

i What's given vs what you carry in

The sheet hands you the **formulas and the Z/t critical values**. What it cannot give you is **which one to use** and **what the number means**. That is the whole exam - the trees on pages 1-2 are the part you rehearse this morning.

★ Exam move - the answer ritual for every Part-B question

(1) Classify - one mean / one proportion / two means (paired or independent?) / regression. **(2) Check conditions** - σ known? CLT met? equal variances? **(3) Pick the tool** from the trees - Z or t, which two-sample test, the right standard error. **(4) Compute** the stat or interval, carrying df. **(5) Decide and write the sentence** in business context, with units. Method marks are awarded at every step - so **show the working even if the arithmetic wobbles**.

You are not being tested on memory - the sheet has the formulas. You are being tested on judgement: name the model, check the conditions, pick the tool, and say what it means. You have practised exactly this. Read the stem twice, breathe, and start with the easy MCQ.

BUSS1020 - WALK IN CALM